

Institute for Organic und Macromolecular Chemistry

Heinrich-Heine-University Düsseldorf

Master thesis

**RAFT-derived Macromolecular Initiators for
Anionic Ring-Opening Polymerization**

Submitted by

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Erklärung

Hiermit erkläre ich, dass ich die vorliegende Abschlussarbeit selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt habe.

Düsseldorf, den 18.09.2025

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Abstract

Polymers are one of the most important classes of materials used today. However, the dependence on oil-based reactants and their accumulation in the environment are problematic. Therefore, the synthesis of polymers based ideally on organism-based compounds and the ability to being degraded easily is an important field of research.

Another important field is the synthesis of block copolymers. These kinds of polymers are especially interesting due to their ability to self-assemble into well-defined nanostructures with tunable properties. By combining distinct polymer blocks within a single material, they enable control over morphology, responsiveness, and functionality. Because of this, block polymers can be used in different fields including nanoparticles and surface coatings.

The combination of traditional oil-based polymers with a bio-based degradable polymer within the same chain could enable the obtention of partly degradable and, therefore, triggerable polymers for use in nanoparticles, for example.

In this Thesis, the synthesis of diblock copolymers based on derivatives of α -lipoic acid, a naturally occurring five-membered cyclic disulfide, whose polymerization is initiated by the use of different vinyl monomer-based thiol-terminated macroinitiators is reported. These macroinitiators are obtained through reversible addition/fragmentation chain transfer polymerization. It was shown that these block copolymers could be synthesized in a “one-pot”-reaction, encompassing the transformation of the RAFT end-group into a thiol and the polymerization of α -lipoic acid methyl ester, with molecular weights from $10 \text{ kg}\cdot\text{mol}^{-1}$ to $40 \text{ kg}\cdot\text{mol}^{-1}$ with dispersities ranging from 1.5 to 1.8. Furthermore, owing to the reducible disulfide bonds in the polymer backbone, the degradability of the second, lipoate block and, thus, the reclaiming of the original first block, could be demonstrated.

Zusammenfassung

Polymere gehören zu den wichtigsten Werkstoffklassen, die heute verwendet werden. Allerdings sind die Abhängigkeit von Reaktionspartnern auf Erdölbasis und deren Anreicherung in der Umwelt problematisch. Daher ist die Synthese von Polymeren, die idealerweise aus Organismus-basierten Verbindungen bestehen und leicht abbaubar sind, ein wichtiges Forschungsgebiet.

Ein weiteres wichtiges Forschungsgebiet ist die Synthese von Blockcopolymeren. Diese Art der Polymere ist aufgrund ihrer Fähigkeit zur Selbstassemblierung zu definierten Strukturen besonders von Interesse. Durch die Kombination unterschiedlicher Polymerblöcke in einem einzigen Material lassen sich dabei Morphologie, Responsivität und Funktionalität steuern. Daher können Blockpolymere in verschiedenen Bereichen eingesetzt werden, darunter Nanopartikel und Oberflächenbeschichtungen.

Die Kombination von traditionellen Polymeren, welche auf konventionellen Monomeren aus Erdöl basieren, mit einem biologisch abbaubaren Polymer innerhalb derselben Kette könnte die Herstellung von teilweise abbaubaren und damit schaltbaren Polymeren ermöglichen, die beispielsweise in Nanopartikeln verwendet werden können.

In dieser Arbeit wird die Synthese von Diblock-Copolymeren auf Basis von α -Liponsäure-Derivaten, einem natürlich vorkommenden fünfgliedrigen Disulfid, beschrieben, deren Polymerisation durch die Verwendung verschiedener Thiol-terminierter Makroinitiatoren, initiiert wird. Diese Makroinitiatoren wurden durch reversible Additions-/Fragmentierungs-Kettenübertragungspolymerisation erhalten. Es konnte gezeigt werden, dass diese Blockcopolymere in einer „Ein-Topf“-Reaktion, welche die Transformation der RAFT-Endgruppe zu einem Thiol und die Polymerisation von α -Liponsäuremethylester umfasst, mit Molekulargewichten von $10 \text{ kg}\cdot\text{mol}^{-1}$ bis $40 \text{ kg}\cdot\text{mol}^{-1}$ und Dispersitäten von 1,5 bis 1,8 synthetisiert werden konnten. Darüber hinaus konnte die Abbaubarkeit des Lipoatblocks, welche von den sich im Polymer-Rückgrat befindlichen Disulfidbindungen stammt, und damit die Rückgewinnung des ursprünglichen ersten Blocks gezeigt werden.

List of abbreviations

AA	Asparagusic acid
AIBN	Azoisobutyronitrile
AIP	Aggregation-induced polymerization
ALA	α -Lipoic acid
aROP	Anionic ring-opening polymerization
ATRP	Atom transfer radical polymerization
cROP	Cationic ring-opening polymerization
CTA	Chain transfer agent
DCC	<i>N,N'</i> -Dicyclohexylcarbodiimide
DHLA	Dihydrolipoic acid
DMAA	<i>N,N</i> -Dimethylacrylamide
DMAP	4-Dimethylaminopyridine
DMF	<i>N,N</i> -Dimethylformamide
DOSY-NMR	Diffusion-ordered nuclear magnetic resonance spectroscopy
DTE	Dithioerythritol
DTT	Dithiothreitol
EDC	1-Ethyl-3-(3-dimethylaminopropyl)carbodiimide
FRP	Free radical polymerization
GPC	Gel permeation chromatography
LAM	Less-activated monomer
LpMe	Methyl lipoate
MALDI-TOF-MS	Matrix-assisted laser desorption/ionization – time of flight – mass spectrometry
MA	Methyl acrylate
MAM	More-activated monomer
MMA	Methyl methacrylate
MWD	Molecular-weight distribution
NHS	<i>N</i> -Hydroxysuccinimide
NMP	Nitroxide-mediated polymerization
NMR	Nuclear magnetic resonance
PDC	Pyruvate dehydrogenase complex
PISA	Polymerization-induced self-assembly
RAFT	Reversible addition-fragmentation chain transfer
ROP	Ring-opening polymerization
RP	Radical polymerization

rROP	Radical ring-opening polymerization
SOSIP	Self-organizing surface-initiated polymerization
St	Styrene
TBD	Triazabicyclodecene
TCEP	Tris-(2-carboxyethyl)phosphine
THF	Tetrahydrofuran
TLC	Thin-layer chromatography

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1 Introduction

1.1 Degradable and Responsive Polymers

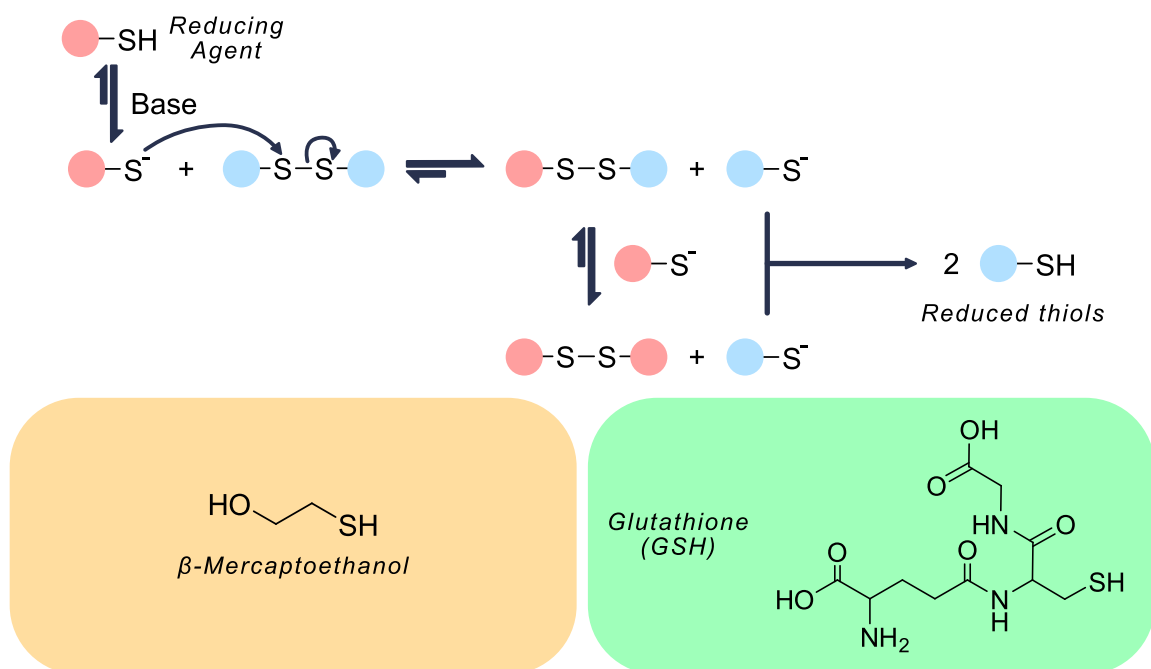
Polymers are one of the most abundant classes of materials. Plastics are used since the 1900s with the introduction of celluloid as a material for movie film. Since then, polymers have many applications, e.g., plastics for packaging, fibers for clothes, and elastomers.^[1] However, due to their longevity, many of these compounds have found their way into the environment. The amount of macro- and microplastics entering the ocean each year was estimated to be 9.5 t, impacting marine life but also human life by proxy.^[2] Therefore, the development of polymers that, if their release is not preventable, show a degree of degradability is a pressing matter.

Another field in which degradable polymers are important is medicine. Materials used for biomedical applications often need to exhibit biocompatibility and, if left in or on the body, also degradability. An example for that would be glycolic- and lactic-acid based sutures, that are used to close wounds and which are then resorbed by the body after a certain amount of time or hydrogels that are supposed to slowly release a pharmaceutical drug into an area of the body.^[3]

Degradable polymers can be subjected to different triggers for degradation, e.g., different values of pH, light and heat. Another one of these triggers is the reduction or oxidation of the polymer.

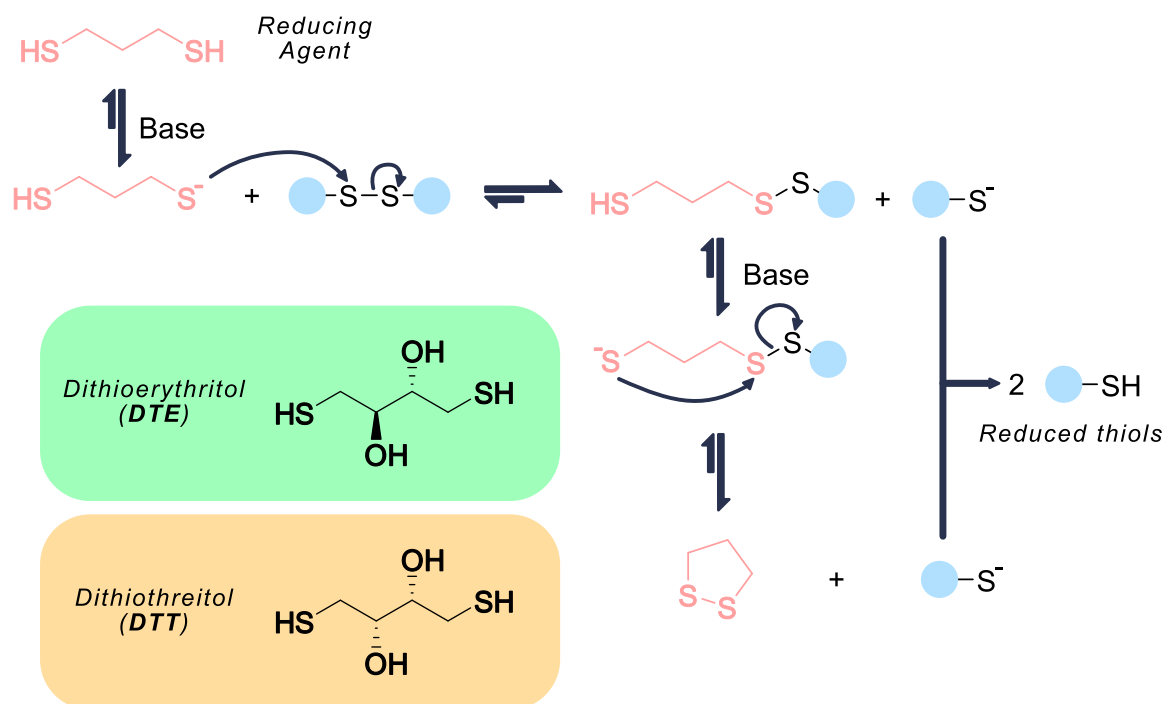
One such redox-responsive link is the disulfide bond, which can be reduced to two thiol groups. The disulfide bond is an example for a dynamic covalent bond, which is easy to cleave but also easily reformed. This dynamic covalent bond is stronger than other supramolecular interactions. Disulfide bonds occur in proteins, where they crosslink two cysteine sidechains. The dynamic nature of this disulfide bridge enables the folding of proteins into their active conformation. Another disulfide containing polymer is rubber. After vulcanization of the rubber with sulfur the formed disulfide and polysulfide bridges between the poly(isoprene) chains enable the elastic behavior of rubber.^[4]

There are various reducing agents to manage the reduction of the disulfide bond. These can be divided into three main categories. Monothiol reducing agents, like glutathione and β -mercaptoethanol, are able to reduce disulfide bonds by a series of disulfide exchange reactions. However, these monothiols need a base as a catalyst, which deprotonates the thiol to the thiolate increasing its nucleophilicity. Also, for the complete reduction of one disulfide bond two monothiols are needed. Glutathione is a notable example of a monothiol reducing agent, because it is the reagent the human body uses to reduce such disulfides.^[5]



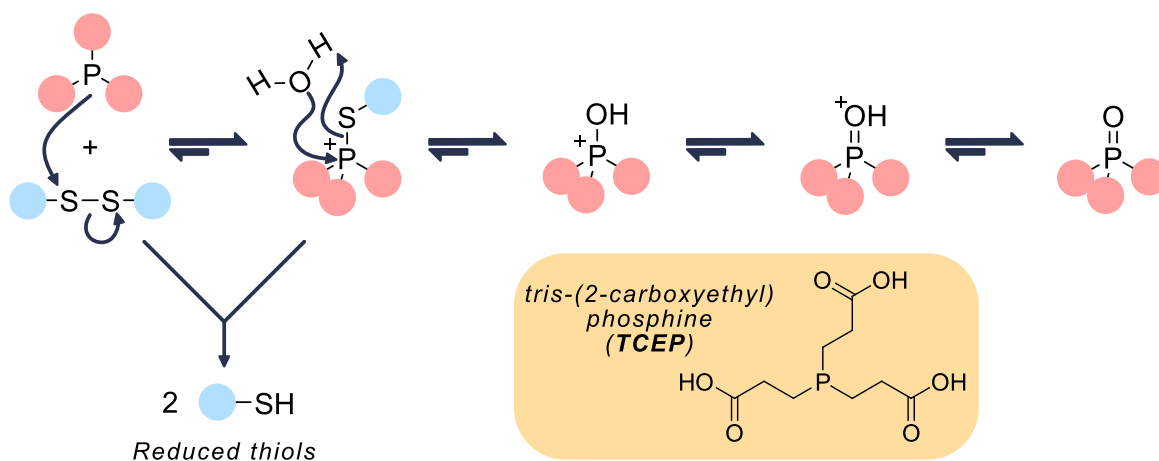
Scheme 1 - Reduction of a disulfide using a monothiol reducing agent along with the structures of commonly used monothiol reducing agents.

The second class of reducing agents is dithiols. First introduced in 1963 by Cleland these dithiols feature two sulfide groups and form stable cyclic compounds when oxidized reducing the amount of reagent needed to only one equivalent.^[6] They also are less toxic and odorous in comparison to the monothiols. However, dithiols only effectively work under basic conditions and only have a short half-life in solution. Because of the ease of synthesizing them, dithioerythritol (**DTE**) and dithiothreitol (**DTT**) are the most used dithiol reducing agents in biochemistry for the reduction of disulfide bridges in proteins.



Scheme 2 - Reduction of disulfides with a dithiol reducing agent along with the most used dithiol reducing agents dithioerythritol and dithiothreitol.

The last group of reducing agents are phosphines. Like thiols, phosphines nucleophilically attack the disulfide bond thus reducing it to free thiols. One of the notable examples include tris-(2-carboxyethyl)phosphine (**TCEP**), which possesses a greater reducing capability than **DTT**. Other advantages include a faster reaction and a wider compatible pH range.^[5]



Scheme 3 - Reduction of disulfides with a phosphine along with the most used phosphine reducing agent **TCEP**.

Polymers with such disulfide bonds in the backbone are called poly(disulfide)s. These polymers are discussed in the next chapter.

1.2 Poly(disulfide)s

Poly(disulfide)s are polymers that contain disulfide bonds in their backbone. The dynamic properties of the disulfide bond enables these polymers to display a high adaptability, stress resistance, self-repair and degradability.^[4]

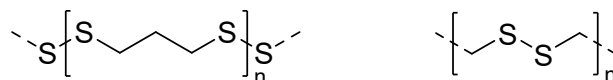
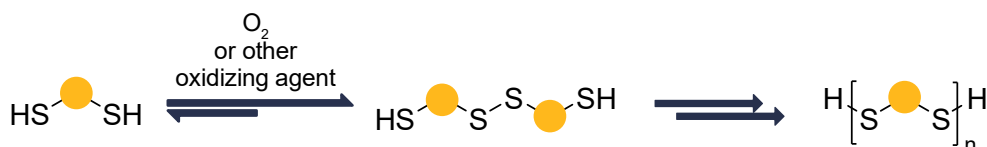


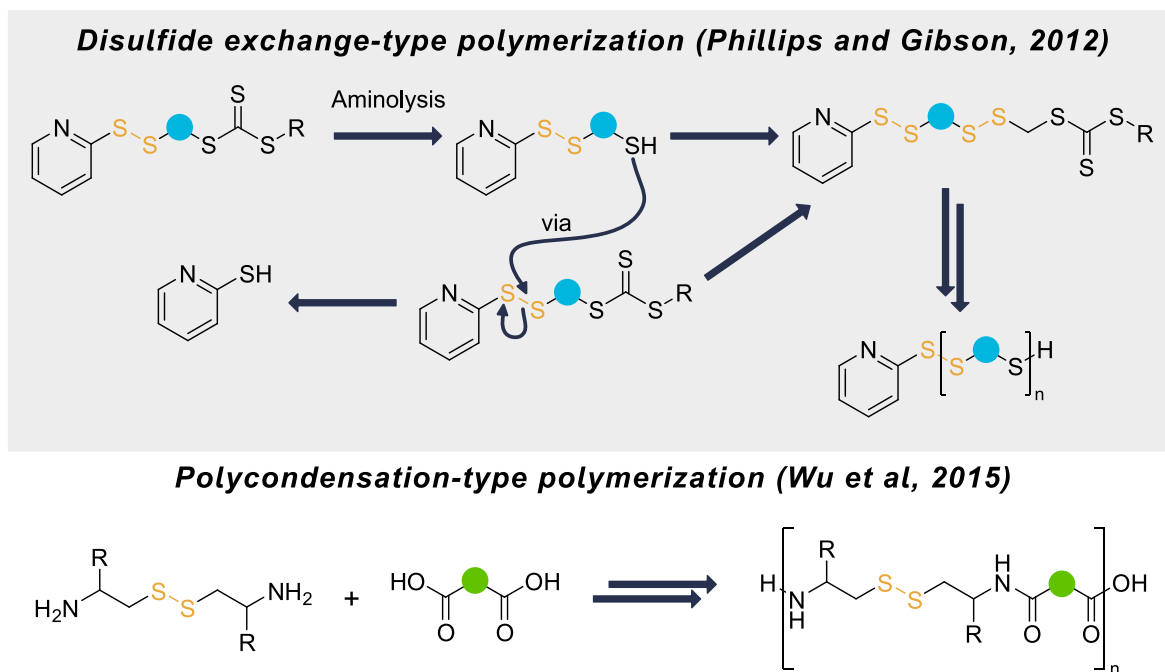
Figure 1 - Structure of poly(disulfide)s; on the left obtained from a ring-opening type reaction, on the right obtained from a polycondensation-type reaction.

Different strategies are used to synthesize poly(disulfide)s, that are detailed in a review from 2021.^[7] Thiols can be oxidized to disulfides. Hence, monomers with two thiol groups can be polymerized to poly(disulfide)s in a step-growth polycondensation-type reaction. The polymerization of dithiols to poly(disulfides) has first been patented in 1944.^[8] Building on that, because polymers made with air as oxidant reach only low molecular weight^[9], many other methods and oxidizing agents have been reported, such as halogens, ferric chloride,^[10] dimethyl sulfoxide,^[11] iodine and hydrogen peroxide^[12].



Scheme 4 - Oxidative polymerization of dithiols using oxygen from air or other oxidizing reagents.

Other polycondensation-type reactions include disulfide-exchange polymerization or the polycondensation of other functional groups. The monomers used either have a terminal disulfide moiety, which is protected with a leaving group and can be attacked with another thiol in a disulfide exchange to obtain the wanted polymer^[13], or they contain a non-terminal disulfide and are polymerized via the formation of amide bonds^[14] or urethane bonds.^[15]



Scheme 5 - Polycondensation-type polymerizations for the obtention of poly(disulfide)s according to Philips and Gibson and Wu et al.^[13,14]

The last major approach for the synthesis of poly(disulfide)s is ring-opening polymerization. Here either multifunctional heterocyclic rings with a disulfide and another polymerizable group, e.g., a lactone, or rings with a disulfide bond as the polymerizable unit can be used. Ring-opening polymerization is the approach taken for the synthesis of poly(disulfide)s in this thesis and will be discussed more in detail later in Chapter 1.3.

As discussed in Chapter 1.1 the disulfide bond present in poly(disulfide)s is a dynamic covalent bond and is easily broken by reduction. Because of the challenges with plastic pollution, new kinds of polymers are being developed that have means of degradation if they find their way into the environment and are easy to recycle. Poly(disulfide)s provide an opportunity of having new materials with inherent degradable properties that also can be tuned and transferred onto other kinds of polymers by means of copolymerization. Recently Okayama et al. published a report on enhanced degradability of vinyl copolymers with lipoic acid.^[16] They found that not only disulfide bonds, which are easily reducible – for that the ratio of disulfide monomer to vinyl monomer must be quite high, which increases the change in properties of the original vinyl polymer – but also C-S bonds can be cleaved under the right conditions. For that, they simply heated polymers with as low as 5 mol% ethyl lipoate in DMF and observed the depolymerization of high-molecular-weight polymers to only low-molecular-weight oligomers. The potential of tuning the degradability of homo- or copolymers with disulfide bonds should be researched further.

Because of the inherent depolymerization ability of poly(disulfide)s they are often combined with other non-degradable materials in order to impart a degree of degradability. It has been used to render vinyl polymers like polyacrylates, that can be used as adhesives, degradable.^[17,18] Also, silicon elastomers could be crosslinked with α -lipoic acid to obtain degradable elastomers.^[19]

The dynamic covalent nature of the disulfide bonds also enables many potential applications for these poly(disulfide)s. Because of the reducibility of the disulfide bonds, the poly(disulfide)s can be utilized for gene and drug delivery. In 2009 Ou et al. reported the synthesis of bio-reducible poly(disulfide amine)s for gene delivery.^[20] Later Hickey et al. demonstrated the synthesis of functionalized poly(disulfide)s from active ester functionalized α -lipoic acid for the uptake of mRNA.^[21] Apart from DNA and RNA the delivery of drugs is an interesting field of research. Bang et al. reported that α -lipoic acid functionalized with positively charged groups could be used to synthesize poly(disulfide)s with possible cell-penetrating properties, which could be used to transport probes or drugs that otherwise are not able to enter cells, through cell membranes.^[22]

Poly(disulfide)s can also be used for smart materials. In 2015 Barcan et al. reported hydrogels generated from water-soluble triblock copolymers with pendant 1,2-dithiolane groups for dynamic crosslinking.^[23] They found that the obtained material showed self-healing abilities, which could be modulated with a change in pH and temperature. Later Zhang et al. in cooperation with Feringa showed that polymers networks made from α -lipoic acid, 1,3-diisopropenylbenzene, and ferric chloride possessed low-temperature processibility, stretchability, self-healing, adhesivity, and reusability.^[24] In a later publication they also demonstrate closed-loop recycling for these kinds of polymer networks.^[25] Hydrogels using cyclic disulfides were also found to be photocurable.^[26] Furthermore, bioadhesives based on α -lipoic acid salts were reported by Ghosh et al. It could be shown that these bioadhesives display similar adhesion strengths as fibrin glue hydrogels.^[27] Adhesives based on α -lipoic acid could also be used for applications in dry/wet environments.^[28]

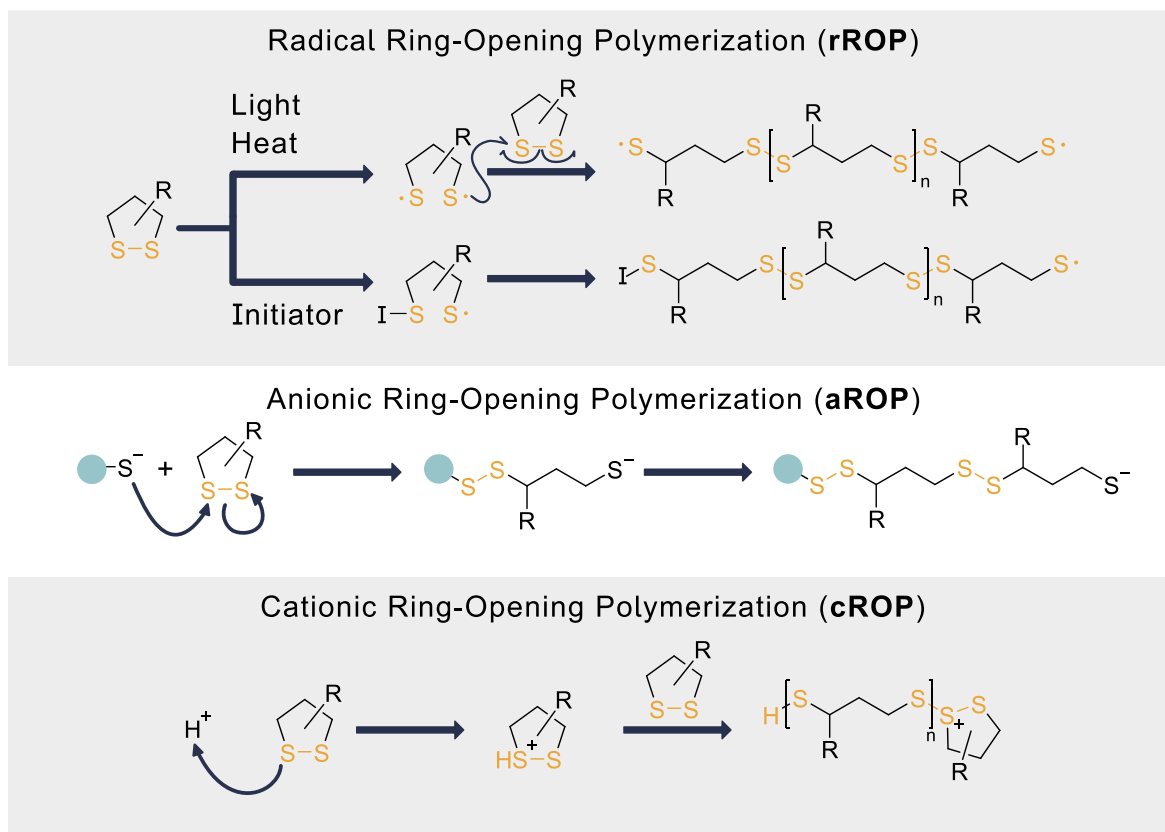
1.3 Cyclic Disulfides

1.3.1 Cyclic Disulfides for Ring-Opening Polymerization

As stated above, **ROP** is one of the methods for the obtention of poly(disulfides). Cyclic disulfides of varying ring sizes were reported to undergo polymerization. In 1948 Davies and Fettes described the reversible polymerization of a seven-membered oxadithiepane,

though the exact mechanism was not explored.^[29] In 1953 Tobolsky and Baysal, inspired by the observation that non-cyclic disulfides react as chain transfer agents in radical polymerization of vinyl monomers, found that cyclic disulfides, e.g., the aforementioned oxadithiepane called diethyl ether disulfide, can form sulfide-linkage containing copolymers.^[30] They reasoned that these were formed by way of homolytic cleavage of the disulfide bond and propagation of the polymer chain via a thiyl radical. Three years later Thomas and Reed, in the attempt to synthesize DL- α -lipoic acid, observed the formation of sticky, colorless polymeric by-products in the last step of the synthesis.^[31] After investigation of the mechanism of the depolymerization of these polymers they concluded the mechanism to be an anionic exchange reaction of a thiolate ion with disulfide linkages. In 1990 Suzuki et al., when examining the copolymerization of lipoamide with vinyl monomers, reasoned in accordance with Tobolsky and Baysal that the polymerization proceeds by attack of the vinyl polymer radical at the disulfide bond of lipoamide.^[32] The group of Endo studied the polymerization of dithiolanes and dithianes via **ROP** extensively. In 2004 they prepared 1,2-dithiane, a six-membered ring, and studied the thermal-induced polymerization, where it was found to be inhibited by radical inhibitors, therefore hinting at a possible mechanism of a radical nature, but also by addition of free thiols.^[33] The resulting polymers were also found to be cyclic in nature and possibly to possess a polycatenane structure. They continued their work in 2006 where they copolymerized 1,2-dithiane with α -lipoic acid, which was found to be similar in structure, and in 2010 where they synthesized α -lipoic acid homopolymers.^[34,35] More details on **ROP** of specifically α -lipoic acid and its derivatives is given later in chapter 1.4.3.

Most of the aforementioned polymerization reactions are radical ring-opening polymerizations (**rROP**). Cyclic disulfides, however, also allow for anionic (**aROP**) and cationic (**cROP**) ring-opening polymerizations besides **rROP**. In the following the mechanisms of **rROP**, **aROP**, and **cROP** are described.



Scheme 6 - Mechanisms of radical, anionic, and cationic ring-opening polymerizations of cyclic, five-membered disulfide monomers.

As shown in Scheme 6 **rROP** proceeds via two reaction pathways. The propagating radical can either be generated by homolytic scission of the disulfide bond, using light or heat, or by the attack of a radical initiator. The so formed radical then continues to attack further cyclic disulfide monomers affording the desired polymer. **aROP**, however, is initiated by the nucleophilic attack of a thiolate anion, which is formed by deprotonation of thiols. The now opened monomer continues to attack more monomers yielding the polymer. Lastly, **cROP** is initiated by the protonation of one of the sulfur atoms in the ring, which then enables other cyclic disulfide monomers to nucleophilically attack the so-called sulfonium ion leading to chain propagation. The application and challenges of these types of mechanisms in the context of polymerization of α -lipoic acid and lipoates is discussed in Chapter 1.4.3.

The ease of **ROP** of dithiolanes or dithianes, which are five- and six-membered rings respectively, is surprising at first, because the rings are found to be resembling cyclopentane and cyclohexane in structure. The C-S-S-angle is found to be nearly tetrahedral with an angle of 107° and though the S-S-bond at $2.05 \text{ \AA}$ is considerably longer than the corresponding C-C-bond it is not sufficient to explain the ease of polymerization. It is therefore hypothesized that the reactivity could be explained by a restricted rotation about the disulfide bond, which results in a high torsional strain of the C-S-S-C bond angle in the

six-membered and especially in the five-membered rings, where a dihedral C–S–S–C angle of often under 35° , in comparison to 90° in a linear disulfide, can be observed. Because of this, the free electron pairs of the sulfur atoms overlap, destabilizing the S–S bond in what is known as closed-shell repulsion.^[36,37]

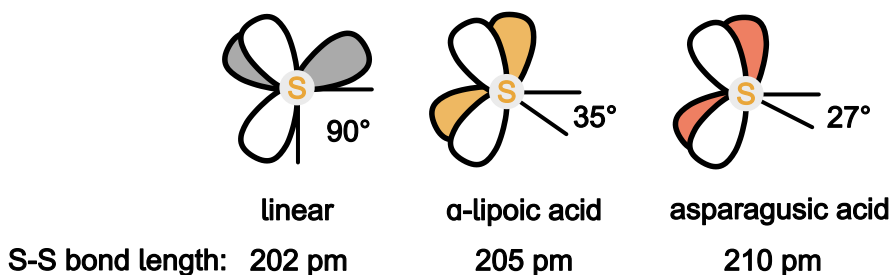
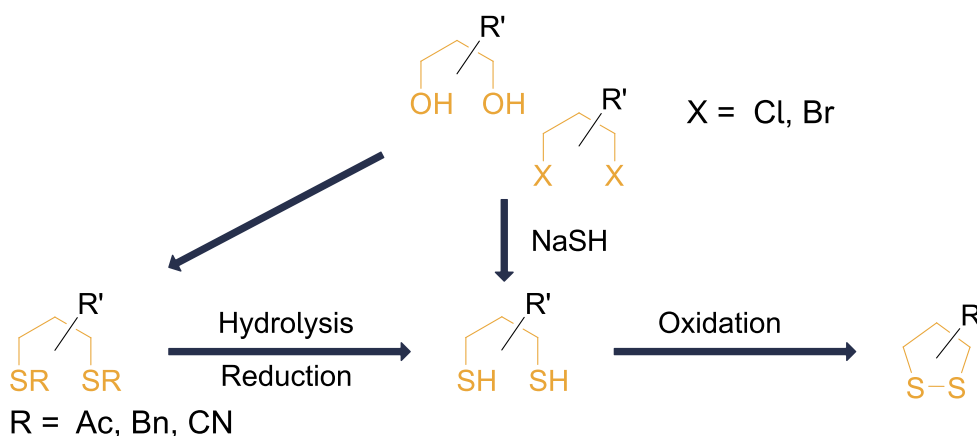


Figure 2 - Torsional C-S-S-C bond angles of linear disulfides, α-lipoic acid, and asparagusic acid.

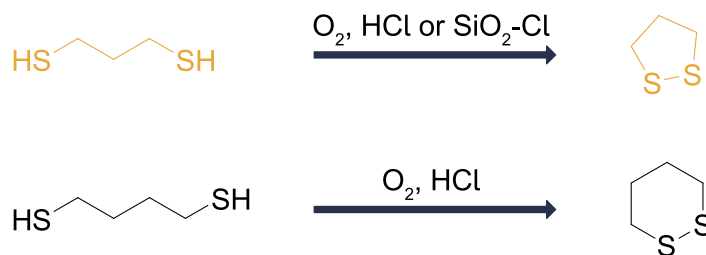
1.3.2 Syntheses of Cyclic Disulfides

1,2-Dithiolanes are commonly synthesized by hydrolysis or reduction of sulfur-containing precursors forming 1,3-dithiols, which are then oxidized to the cyclic disulfide.^[37] These precursors are often obtained from 1,3-dithiols or 1,3-dihalides.



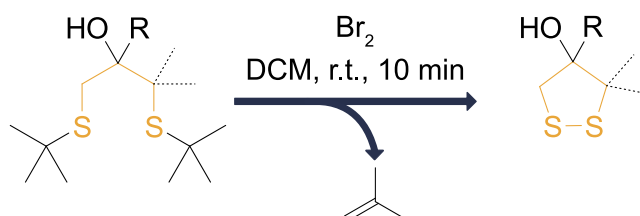
Scheme 7 - General synthetic approach for 1,2-dithiolanes.

1,2-Dithiolane, which is the simplest five-membered cyclic disulfide, can be synthesized from 1,3-propanedithiol by oxidation with air under either acid catalysis or heterogenous catalysis by silica chloride. Similarly, 1,2-Dithiane can be synthesized from 1,4-butanedithiol.^[38,39]



Scheme 8 - Synthesis of 1,2-dithiolane and 1,2-dithiane from the corresponding dithiol by oxidation from oxygen.
[38,39]

Due to the often harsh conditions for the hydrolysis or reduction of these precursors, limiting compatibility with different functional groups, Scheutz et al. developed a one-step method for the synthesis of functionalized 1,2-dithiolanes from 1,3-*tert*-butyl thioethers.^[37]



Scheme 9 - Synthesis of 1,2-dithiolanes from 1,3-*tert*-butyl thioethers according to Scheutz et al.^[37]

Using this method, 1,2-dithiolanes with different substituents and side chains are available, increasing possible monomers for ring-opening polymerization.

In the following chapter the cyclic disulfide used in this thesis, α -lipoic acid, is introduced and the current state of research in lipoic acid polymerization is shown. More syntheses of specifically α -lipoic acid are also discussed in the next chapter.

1.4 Lipoic Acid

α -Lipoic acid (**ALA**) is a naturally occurring molecule, that combines a dithiolane ring with a carboxyl group linked by an alkyl chain. **ALA** has one chiral center at C6 resulting in *S*-**ALA** and *R*-**ALA**, of which only the *R*-isomer is naturally occurring and biologically active.^[40–42]

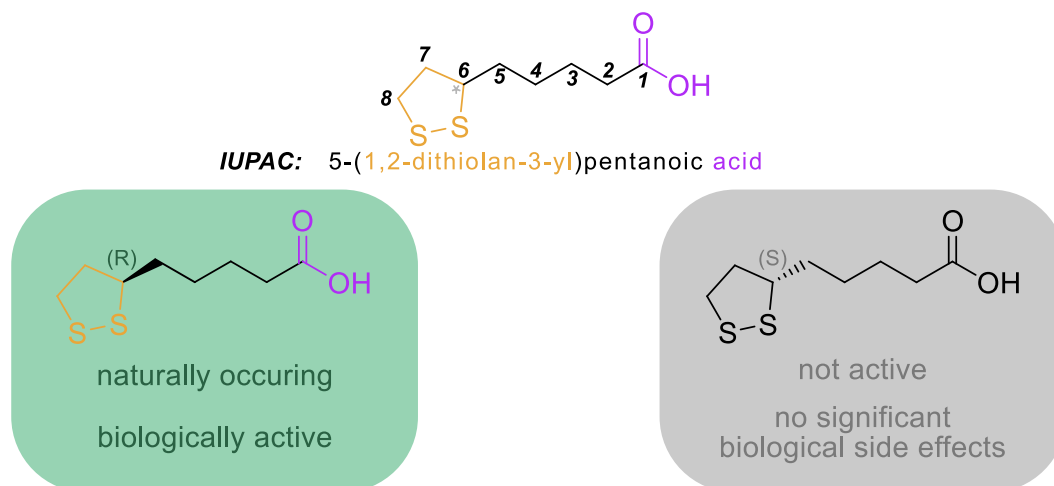
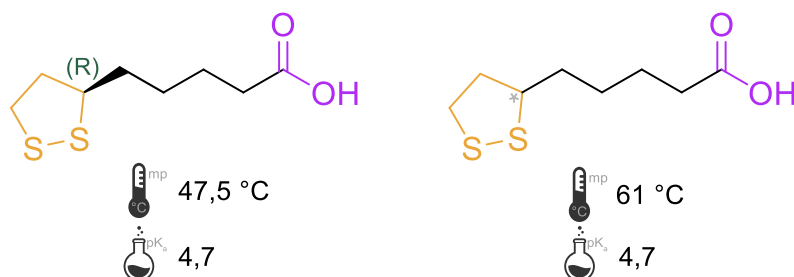


Figure 3 - Structure of Lipoic Acid.

The name “ α -lipoic acid” stems from the fact that **ALA** is highly soluble in “fat solvents”, involved in the biosynthesis of acetate, which in turn is a precursor to fatty acids. The “ α ” indicates that it is the first compound in a series of different compounds showing similar biological activities.^[43] Other names for **ALA** include thioctic acid and 6-thioctic acid. *R*-**ALA** has a melting point of 47.5 °C, while racemic **ALA** has a melting point of 61 °C. It also shows acidity with a pK_a of 4.7.^[43,44]

Figure 4 - Properties of *R*-**ALA** and *rac*-**ALA**.

ALA was considered a vitamin before it was discovered that plants and animals can synthesize it themselves. It functions as a coenzyme for pyruvate dehydrogenase and ketoglutarate dehydrogenase and is involved in the metabolism of glucose and lipids and in gene transcription. In humans **ALA** is synthesized in small amounts from fatty acids and cysteine. It is, however, not enough to supply the required amount and therefore must be ingested. **ALA** can be found in many vegetables, fruits, meats, and entrails. **ALA** displays antioxidant properties and can regenerate other antioxidants. In its reduced form, 6,8-dithiooctanoic acid, it is also called dihydrolipoic acid (**DHLA**). It is also able to chelate

metals and therefore remove heavy metals from the blood stream due to its disulfide bonds and quench radicals.^[45,46]

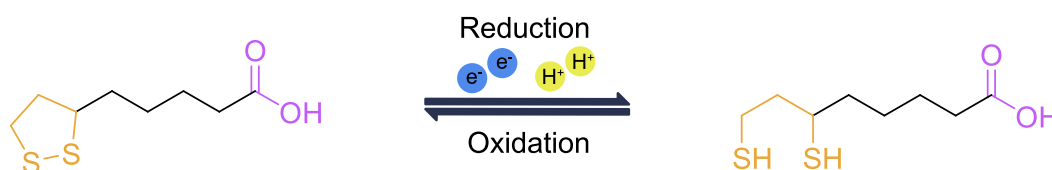


Figure 5 - Reduction of **ALA** to **DHLA**.

ALA has various potential pharmaceutical uses due to its antioxidant properties. In clinical trials **ALA** has shown benefits in particular diseases, e.g., diabetes, obesity, schizophrenia, multiple sclerosis, abnormalities in pregnancies, and organ transplantation.^[46]

1.4.1 Origin and Syntheses

The isolation of **ALA** was first reported in 1951 by Reed et al. from bovine liver.^[43] In living organisms **ALA** is found attached to the ϵ -amino group of specific lysine residues in the enzymes for which it serves as a cofactor.^[47] The biosynthesis of **ALA** in *Escherichia coli* is shown in Figure 6.

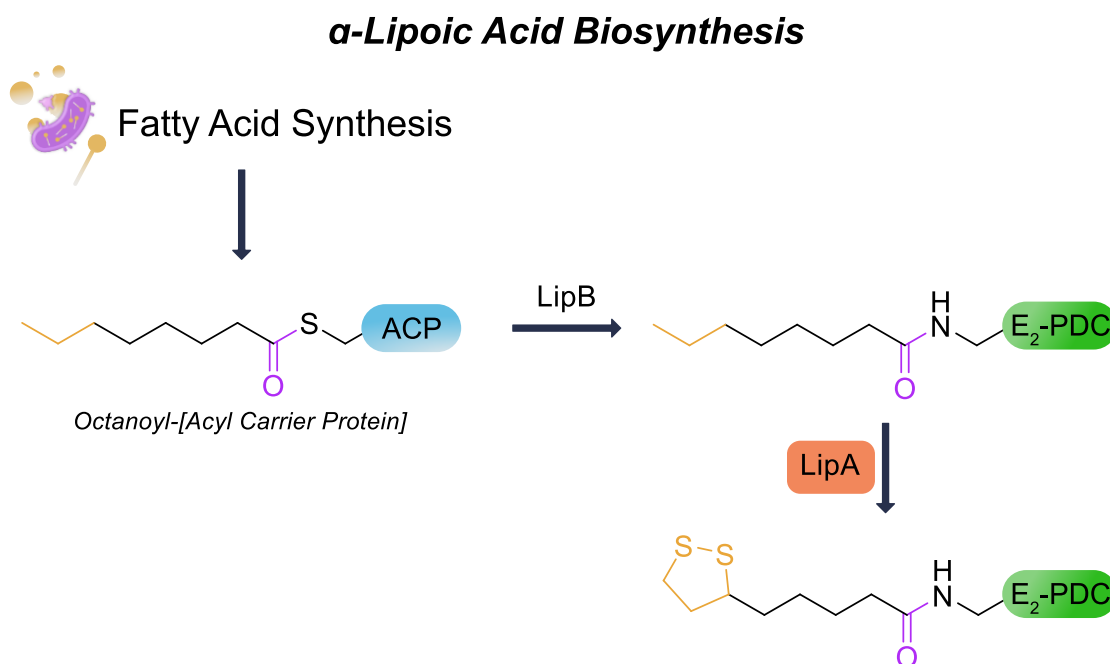
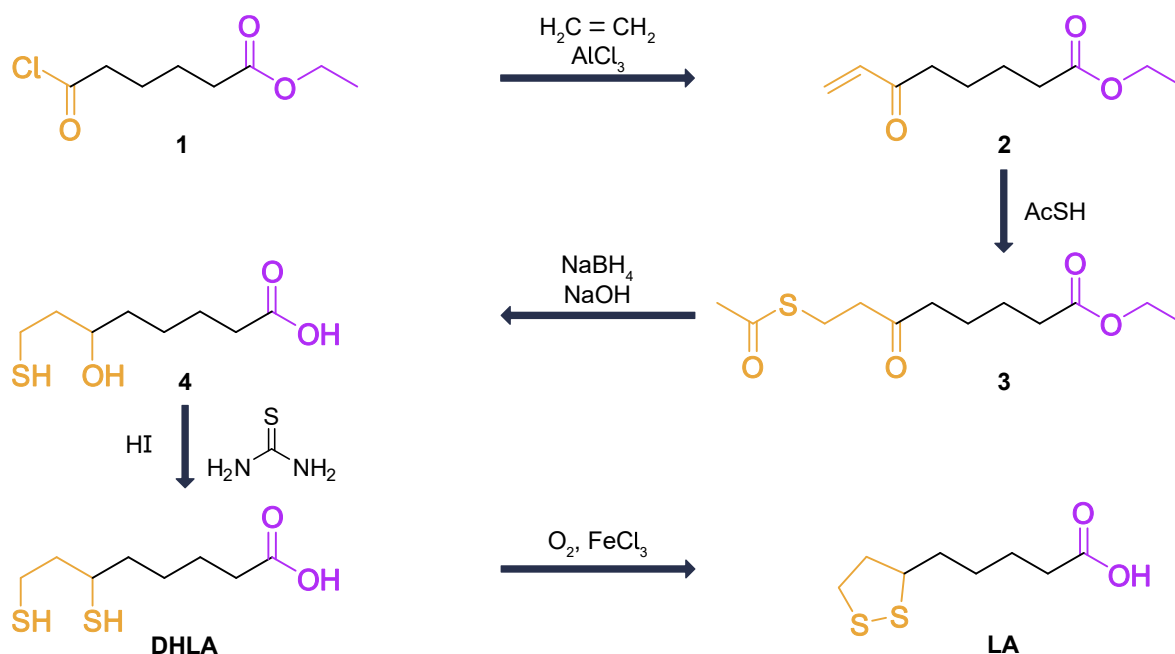


Figure 6 - Biosynthetic pathway for the synthesis of **ALA** from octanoic acid derived from fatty acid synthesis.^[48]

The lipoyl group can be synthesized starting from octanoic acid derived from fatty acid synthesis. The octanoyl group bound to the acyl carrier protein (**ACP**) obtained from fatty acid synthesis is transferred to lipoyl-accepting domains of certain enzymes, e.g., the E2 subunit of the pyruvate dehydrogenase complex (**PDC**), by lipoyl (octanoyl) transferase **LipB**. There, lipoyl synthase **LipA** catalyzes the insertion of sulfur into the octanoyl chain.^[48] The mechanism is of radical nature where **LipA** abstracts two hydrogen atoms at C6 and C8 allowing for sulfur insertion.^[49]

Although **ALA** is a naturally occurring compound, since it is bound to certain enzymes, it is therefore difficult to obtain in large quantities. In commercial meats its content was determined to be 1,3 mg/kg at the maximum in bovine kidney.^[50] Thus, synthetic methods had to be developed to acquire larger quantities. In 1952 the first total synthesis of **ALA** was reported by Bullock et al. from ethyl adipyl chloride, which is shown in Scheme 10.^[44,51]



Scheme 10 - First reported synthesis of α-lipoic acid after Bullock et al.^[44]

As shown in Scheme 10 the Lewis acid-catalyzed insertion of ethene using aluminum trichloride into the acyl chloride bond of compound **1** yields the α,β-unsaturated ketone **2**, which is susceptible to Michael addition. After thiol-ene addition of thioacetic acid and subsequent reduction using sodium borohydride, the resulting 6-hydroxy-8-mercaptooctanoic acid **4** can be converted into **DHLA** with an excess of hydroiodic acid and thiourea. Finally, oxidation of **DHLA** with oxygen and ferric chloride as catalyst yields the desired **ALA**.

Many more different syntheses of lipoic acid have been described. Recently Wang et al. published a detailed review of synthetic methods for racemic lipoic acid as well as

enantiopure (*R*)-lipoic acid.^[52] According to the review the synthetic methods for racemic lipoic acid can be divided into three main pathways as seen in Figure 7.

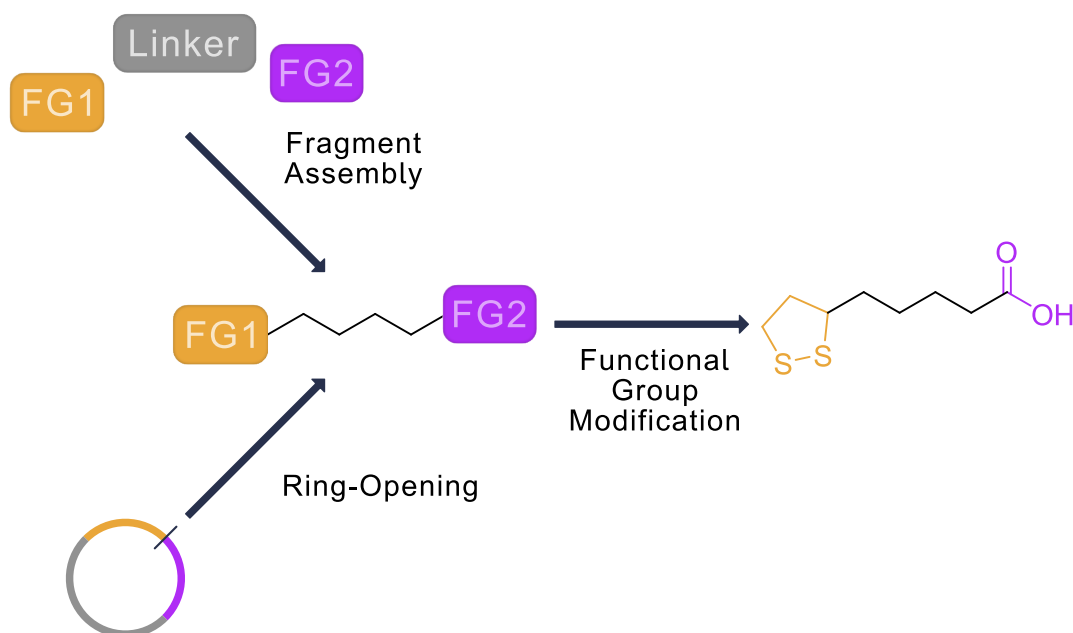
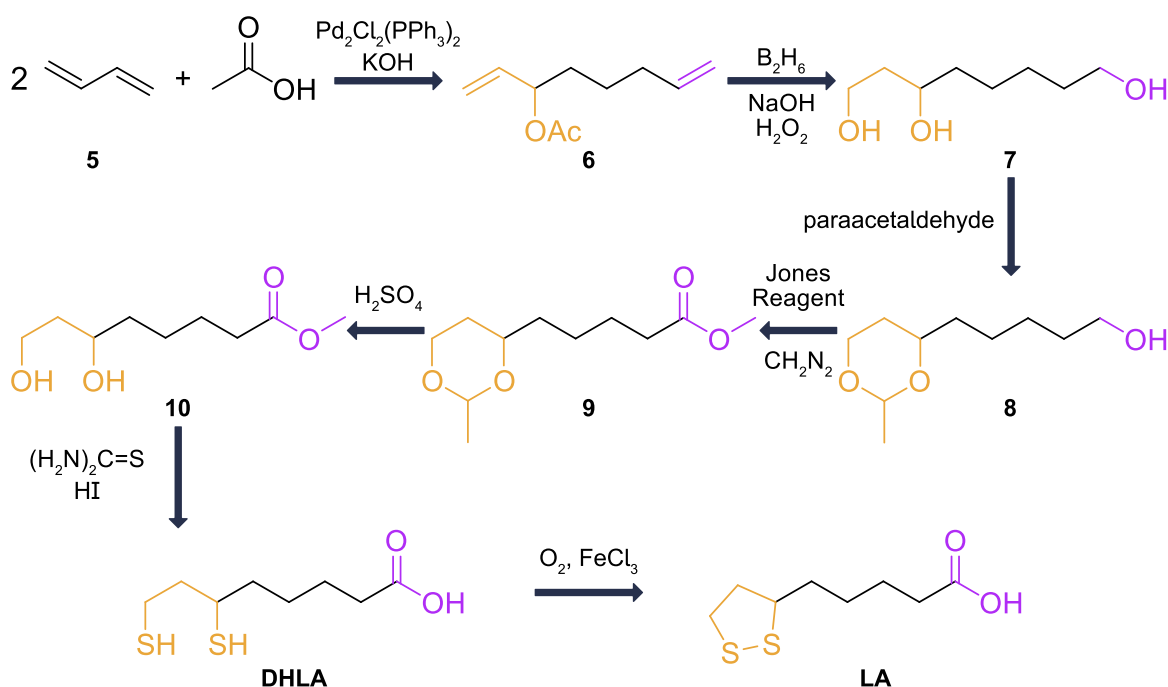


Figure 7 - Approaches for the synthesis of lipoic acid; adapted after Wang et al.^[52]

As depicted in Figure 7 the main pathway for the synthesis of **ALA** is functional group modification of various precursors, which are either commercially available or must be synthesized by one of the other pathways, namely fragment assembly or ring-opening. The first reported synthesis by Bullock et al. shown above is an example for the functional group modification approach. In the following more syntheses of **ALA** are described exemplarily, showing the two other pathways.

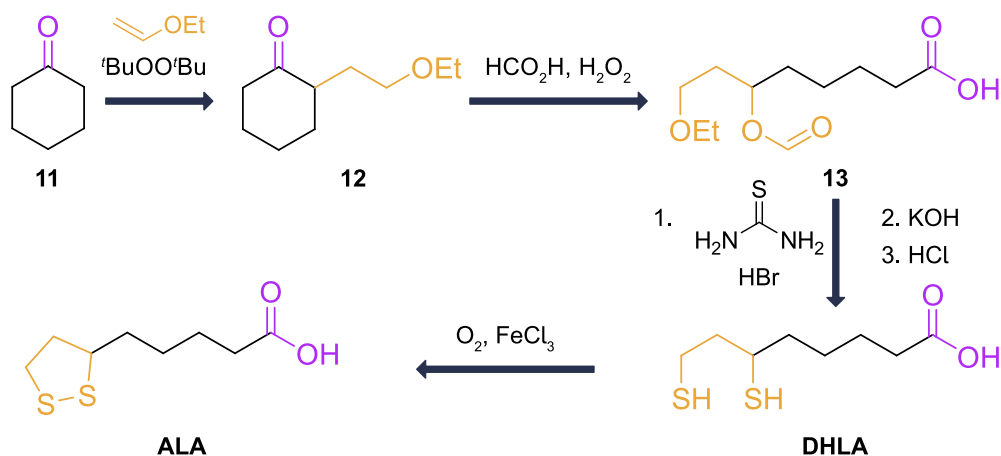
In 1978 Tsuji et al. reported a fragment assembly synthesis of lipoic acid based on the telomerization of butadiene with acetic acid, which is shown in Scheme 11.^[53]



Scheme 11 – Fragment assembly synthesis of racemic **ALA** via Pd-catalysed telomerization of butadiene and acetic acid, as reported by Tsuji et al.^[53]

The palladium-catalysed telomerisation of butadiene **5** with acetic acid affords 3-acetoxy-1,7-octadiene **6** and its isomer 1-acetoxy-2,7-octadiene, which can rearrange with the same palladium-catalyst to the desired isomer **6**. Hydroboration with borane then yields 1,3,8-octanetriol **7**, whose hydroxyl groups in position 1 and 3 are then protected using paracetaldehyde to specifically oxidize the remaining hydroxyl group to the desired carboxyl group in a Jones oxidation with chromium trioxide and sulfuric acid. The obtained acid is then methylated using diazomethane to ester **9** to avoid lactonization as a side reaction. After deprotection of the 1,3-diol the hydroxyl groups are transformed into thiols using thiourea and hydroiodic acid, which also leads to the hydrolysis of the methyl ester. The obtained **DHLA** is finally oxidized by oxygen and ferric chloride as catalyst to yield racemic **ALA**.

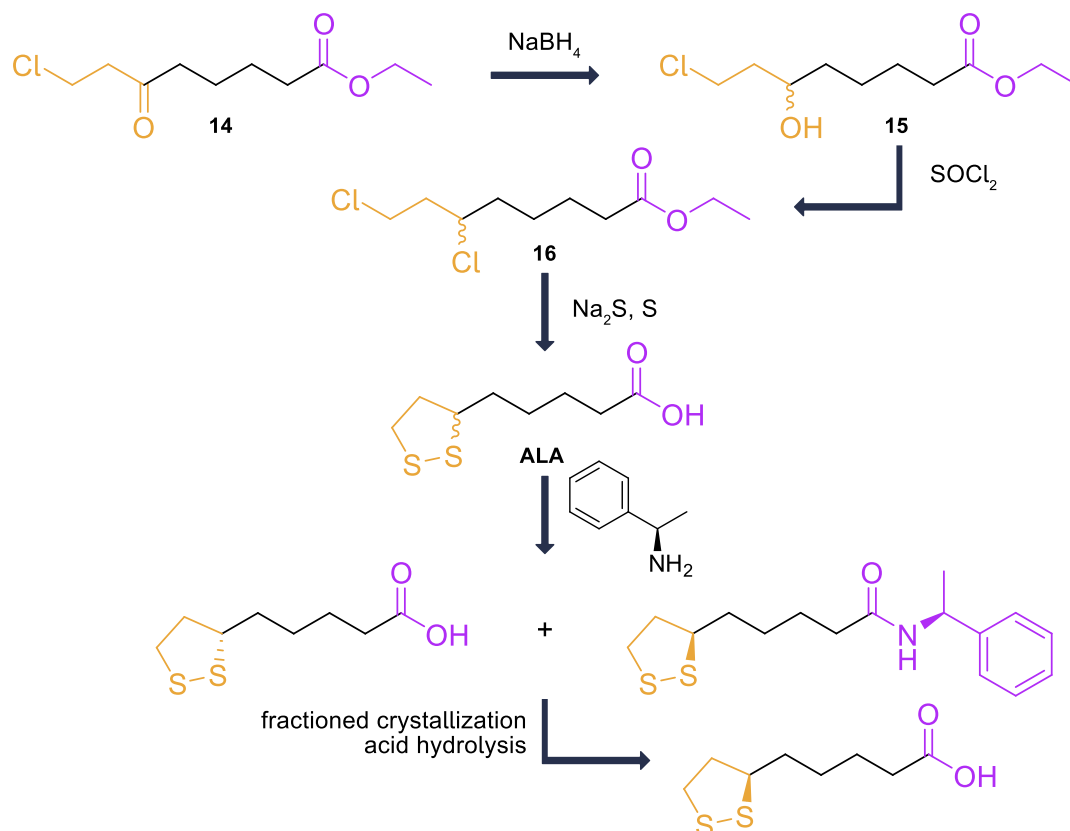
In 1999 Paust et al. built on earlier work by Segre et al.^[54], developing a synthetic route centered around Baeyer-Villiger oxidation, which facilitates ring opening of the intermediate to yield a precursor of **ALA**.^[55] The synthesis is shown in Scheme 12.



Scheme 12 – Ring-Opening synthesis of **ALA** via Baeyer-Villiger oxidation as the key step, as reported by Balkenhohl and Paust.^[55]

As shown in Scheme 12 the substrate for the Baeyer-Villiger oxidation **12** is synthesized from cyclohexanone **11** and vinyl ethyl ether. **12** then undergoes the Baeyer-Villiger oxidation and ring opening to **13**, which gives, after reaction with thiourea and hydrobromic acid, the reduced **DHLA**. Oxidation with oxygen and ferric chloride as catalyst yields **ALA**.

In the industry (Scheme 13) the synthesis of **ALA** starts with ethyl 8-chloro-6-oxooctanoate **14** which is reduced by sodium borohydride to give the corresponding racemic secondary alcohol **15**. The hydroxyl group is then substituted to a chloride by thionyl chloride to yield **16**. To afford the racemic **ALA** the chloride **16** is reacted with sodium sulfide and elementary sulfur.^[56,57]



Scheme 13 - Industrial synthesis and chemical resolution of (*R*)-α-lipoic acid after Zhang et al.^[56,57].

There are also several approaches for the synthesis or resolution of *R*-**ALA** including chemical resolution, enzymatic resolution, and asymmetric syntheses with chiral auxiliaries or chiral catalysts.^[52,57] As shown in Scheme 13, to separate *R*-**ALA** from *S*-**ALA** the racemic mixture is treated with (*R*)-(+)-α-methylbenzylamine, which selectively reacts with *R*-**ALA** to give the corresponding amide. The mixture of the amide and *S*-**ALA** can then be separated by fractionated crystallization and the desired *R*-**ALA** can be obtained.^[56,57] Other approaches are shown in Figure 8.

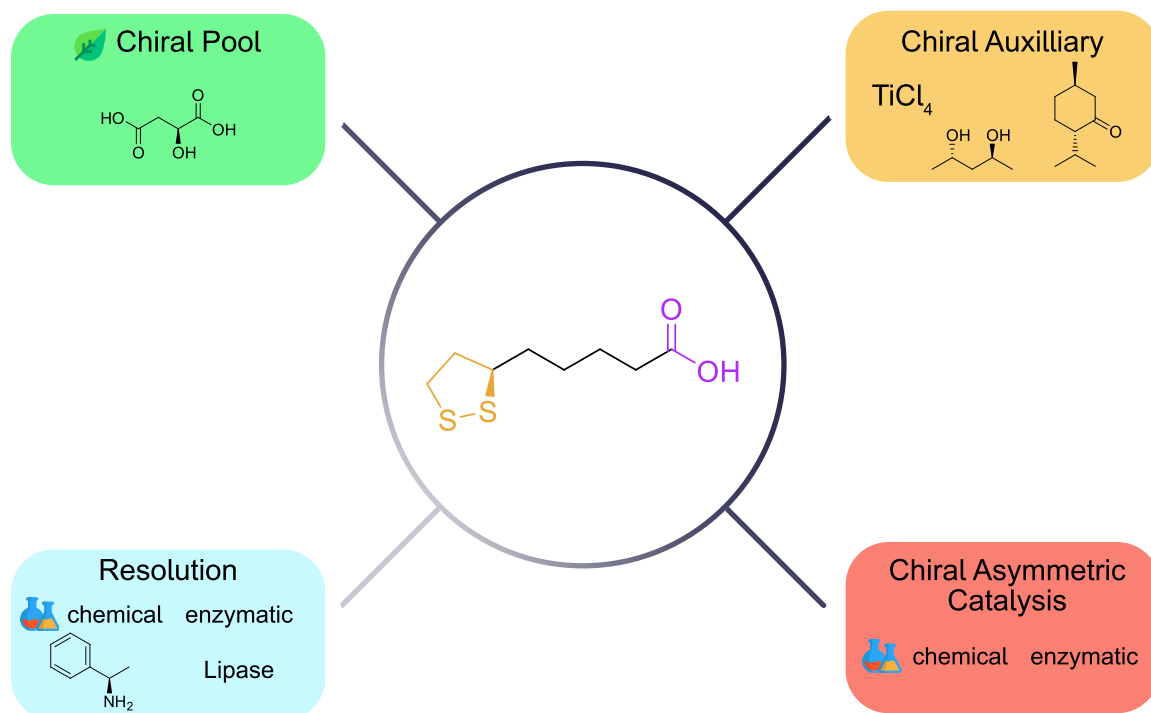


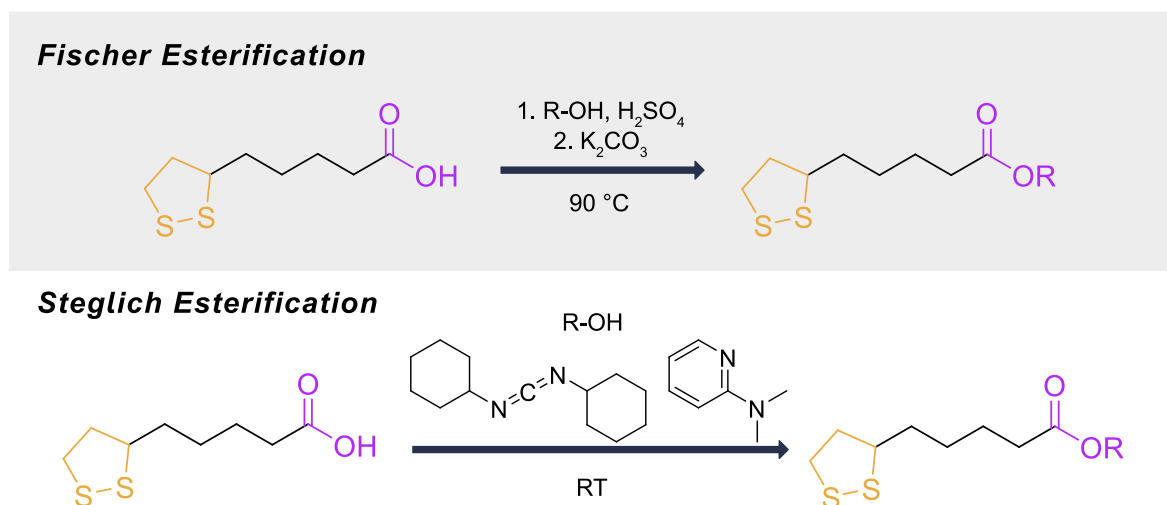
Figure 8 - Approaches to the synthesis or resolution of enantiopure lipoic acid.

Apart from the chemical resolution of **ALA** with (*R*)-(+)- α -methylbenzylamine there are also enzymatic resolution approaches utilizing the stereospecific esterification of **ALAs** carboxylic acid group. For that, lipases are routinely used.^[52,58,59] The chemical resolution as well as the enzymatic resolution can also be utilized to resolve racemic compounds used to synthesize **ALA**.^[60,61] A different approach is to take molecules that are enantiomeric in nature from the “chiral pool” and use these to synthesize the desired enantiopure **ALA**. Examples for used materials include 2-hydroxysuccinic acid^[42], 3,4,6-tri-*O*-acetyl-D-glucal,^[62,63] and D-mannitol^[64]. Furthermore, syntheses using chiral auxiliaries were developed. Johnson et al. reported the use of *R,R*-2,4-pentanediol as a chiral auxiliary in a aldol-type reaction catalyzed by titanium tetrachloride as the activating Lewis acid.^[65,66] Moreover, menthone could be used by Menon et al. as a recyclable chiral auxiliary.^[67] Lastly, asymmetric catalysis approaches were also reported using chemical as well as enzymatic asymmetric catalysis.^[68–71]

1.4.2 Functionalization and Derivatives

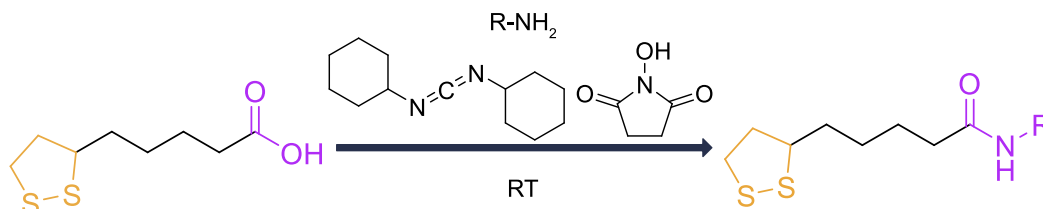
Because of its carboxylic acid group **ALA** can be readily functionalized. Common modifications of **ALA** include esters and amides. Esters can be synthesized using standard Fischer esterification, which includes dissolving **ALA** in the alcohol, which should form the ester, adding sulfuric acid as a catalyst and heating the mixture. Because **ALA** and the ester can polymerize under acidic conditions, it is important to keep the temperature far over the ceiling temperature, so that any polymerization is suppressed.^[72] Under more mild

conditions **ALA** can also be esterified using Steglich esterification, utilizing *N,N'*-dicyclohexylcarbodiimide (**DCC**), dimethylaminopyridine (**DMAP**), and the desired alcohol.^[73]



Scheme 14 - Fischer and Steglich esterifications of **ALA**.

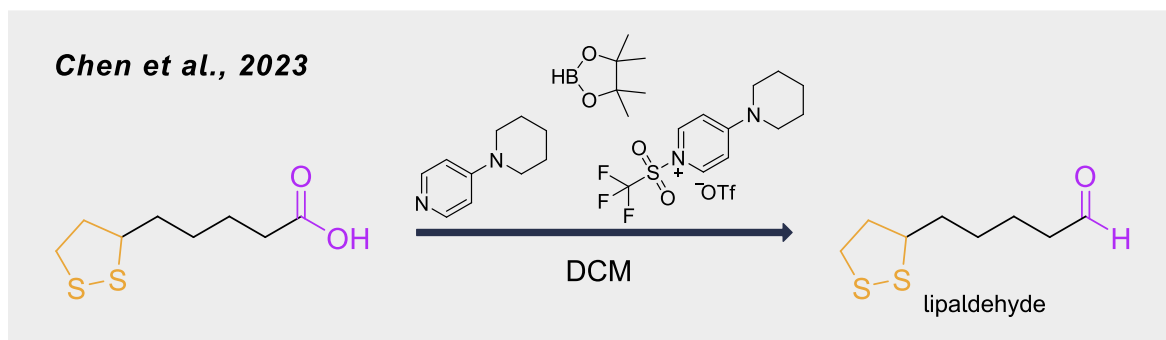
Amides of **ALA** are accessible through a similar way. Utilizing **DCC** or 1-ethyl-3-(3-dimethylaminopropyl)carbodiimide (**EDC**), *N*-hydroxysuccinimide (**NHS**), and the desired amine, **ALA** can be transformed into the corresponding amide.^[74]



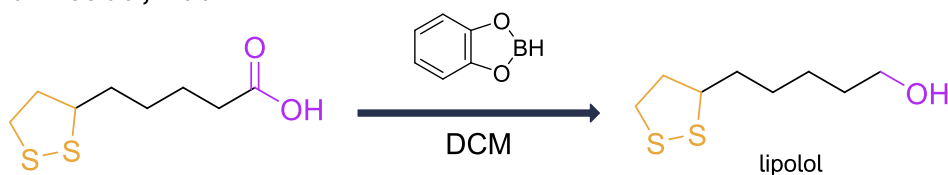
Scheme 15 - Amidation of **ALA** using **DCC** and **NHS**.

Using both approaches **ALA** can be functionalized with a multitude of different sidechains.

Although many **ALA** derivatives were used to perform **ROP**, the use of derivatives, where the carboxylic acid is reduced or fully removed, is not explored. Using 4-piperidin-1-ylpyridine, triflyl 4-piperidin-1-ylpyridine and pinacol borane, **ALA** can be reduced to the aldehyde.^[75] Using catecholborane in dichloromethane directly yields lipolol, the corresponding alcohol.^[76]



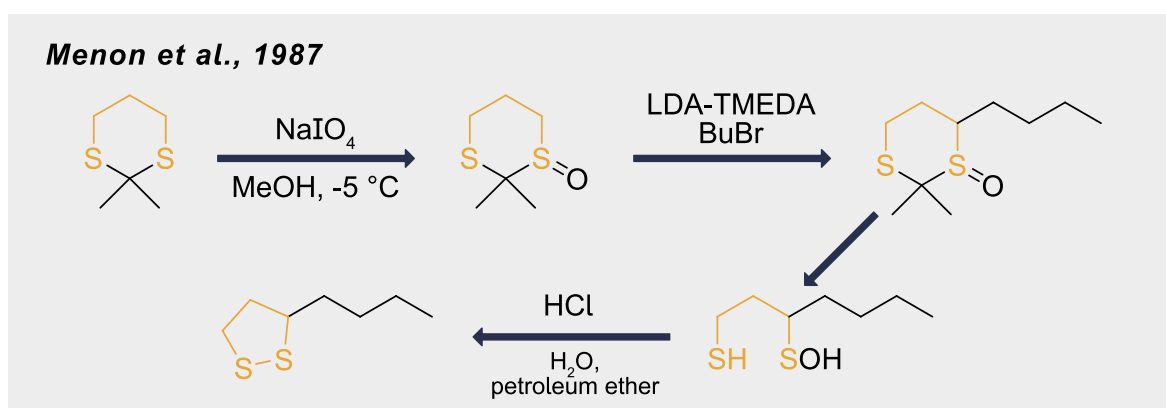
Koufaki et al., 2007



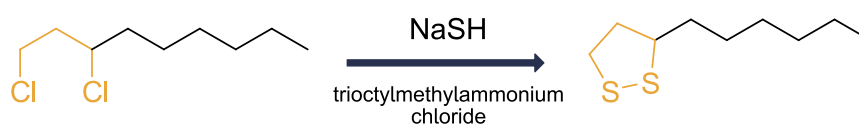
Scheme 16 - Reduction of **ALA** using different boranes to obtain lipaldehyde and lipolol.^[75,76]

Lipolol can be further functionalized to the corresponding azide using mesyl chloride to obtain the mesylate and then substitution with sodium azide.^[76] The aldehyde can be converted with the Bestmann-Ohira reagent in slightly basic conditions to obtain the corresponding alkyne.^[77]

Although the synthesis of the fully reduced dithiolane derived from **ALA** is not documented in the literature, there are syntheses for similar products, i.e., 3-butyl-1,2-dithiolane and 3-hexyl-1,2-dithiolane.^[67,78]



Yoshikawa et al., 2001



Scheme 17 - Syntheses of 1,2-dithiolanes similar to **ALA**-derived alkyl-dithiolanes.^[67,78]

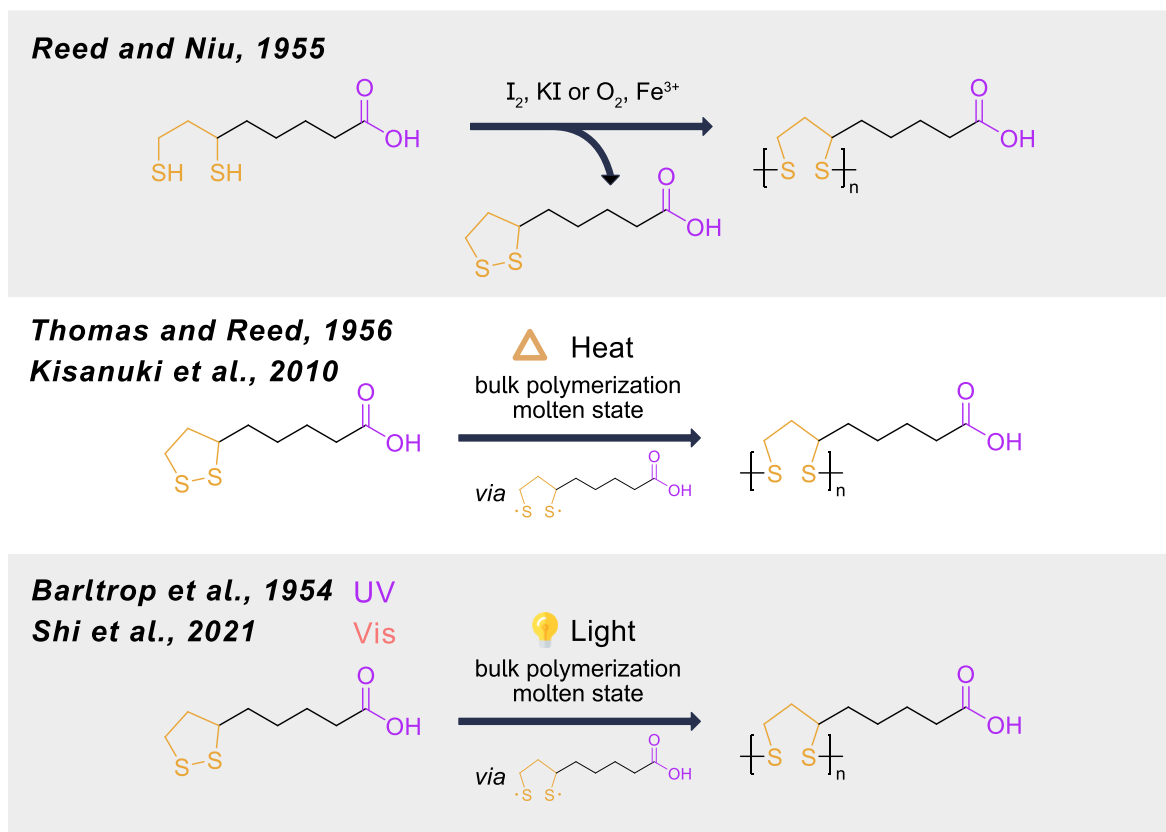
1.4.3 Lipoates as Monomers for Poly(disulfides)

1.4.3.1 Radical Ring-Opening Polymerization (*rROP*)

Polymeric material made from **ALA** was first described in 1955 by Reed and Niu as a byproduct of the oxidation of 6,8-dithiioctanoic acid to **ALA**.^[79] The polymeric material exhibited ultraviolet absorption at 250 nm consistent with spectra from other saturated organic disulfides. It also could be reduced to monomeric **ALA** and was rapidly hydrolyzed by dilute sodium hydroxide. Later, based on the knowledge that styrene and vinyl acetate could be copolymerized with 1-oxa-3,4-dithiacycloheptane, Suzuki et al. used lipoamide in a radical copolymerization with vinyl monomers yielding polymers with sulfide and disulfide linkages in its main chain.^[32] After Endo et al. first reported the synthesis of poly(1,2-dithiane) by thermally induced polymerization in 2004^[33] they followed up with the copolymerization of 1,2-dithiane with **ALA**, which was reported to yield polymers of cyclic and polycatenane structure.^[34]

The first report of purposefully obtaining homopolymers of **ALA** was published by Thomas and Reed in 1956. However, only in 2010 by Kisanuki et al., following the work of Endo, poly(**ALA**) was obtained by thermally induced polymerization without initiator.^[35] The polymer exhibited a glass transition temperature of $-11\text{ }^{\circ}\text{C}$ without observation of a melting point. Additionally, the polymer could successfully be modified to the corresponding methyl ester with (trimethylsilyl)diazomethane and be decomposed by UV irradiation. In 2021 Raeisi and Tsarevsky published an article on kinetic and thermodynamic aspects of the *rROP* of different **ALA** derivatives.^[80] The *rROP* of these derivatives was studied in bulk and in solution. It was found that monomer conversion decreased with higher temperatures and lower concentrations, with the ceiling temperature being $139\text{ }^{\circ}\text{C}$. The polymers degraded at temperatures above that but also upon addition of radical initiator in anisole and in the presence of reducing agents.

Apart from thermally induced polymerization or polymerization induced by a radical initiator **ALA** polymerization can also be initiated by light. It was known, that scission of the dithiolane ring occurs under UV light, however because of the strained nature of the disulfide-containing five-membered rings the absorption spectrum is significantly red-shifted allowing for excitement and scission of the ring using visible light. Shi et al. made use of this property to obtain a poly(**ALA**) network under irradiation of molten **ALA** with light at 420 nm without the use of additional reagents.^[81–83]



Scheme 18 - First report of **ALA** polymers by oxidation of **ALA** and radical ring-opening polymerizations of **ALA** initiated by heat or light.

Polymerization of **ALA** under the previously mentioned conditions was mostly only able to produce poly(**ALA**) with no control over molecular weight and polydispersity. Because of this Albanese et al. reported the controlled co-polymerization of **ALA** and different acrylates and acrylamides using reversible addition-fragmentation chain transfer polymerization.^[18] It was shown that trithiocarbonates and dithiobenzoates as chain-transfer reagents were, in contrast to xanthates or dithiocarbamates, able to produce polymers with controlled molecular mass (6 – 56 kDa) and dispersity (1.1 – 1.2). Owing to the formation of **ALA** dyads the resulting polymers were able to rapidly degrade under reducing conditions.

The same group later increased the scope of **ALA** polymerization methods by miniemulsion polymerization.^[84] Compared to polymerizations in bulk or in solution the polymerization in emulsion offers many advantages, like low viscosity, better heat transfer and increased molecular weights, which is why this kind of polymerization is often used in industry. To use these advantages the aqueous miniemulsion copolymerization of **ALA** or ethyl lipoate (**EtLp**), which presents better solubility in the micelles, with *n*-butyl acrylate was described. Using that method polymerizations could be scaled to obtain 500 g of product with a molecular mass up to 270 kDa from a 4 L reactor.

Most recently Fang et al. described an electrochemical approach to polymerization of **ALA** and derivatives.^[85] **rROP** was achieved through oxidation of **ALA** at the anode. The formed radical cation then initiates **rROP**. Polymers could be obtained without inert atmosphere under ambient conditions with conversion of up to 81 % within 30 min and molecular weights of up to 75 kDa.

1.4.3.2 Anionic Ring-Opening Polymerization (**aROP**)

The first instance of **aROP** of **ALA** was described in 2011 by the group of Matile, who used reversible disulfide exchange as a method of **aROP** for self-organizing surface-initiated polymerization (**SOSIP**) of derivatives of asparagusic acid (**AA**) to obtain ordered and oriented systems on transparent oxide surfaces.^[86] They later followed up on their work by applying their knowledge of **SOSIP** to substrate-initiated polymerization of **ALA** derivatives. After deprotonation of substrates such as probes, drugs or proteins with thiol groups the resulting thiolate anions can initiate **aROP** to obtain cell-penetrating poly(disulfide)s.^[22] In a later communication by the same group the ability of **ALA** and **AA** for substrate-initiated polymerization and **SOSIP** were compared.^[87] It was found that propagators derived from **ALA** are well suited for substrate-initiated polymerization in solution, whereas propagators based on **AA** are too reactive for controlled polymerization in solution and are therefore only suited for **SOSIP**. This behavior of **AA** is attributed to its lower torsion angle of 27° against the 35° of **ALA**, therefore increasing reactivity, because of the increase in unfavorable interactions between the sulfur lone pairs.

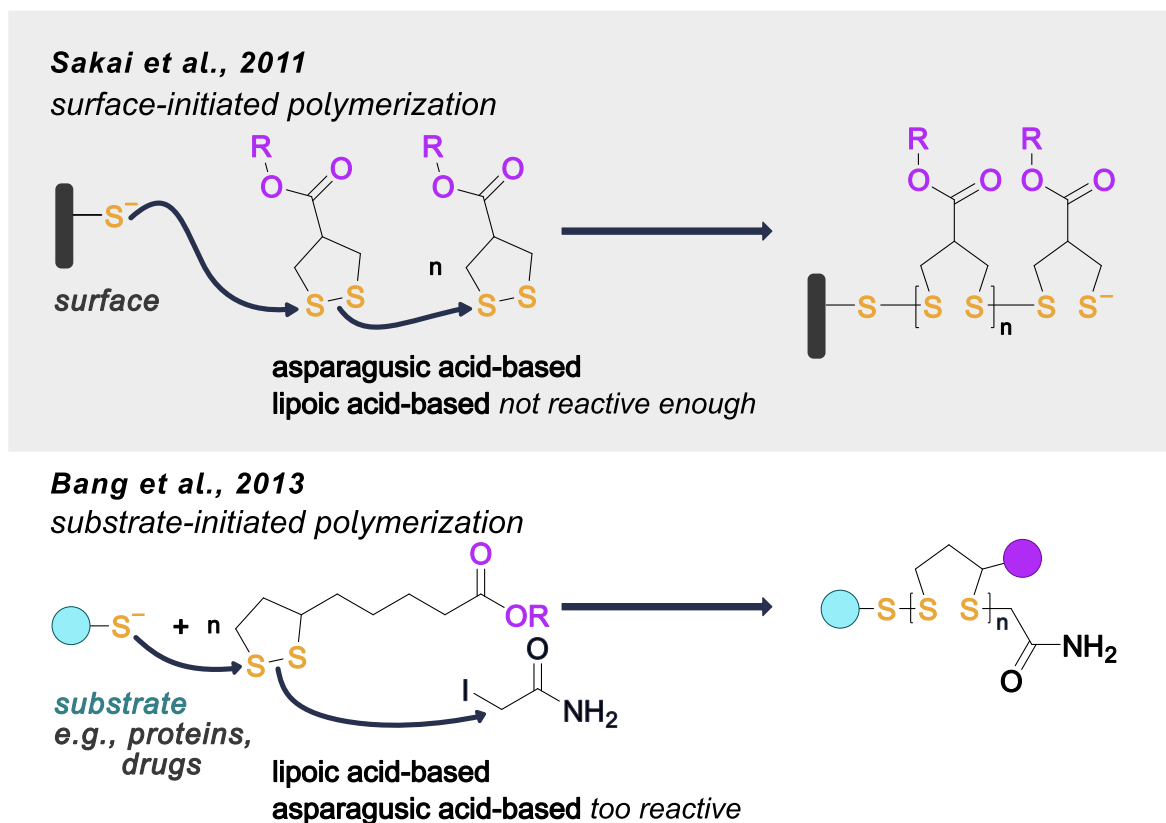


Figure 9 - Surface- and substrate-initiated **aROP** of **AA** derivatives and **LA** derivatives, according to Sakai et al. and Bang et al.^[86,22]

Expanding on the findings of the group of Matile, the group of Moore reported the architecture-controlled polymerization of methyl lipoate (**LpMe**).^[73] Owing to different nucleofugalities of the initiating thiolate anion, either linear or cyclic poly(**LpMe**) could be obtained. It was found that initiators that are better leaving groups, e.g., thiophenol and other aromatic thiols, promote backbiting and therefore ring formation. Meanwhile, alkyl thiols are worse leaving group and thus promote the formation of linear poly(**LpMe**). With this approach poly(**LpMe**) with a high degree of polymerization of 2800 and low dispersity (1.4 - 1.5) could be obtained after ten minutes. Since this polymerization approach requires strong bases, only esters like **LpMe** or other derivatives of **ALA** can be used. Consistent with other **ROPs**, it was found that the synthesis of poly(**LpMe**) is favored at higher concentrations. The need for high concentrations is a consequence of a rather low ring strain. However, the high viscosity of the monomer and the increase of the viscosity of the reaction mixture make agitation difficult at higher concentrations. Also, polymerization is favored at lower temperatures. The ceiling temperature of the **aROP** was determined to be around 300 K. Because of this, in 2020 Lu et al. reported the cryopolymerization (**cryoROP**) of **ALA** derivatives initiated by polymers.^[88] It could be shown that polymers of around 55 kDa were obtained at $-30\text{ }^{\circ}\text{C}$ with an initial monomer concentration of only 100 mM at a pH of 7.5 to 8.5. However, monomer conversion was quite low with it being only around

8 %. Although polymers were successfully obtained, many proteins are deactivated in such conditions. Therefore, other methods were developed. Lu et al. reported the aggregation-induced polymerization (**AIP**) of **ALA** derivatives at room temperature initiated by the aggregation of these compounds, which were designed to contain the hydrophobic **ALA** alongside a hydrophilic chain coupled to the carboxylic acid of **ALA**.^[89] Because of the aggregation of the monomers the local concentration was increased, allowing a polymerization that behaves similarly as in bulk. Using this approach monomer concentration could be 80 times lower than normal.

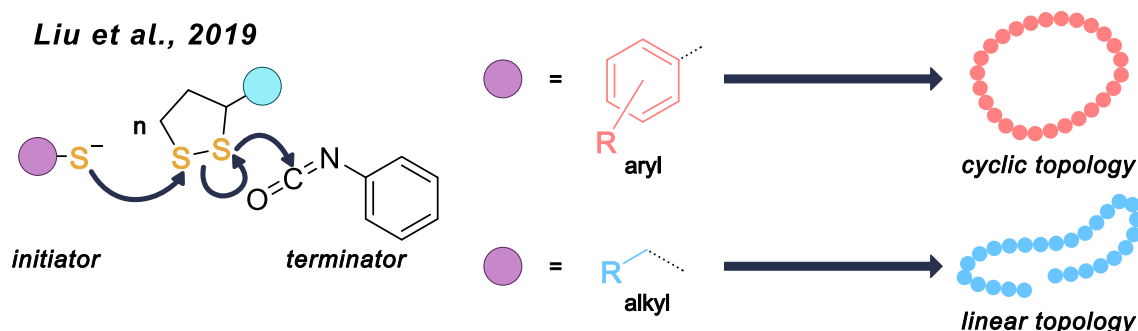
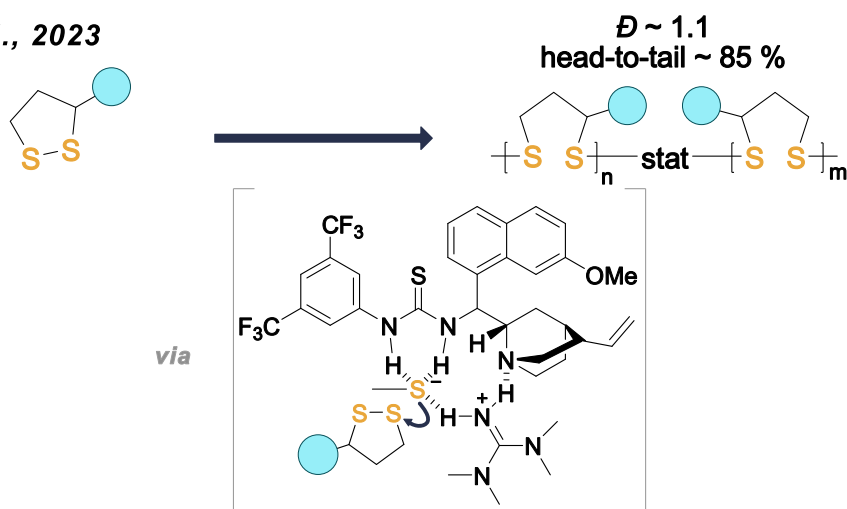


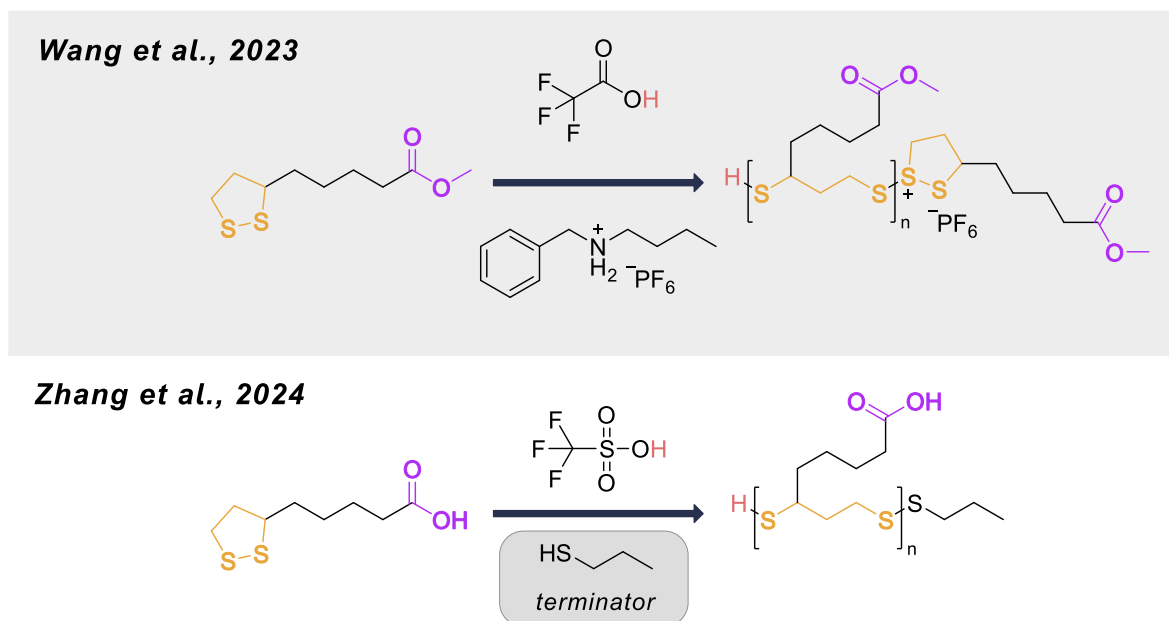
Figure 10 - Architecture-controlled polymerization of **ALA** derivatives. Arylthiolate initiators lead to polymers with cyclic topology due to better nucleofugality, according to Liu et al.^[73]

aROP can present different challenges. Because of the high reactivity of the active chain end, i.e., the thiolate, it can readily react with disulfides bonds already formed in the polymer in a chain transfer-type reaction that resembles thiol exchange polymerization, leading to a highly regiochemically irregular polymer backbone. To mitigate this, Du et al. used an anion-binding ligand to form a complex with the reactive thiolate anion, improving its selectivity towards ring-opening, i.e., propagation, against chain transfer and, due to the sterical hinderance of the ligand, improving regioselectivity.^[90] It was further explained that the narrow dispersity of 1.1–1.2 was reached because of the reversible termination of the reactive chain end by a proton and the hydrogen bonding to the ligand, improving selectivity for ring-opening. Furthermore, a head-to-tail to head-to-head ratio of 0.85 could be reached.

Du et al., 2023Figure 11 - Anion-binding catalysis for **aROP** of lipoates.^[90]

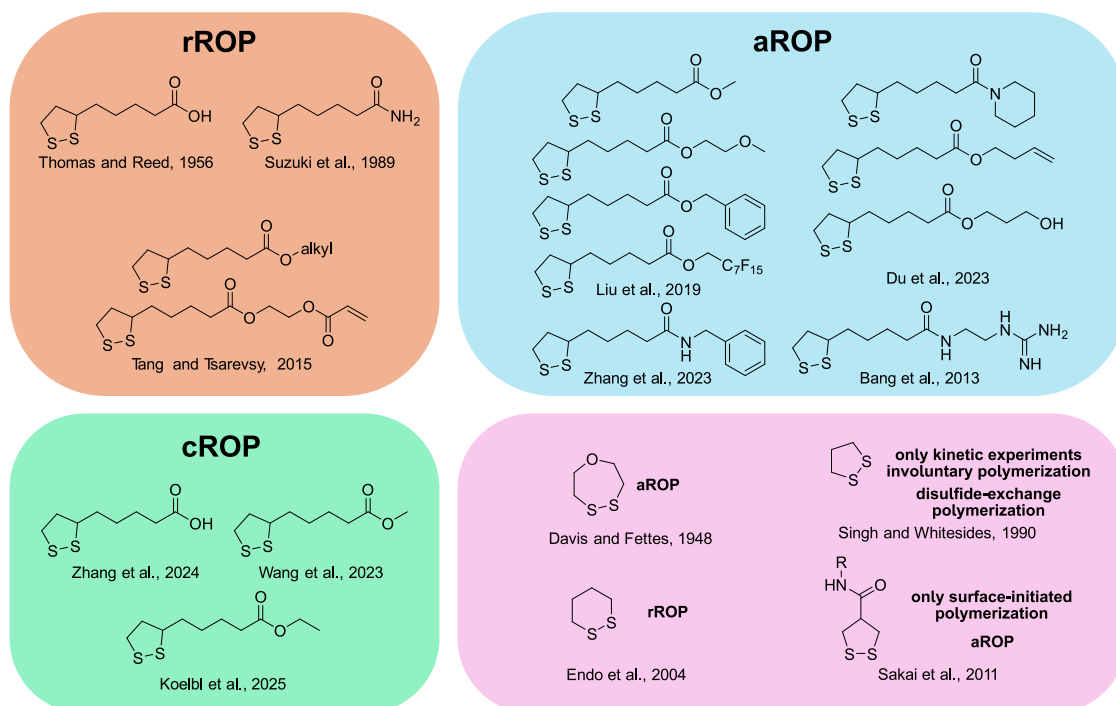
1.4.3.3 Cationic Ring-Opening Polymerization (**cROP**)

The newest reported fundamental polymerization mechanism for **ALA** derivatives is **cROP**. In 2023, Wang et al. reported the acid-catalyzed polymerization of **LpMe**.^[91] It was shown that after protonation of one of the sulfur atoms, forming a sulfonium cation, the nucleophilicity of the other sulfur atom is sufficient for the ring to be attacked thus enabling ring-opening. The polymerization could be carried out under air and at room temperature and showed completion after a few minutes. With this approach, polymers of over 1000 kDa could be obtained. It was also reported that non-coordinating salts facilitate **cROP**, increasing the conversion. Furthermore, the polymerization was shown to be reversible upon addition of the acid and non-coordinating salts and heating the mixture resulting in a polymer-to-monomer conversion of 89 %. Zhang et al. later built on these findings to obtain poly(**ALA**) with targetable molecular mass, achievable by the ratio of acid to monomer, which could later be depolymerized and repolymerized.^[92]

Figure 12 - Reported **cROP** of **LpMe** and **ALA**, according to Wang et al. and Zhang et al., respectively.^[91,92]

1.4.3.4 Polymerized α -Lipoic Acid Derivatives

An overview of ALA derivatives and similar molecules, i.e., 1,2-dithiolanes, 1,2-dithianes, or other disulfide-containing monomers that were successfully polymerized is given in Figure 13.

Figure 13 - Overview of different **ALA** derivatives and related molecules for **ROP**.^[22,29,31–33,38,72,73,86,90–93]

Of special interest in this thesis are **ALA** derivatives suitable for **aROP**. That includes **LpMe**, the monomer used for polymerization in this thesis.

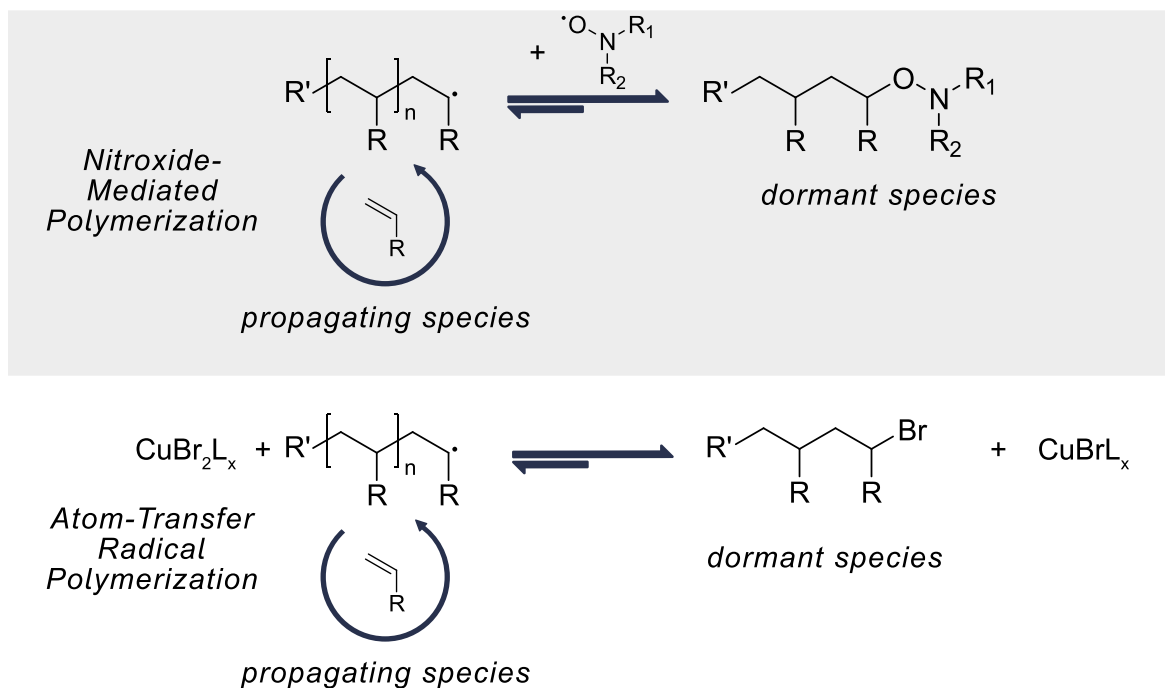
1.5 Methods

1.5.1 Reversible Addition-Fragmentation Chain Transfer Polymerization

Radical polymerization (**RP**) is one of the most used polymerization methods of commercially available polymers. Its main advantages include compatibility with a wide range of functional groups and therefore a large number of available monomers, e.g., styrenes, (meth)acrylates, (meth)acrylamides, butadiene, and vinyl acetate. Furthermore, **RP** is usable in a variety of reaction conditions. Besides polymerization of bulk materials, i.e., polymerization of monomers without solvent, **RP** can be applied in solution, to lower the viscosity of the reaction mixture, facilitate better heat transfer, and avoid autoacceleration or the gel effect. Also, **RP** can be used for suspension, emulsion, and precipitation polymerizations. Lastly, **RP** is easy and inexpensive to implement compared to other methods.

However conventional **RP**, or free **RP** (**FRP**) is known to have some disadvantages. Besides the aforementioned autoacceleration in bulk, which can be avoided, it also only shows limited control over molecular weights and dispersity of the synthesized polymers. Polymers synthesized using **FRP** typically show a dispersity $\bar{D}$ of over 2.0 if termination occurs by disproportionation or over 1.5 if termination occurs by recombination.

To control radical polymerizations, i.e., to control the molecular weight of the polymers and achieve a narrow molecular-weight distribution with dispersity values ideally below 1.05, several methods have been developed, which utilize the reversible deactivation of active chain ends to suppress chain termination. These polymerization methods are called reversible-deactivation radical polymerizations (**RDRP**). Most notable are nitroxide-mediated polymerization (**NMP**), atom-transfer radical polymerization (**ATRP**), and reversible-addition chain transfer polymerization (**RAFT**).^[94]



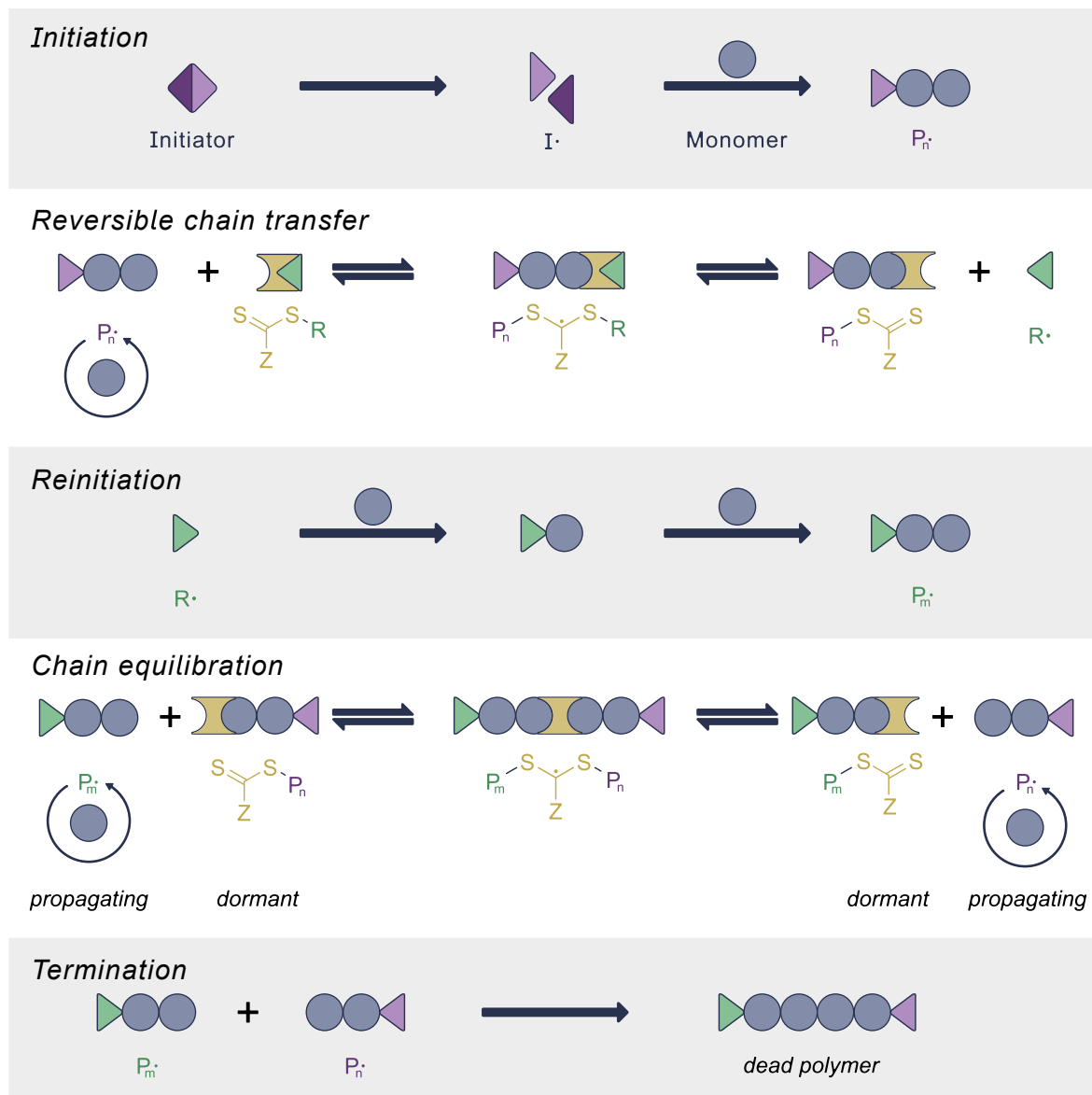
Scheme 19 - Reversible deactivation of propagating species in **NMR** and **ATRP**.

NMP, the first developed **RDRP** method, which was found to achieve similar results as anionic polymerization, utilizes a nitroxide as a radical trap, to reversibly deactivate growing chains so that only a few chains can grow simultaneously. The reaction equilibrium lies on the side of the dormant chains, so that the instantaneous radical concentration is kept low and the termination is hindered. Its development enabled the synthesis of polymers with controlled molecular weight but also new materials, which were not accessible by radical polymerization before, like block copolymers, star and graft polymers, or dendritic structures. Major advantages of **NMP** are the simple application, especially when using an alkoxyamine initiator, and the reaction, and therefore the polymers, being metal-free. However, **NMP** only has a limited scope of monomers and cannot be used with vinyl acetate, except in copolymerization.^[95–97]

ATRP, which was developed in 1995, functions similarly to **NMP**, however the initiation and reversible deactivation is an inner-sphere electron transfer and is carried out using a metal complex (mostly copper) and an initiator, that is mostly an organic halide (often an alkyl bromide). It also shows control over molecular weight and its distribution and shows more versatile applications than **NMP**. Furthermore, **ATRP** enables targeted end-group functionalization. However, the presence of transition group metals in the final polymer is a major disadvantage.^[98,99]

The newest **RDRP** method, **RAFT**, was first introduced in 1998. In **RAFT** polymerization thiocarbonylthio compounds are used to induce reversible chain transfer of the growing

chain and therefore decreasing the number of propagating chains in the reaction mixture. The mechanism of **RAFT** polymerization is shown in Scheme 20.



Scheme 20 - Mechanism of **RAFT** polymerization.

As shown in Scheme 20 the **RAFT** polymerization mechanism starts similarly to the mechanism of **FRP**. Initiation and termination follow the same reactions as in **FRP**. However, shortly after the reaction is initiated, the propagating radical $P_n\cdot$ undergoes addition to the thiocarbonylthio compound followed by fragmentation of a new radical $R\cdot$, which can initiate a new propagating chain $P_m\cdot$. The propagating chain is in rapid equilibrium with the dormant species thus providing the same probability for all chains to grow equally leading to low dispersity polymers. Because of that and the limited number of active chains the amount of chains that are terminated is kept to a minimum. After completion of the

polymerization or if the polymerization is quenched most of the chains are dormant and can be isolated and stored.^[100]

For successful application of **RAFT** polymerization, the utilized-chain transfer agent (**CTA**), or **RAFT** agent, must be specifically selected for every monomer. The choice of the **CTA** depends on several factors, primarily the reactivity of the monomer and the desired polymer architecture. For that, many **CTAs** have been developed.

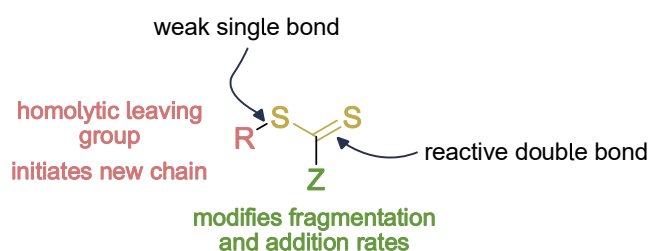


Figure 14 - Structure of chain-transfer agents for **RAFT** polymerization and requirements to the different groups of the **CTA**.

The general structure of the **CTA** consists of a thiocarbonylthio moiety with different R- and Z-groups. R is a free-radical leaving group and has an impact on the rate of fragmentation. It also must be a sufficiently good initiator to reinitiate polymerization. The Z-group on the other hand can be modified to influence the rate of addition and fragmentation by activating or deactivating the $C=S$ double bond and modifying the stability of the intermediate radical. For efficient **RAFT** polymerization the **CTA** must have a reactive $C=S$ double bond and a weak $S-R$ bond so that the intermediate radicals fragmentation is fast, without side reactions, and progresses the polymerization in favor of the product.^[101]

Different classes of **CTAs**, such as dithiobenzoates, trithiocarbonates, dithiocarbamates and xanthates, are best suited for different classes of monomers based on the rates of fragmentation and addition in the **RAFT** equilibrium. Figure 15 shows the most common R- and Z-groups for different classes of monomers, namely less activated monomers (**LAMs**) and more activated monomers (**MAMs**).

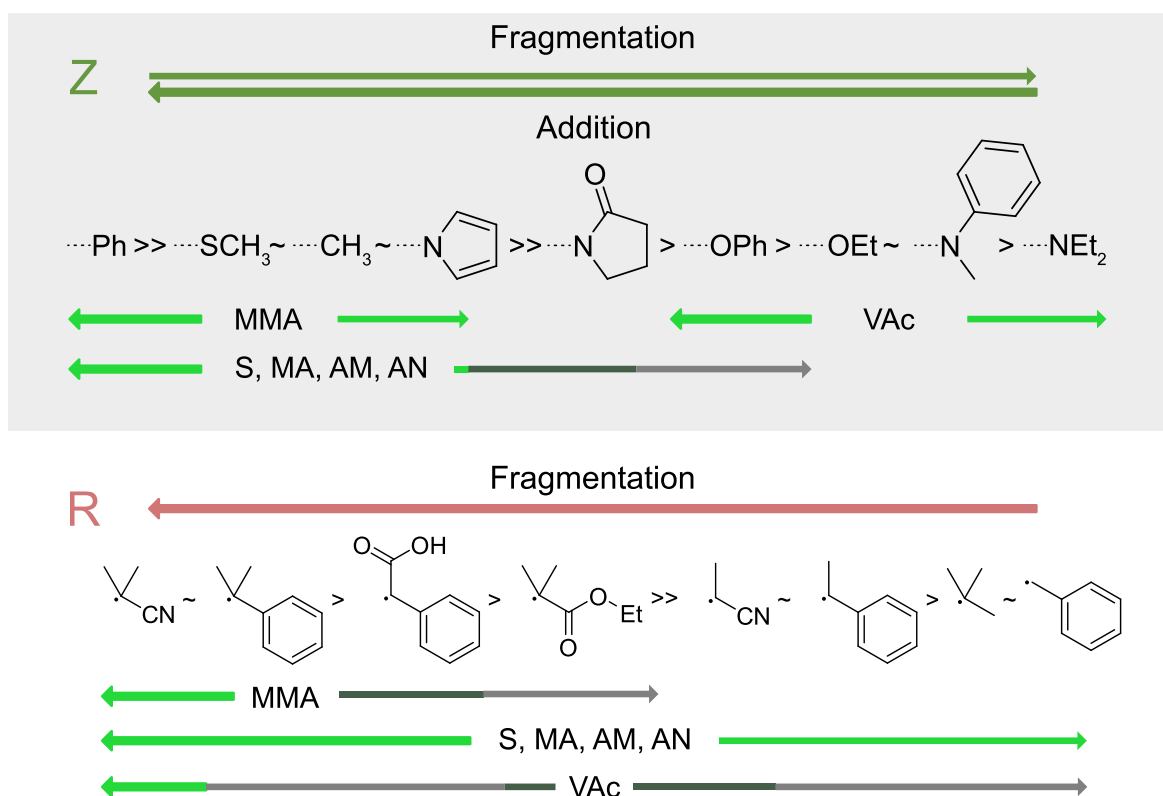
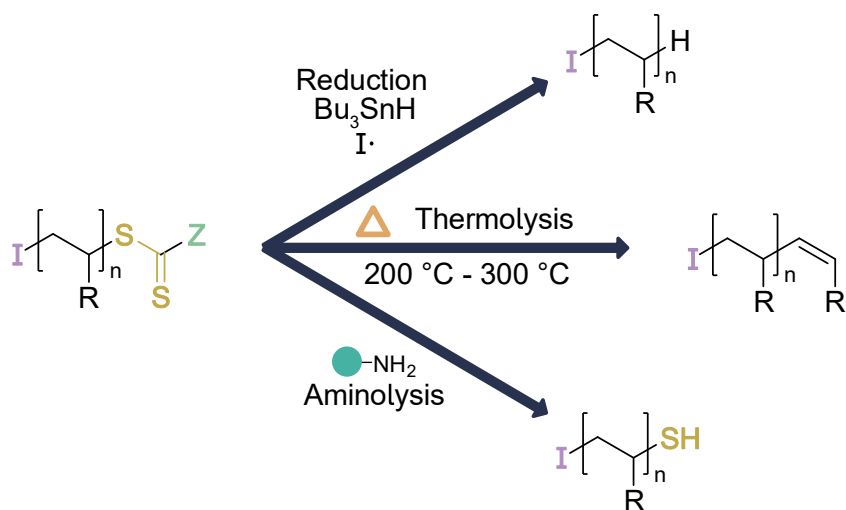


Figure 15 - Different Z- and R- groups and their effect on the rate of fragmentation and addition and the compatibility of these groups with different monomers (**S** = styrene, **MMA** = methyl methacrylate, **MA** = methyl acrylate, **AM** = acrylamide, **AN** = acrylonitrile, **VAc** = vinyl acetate).

MAMs, such as styrenes, acrylates, and acrylamides tolerate a wide variety of R-groups, are however limited in the choice of the Z-group, which should stabilize the intermediate radical. Therefore Z-groups like phenyl, thioalkyl, alkyl, or other aromatic groups are beneficial in accelerating the fragmentation. Methacrylates are compatible with the same Z-groups, however the R-group needs to be specifically chosen to effectively reinitiate the polymerization. An example would be a 2-cyanopropyl group, whose radical after fragmentation is stabilized by the electron withdrawing cyanide group enabling a fast fragmentation. **LAMs**, like vinyl acetate, however only tolerate **CTAs** with electron-withdrawing Z-groups, like dithiocarbamates or xanthates, and also relatively stable R-radicals to destabilize the intermediate radical leading to faster fragmentation.^[101]

Since polymers made by **RAFT** polymerization retain the thiocarbonylthio end-group, it opens up the possibility for ω -chain end modification. The thiocarbonyl group can be cleaved by different methods. Reduction with tributyltin hydride and a radical initiator leads to the complete cleavage of the thiocarbonylthio moiety leading to the corresponding aliphatic polymer, whereas thermolysis affords a terminal insaturation. Lastly, nucleophilic attack with an amine or alcohol on the thiocarbonyl group yields a polymer with a thiol end-group.^[102]



Scheme 21 - ω -end-group functionalization by reduction, thermolysis, and aminolysis.

The aminolysis of **CTA**-terminated polymers will be made use of in this thesis to obtain thiol-terminated macroinitiators for the initiation of **aROP** of **ALA** derivatives.

2 Motivation

Plastic waste and pollution of the environment, especially pollution of the seas, is one of the biggest cases of environmental damage in our time. Additionally, resources to produce these plastics are mostly derived from crude oil, which is of limited supply, and damage the environment further when they are being manufactured. Therefore, alternative classes of polymers are being investigated, that are at best biobased, biodegradable and not harmful for the environment, humans and animals, which would not only enable closed-loop recycling of plastics and their degradation in the environment, but also the application of new kinds of polymers in biomedical and material science contexts.

The combination of traditional oil-based chemistry for the manufacture of polymers and new biobased, biodegradable materials is a means of bridging both worlds, thus opening the possibility for traditional materials that are imparted with a degree of degradability and reducing the amount of limited resources being used.

In this thesis α -lipoic acid (**ALA**) is explored for the synthesis of these combination polymers. Although **ALA** is found in nature and could thus technically be biobased, the feedstock is rather small and, therefore, also being produced from oil-based resources. However, the dithiolane moiety, which is a five-membered ring with a disulfide bond, enables polymers synthesized from it to be easily degraded by reduction of the formed disulfide bonds in the polymer backbone.

Here **ALA** is used to synthesize new block copolymers initiated by **RAFT**-derived vinyl polymers. The lipoate block is meant to be degradable by reduction, which would enable regaining the original vinyl polymers. If the polymerization of the second block can be effectively controlled this could for example be utilized for surface coating, forming different domains of degradable and non-degradable regions. The degradable regions could then be depolymerized in a controlled manner, regaining the monomer and leaving the vinyl polymer coating behind. Another application could be the formation of redox-responsive nanoparticles, which would enable them to change in size or morphology due to the degradation of the second block.

The goal of this work is to establish the polymerization of an **ALA** derivative, namely methyl lipoate (**LpMe**), initiated by aminolyzed **RAFT**-derived vinyl polymers of different monomers and the characterization of these polymers. Additionally, the influence of different reaction conditions on the size and dispersity of the polymers is explored. Lastly, the degradability of the polymers is to be investigated.

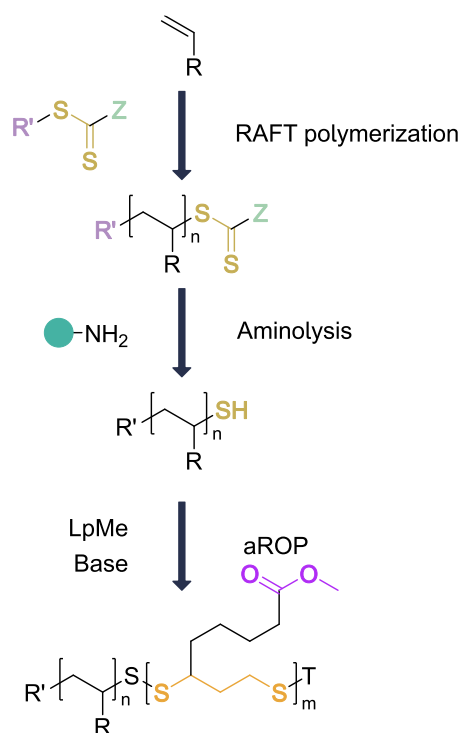


Figure 16 - General reaction scheme for the thiol-terminated vinyl polymer initiated **aROP** of **LpMe**.

3 Results and Discussion

3.1 Preliminary Proof of Concept for the Initiation of Anionic Ring-Opening Polymerization using RAFT-derived Macroinitiators

To investigate the feasibility of thiol end-group initiated **aROP** of **ALA** derivatives, first, methyl acrylate (**MA**) was polymerized using 2-(dodecylthiocarbonothioylthio)-2-methylpropionic acid (**DDMAT**) as the **CTA** and azoisobutyronitrile (**AIBN**) as initiator to obtain a **RAFT**-derived macroinitiator.

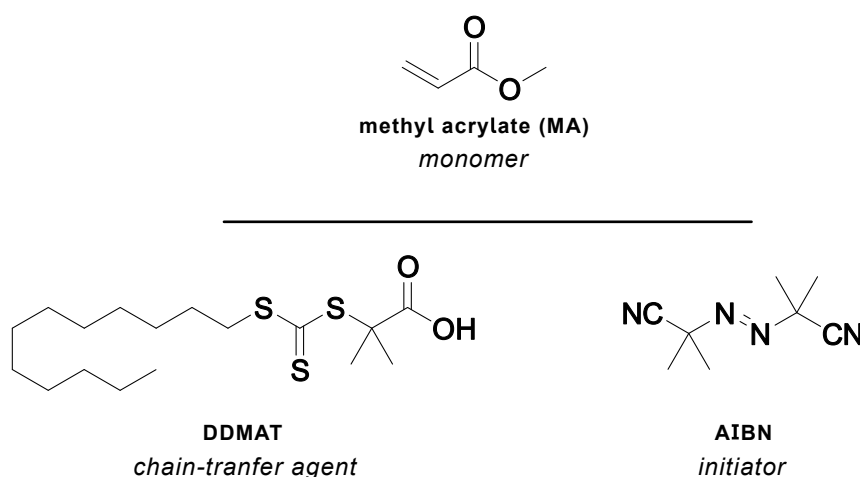


Figure 17 - Structures of methyl acrylate (**MA**) as monomer, **DDMAT** as **CTA** and **AIBN** as initiator.

The choice of the **CTA** was based on compatibility with the monomer (see Chapter 1.5.1) and availability at the time. **MA** was chosen because of its fast reaction times. A chain length of 20 and a conversion of 67 % was targeted, so the monomer to CTA to initiator ratio was set to 30:1:0.1.

After purification the obtained polymer was analyzed using gel permeation chromatography (**GPC**) and matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (**MALDI-TOF-MS**). The obtained polystyrene displayed a number-average molecular weight (M_n) of $2.5 \text{ kg} \cdot \text{mol}^{-1}$, which corresponds to a chain length of 25, and a polydispersity of 1.1 (Figure 18). The **MALDI-TOF-MS** spectrum showed a repeating mass of 86.01 m/z, corresponding to the mass of the repeating methyl acrylate unit with a theoretical mass of 86.04 m/z. The mass of 2107.8 m/z can be assigned to the theoretical mass of the sodium adduct of the desired poly(methyl acrylate) with a chain length of 20 of 2107.9 m/z (see Appendix 3.1 Figure 44). Combining these results, it can be reasoned that the desired polymer was successfully synthesized.

Following that, the synthesized poly(methyl acrylate) was aminolyzed using ethanolamine. Both the polymer solution in tetrahydrofurane (**THF**) and the ethanolamine solution were degassed with argon. The aminolyzed polymer was purified after complete aminolysis and analyzed using **GPC**. The molecular-weight distribution (**MWD**) along with the theoretical **MWD** of the aminolyzed poly(methyl acrylate), which was calculated by subtracting the molecular weight of the dodecylthio leaving group and scaling to the corresponding value of the experimental **MWD**, is shown in Figure 18.

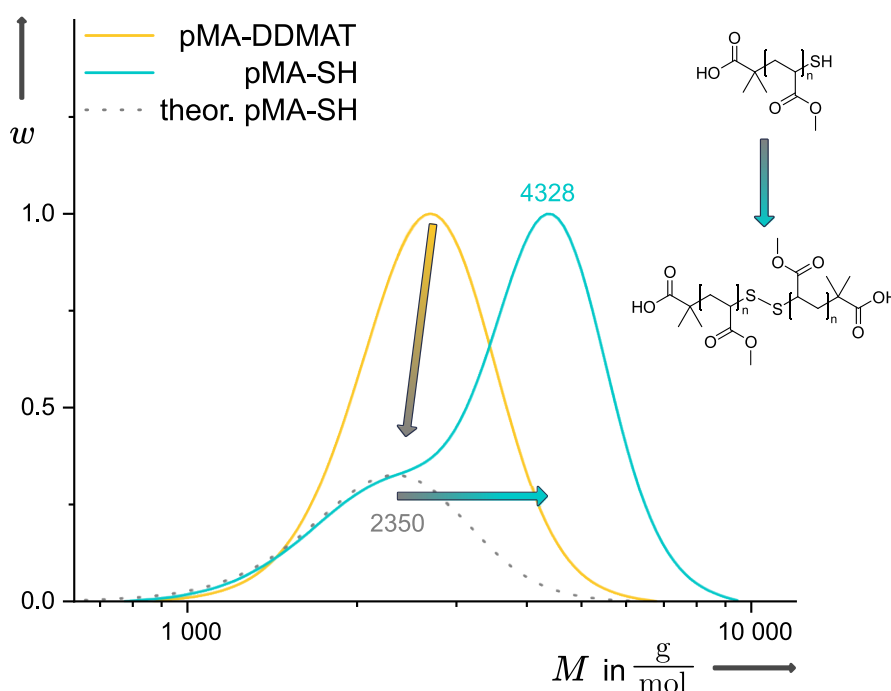


Figure 18 – Molecular-weight distributions of **pMA-DDMAT** (orange) and the aminolyzed product (blue) with the theoretical molecular-weight distribution of the thiol-terminated polymer (grey).

As shown in Figure 18, the monomodal **MWD** of the original polymer appears to become multimodal after aminolysis. The multimodal **MWD** of the aminolyzed product displays two populations with the molecular weight of the higher-weight population being approximately double the molecular weight of the lower-weight population. This could suggest an incomplete aminolysis or the oxidation of the formed thiols to disulfides, only leaving a small amount of free thiols.

The completion of the aminolysis was confirmed using UV-Vis spectroscopy by the loss of the absorption band at 310 nm, which corresponds to the π - π^* transition of the thiocarbonyl group. The UV-Vis spectrum is shown in Figure 19.

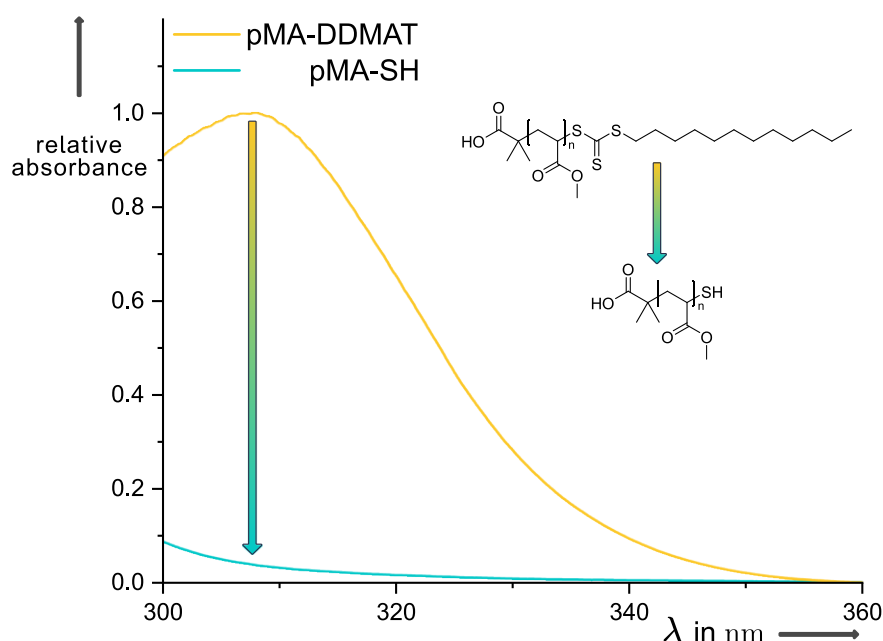


Figure 19 - UV-Vis spectra of **pMA-DDMAT** and the aminolyzed polymer in THF.

To further confirm that not only polymers based on **MA** display this behavior, the same procedure was done with polystyrene. The obtained **RAFT** polymer had a M_n of $3.1 \text{ kg} \cdot \text{mol}^{-1}$, a chain length of 27, and a dispersity of 1.1. The loss of absorption in the UV-Vis spectrum and the shift of the molecular weights are shown in Figure 20.

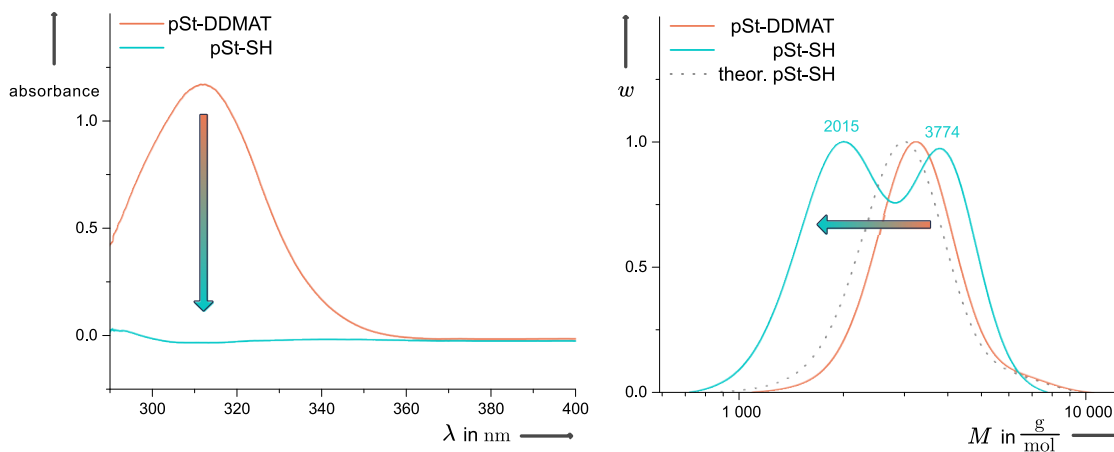


Figure 20 - UV-Vis spectra and molecular-weight distributions of **pSt-DDMAT** (red) and **pSt-SH** (blue).

The loss of the absorption band at 310 nm indicates that aminolysis was complete. However, the molecular-weight distribution again shows a bimodal distribution with one peak being at the approximate double weight than the other indicating that thiols which were formed during the aminolysis were oxidized. Unexpectedly, the low-molar-mass population is strongly shifted towards lower masses (that is, higher elution volumes) as compared to

the pristine **RAFT** polymer. The free carboxylic acid included in the R group of **DDMAT** could cause interactions with the **GPC** column material, yet so should it do in the pristine RAFT polymer. One could assume then a strong initial solvation due to the Z-group dodecyl chain. Nevertheless, in order to avoid potential **GPC** issues and to lower the degree of complexity for the experiment design of the **aROP** of the **ALA** derivative macroinitiators without the acid group were synthesized and used in further experiments.

However, for a first preliminary proof of concept for the **RAFT**-derived macroinitiator-initiated polymerization of **LpMe**, the obtained thiol-terminated polystyrene could be used.

The polymerization was carried out under inert atmosphere in dry **THF**. Since **aROP** is not compatible with protic conditions, **ALA** had to be modified by esterification to prevent termination of the thiolate anions by protonation. Thus, **LpMe** was chosen because of the ease of analysis of the pendant methoxy group using **NMR** spectroscopy.

As base for the deprotonation of the thiol-terminated polymers, triazabicyclodecene (**TBD**) was chosen, since **aROP** initiated with low-molecular-weight thiols was reported to be effectively started with TBD as a base.^[73] Termination of the active thiolate anions was carried out using methyl iodide.

The obtained polymer after purification by precipitation in methanol was analyzed using **GPC**. The molecular-weight distribution is shown in Figure 21.

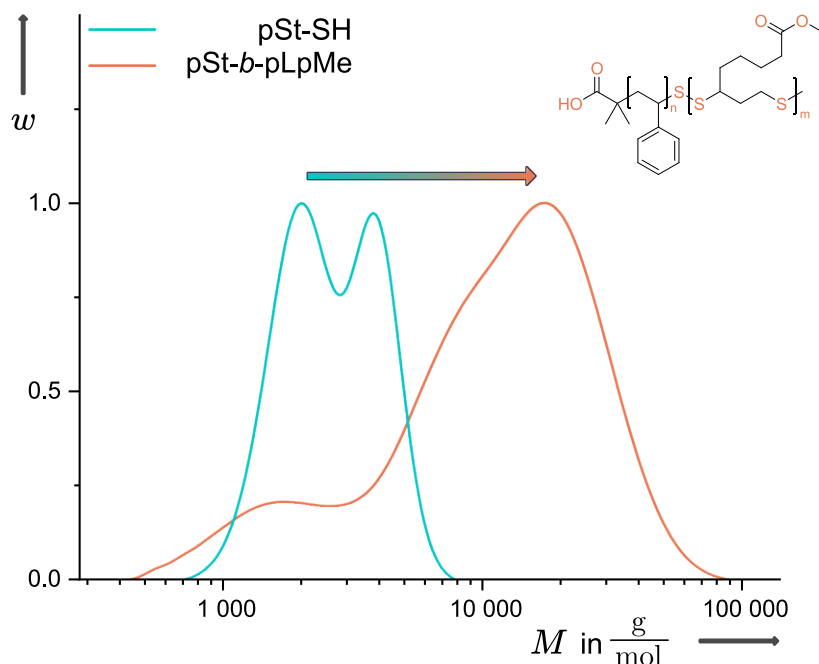


Figure 21 – Molecular-weight distributions of **pSt-SH** (blue; with oxidized thiols) and **pSt-b-pLpMe** (red).

The molecular weight of the sample increased as expected after polymerization of the second block from M_n 2.4 kg·mol⁻¹ to M_n 6.0 kg·mol⁻¹, M_p 17.2 kg·mol⁻¹. However, because of the polymer chains, which were oxidized to disulfides after aminolysis, only a part of the macroinitiator could initiate the polymerization of **LpMe**. Thus, the molecular-weight distribution of the obtained block copolymers is multimodal, which could indicate the formation of low-molecular-weight **LpMe** oligomers. Additionally, the asymmetry of the higher-molecular-weight fraction of the polymer could also be explained by the occurrence of thiol-exchange-type reactions between the disulfides from the oxidized initiators present in the reaction mixture and the active thiolate chain end of propagating **LpMe** monomers.

To avoid the formation and, thus, the presence of the disulfides formed by oxidation of the macroinitiator thiol end-group, it was reasoned that combining the aminolysis of the macroinitiators precursor with the polymerization of **LpMe** in a “one-pot” reaction under sufficiently dry and inert conditions would suppress the occurrence of the disulfides and simplify the experiment design by skipping the purification of the thiol-terminated macroinitiator.

Also, to alleviate the effects of a carboxylic acid end-group, mainly the possibility of interactions with the GPC column material and the additional complexity for the polymerization of **LpMe**, new macroinitiators were synthesized without the carboxylic acid group.

3.2 Synthesis of the Macromolecular Initiators using RAFT polymerization

For the synthesis of the macromolecular initiators for **aROP** of **LpMe** four vinyl monomers stemming from four different monomer classes were chosen to study the compatibility with the aminolysis-**aROP** strategy. The monomers chosen were styrene, methyl methacrylate, methyl acrylate, and *N,N*-dimethylacrylamide, because of their variety in reactivity.

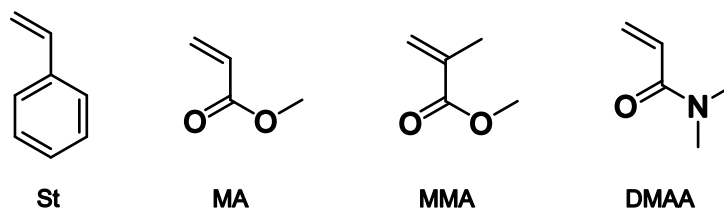


Figure 22 - Monomers chosen for **RAFT**-derived macromolecular initiators for **aROP**.

The monomers were polymerized in bulk using a trithiocarbonate **CTA** to obtain low-dispersity and low-molecular-weight polymers. The targeted range of the degrees of polymerization was 20 to 30. To ensure good end-group fidelity the polymerizations were

quenched at approximately 70 % conversion. Therefore, the ratio of the monomer to the CTA and initiator was set to 30:1:0.1.

The conversion, as well as the degrees of polymerization, which were measured using nuclear magnetic resonance (**NMR**) spectroscopy, of the synthesized oligomers are shown in Table 1.

Table 1 - Obtained **RAFT** polymers as macromolecular initiator precursors, their monomer conversion, and degree of polymerization measured with **NMR** spectroscopy.

Polymer	Conversion	DP_n (NMR)
pSt-CPDT	80 %	26
pMA-CPDT	87 %	25
pMMA-CPDT	69 %	24
pDMAA-CPDT	84 %	27

The macromolecular initiator precursors were also characterized using **GPC** in **THF**. The molecular-weight distributions of the polymers synthesized using styrene, methyl acrylate and methyl methacrylate are shown in Figure 23. The macroinitiator precursor based on **DMAA** could not be analyzed using **THF-GPC**, because of interactions between the sample and the column. To alleviate this, the macroinitiator precursor would have to be analyzed using **GPC**, that is operated with *N,N*-dimethylformamide (**DMF**).

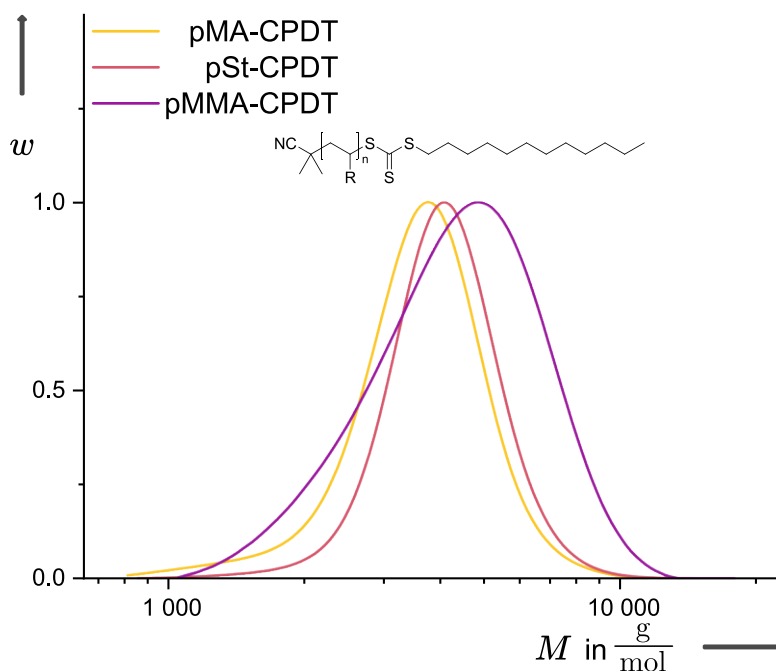


Figure 23 – Molecular-weight distributions (UV) of macroinitiator precursors based on methyl acrylate, styrene, and methyl methacrylate.

As seen in Figure 23 the polymers were obtained with monomodal and, in the case of styrene and methyl acrylate, symmetric molecular-weight distributions. Table 2 shows the number- and weight-average molecular weights and the corresponding dispersity values of analyzed oligomers.

Table 2 – Number- and weight-average molecular weights and dispersity values of the polymers measured using THF-GPC.

Polymer	M_n in $\text{kg}\cdot\text{mol}^{-1}$ (UV)	DP_n (GPC)	$\bar{D}$ (UV)
pSt-CPDT	3.9	34	1.1
pMA-CPDT	3.3	35	1.1
pMMA-CPDT	3.9	36	1.2
pDMAA-CPDT	—	—	—

Lastly, end-group analysis using matrix-assisted laser desorption/ionization time of flight mass spectrometry (**MALDI-TOF-MS**) was done to ensure a good end-group fidelity and to confirm the obtention of the desired end-functional polymers. Figure 24 shows the linear regression of a few signals obtained from the corresponding **MALDI-TOF-MS** spectra.

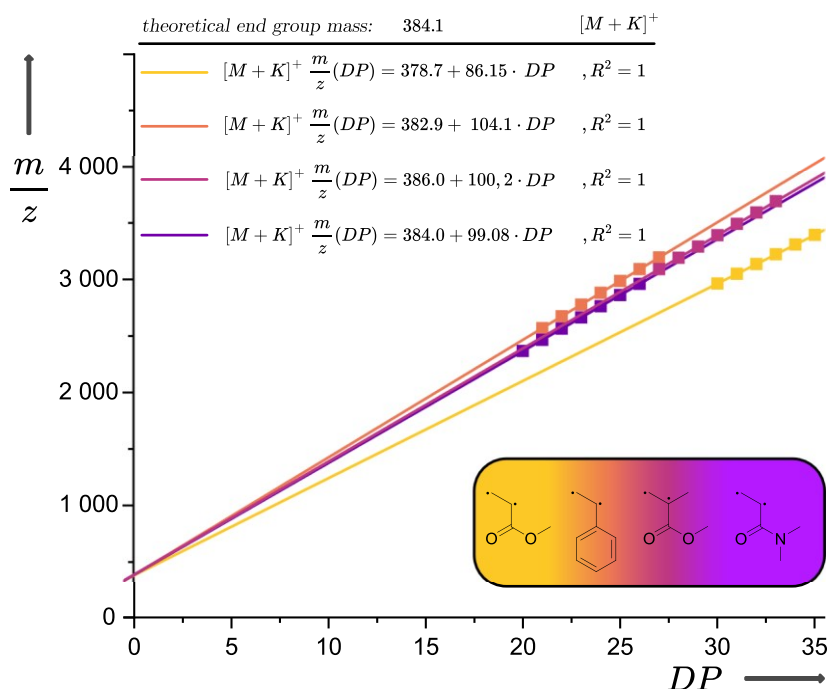


Figure 24 – End-group analysis by linear regression of signals in **MALDI-TOF-MS**.

As indicated in Figure 24, the signals of the potassium adduct of the corresponding macroinitiator $[M+K]^+$ approximately intercept the y-axis at the theoretical mass to charge ratio of the end-groups, which is m/z 384.1 $[M+K]^+$. Due to the low resolution of the spectra of the **MA** and **MMA**-based macroinitiators, measured using linear time-of-flight detection (see appendix, Chapter 6.2) the deviation from the theoretical mass is highest. Opposite to that, the mass spectrum of the **DMAA**-based macroinitiator displays the lowest deviation from the expected value because of the high resolution of the spectrum. However, based on the combination of results from **NMR** and UV-Vis-spectroscopy the error in determined end-group mass is likely to be caused by the low resolution of the spectra. It can, therefore, be assumed that the desired RAFT-derived precursors for the macroinitiators could successfully be obtained.

3.3 Ring-Opening Polymerization of Methyl Lipoate through Combined Aminolysis and Polymerization

In order to prevent oxidation of the aminolyzed macroinitiators to disulfides and the termination of thiolate anions by water, aminolysis and ring-opening polymerization were performed in a “one-pot” reaction under inert atmosphere. For that, separate stock solutions of the **RAFT** polymer, *n*-butylamine, and triazabicyclodecene (**TBD**) were prepared in dry *N,N*-dimethylformamide (**DMF**) by degassing with argon gas and drying over molecular

sieves. Additionally, **LpMe** was degassed and dried over molecular sieves by three cycles of evacuation and backfilling with argon gas.

DMF was chosen as the solvent for **aROP** for its ability to dissolve all RAFT polymers as well as all other reagents needed for **aROP**. Additionally, although **DMF** displays a higher cut-off in UV-vis spectroscopy than **THF** the presence of inhibitor in **THF** and the lack thereof in **DMF**, make the latter better suited for the verification of complete aminolysis using UV-Vis spectroscopy.

Although ethanolamine was used for aminolysis in the preliminary experiments (see Chapter 3.1), here it is replaced with *n*-butylamine, because unlike ethanolamine it does not show hygroscopy. Residual water in ethanolamine could protonate and, therefore, terminate active thiolate chain ends.

TBD was chosen as the base because **aROP** of **LpMe** was found to be highly dependent on the pK_a of the conjugate acid.^[73] To reach high conversions, over 70 %, a base whose conjugate acid has a pK_a over 18 is necessary. In acetonitrile the conjugate acid of **TBD** exhibits a pK_a of 26.03, which was found to be well correlated with the pK_a in **DMF**.^[103] This is sufficient to quantitatively deprotonate the thiol initiators and initiate **aROP**.

Lastly, termination of the active polymer chains was performed using 2-iodopropane. In comparison to methyl iodide, which was used before, 2-iodopropane, because of its higher vapor pressure, allows for more accurate dispensing. In addition, the characteristic signal of the isopropyl group should allow for easier analysis in **NMR** spectra. Although 2-iodopropane could undergo elimination of hydrogen iodide under strong basic conditions, because of the soft nucleophilicity of the thiolate anions, which matches the soft electrophilicity of the polarized carbon of isopropyl iodide, substitution should be favored in comparison to elimination. Finally, the terminator being added in large excess should permit quantitative capping.

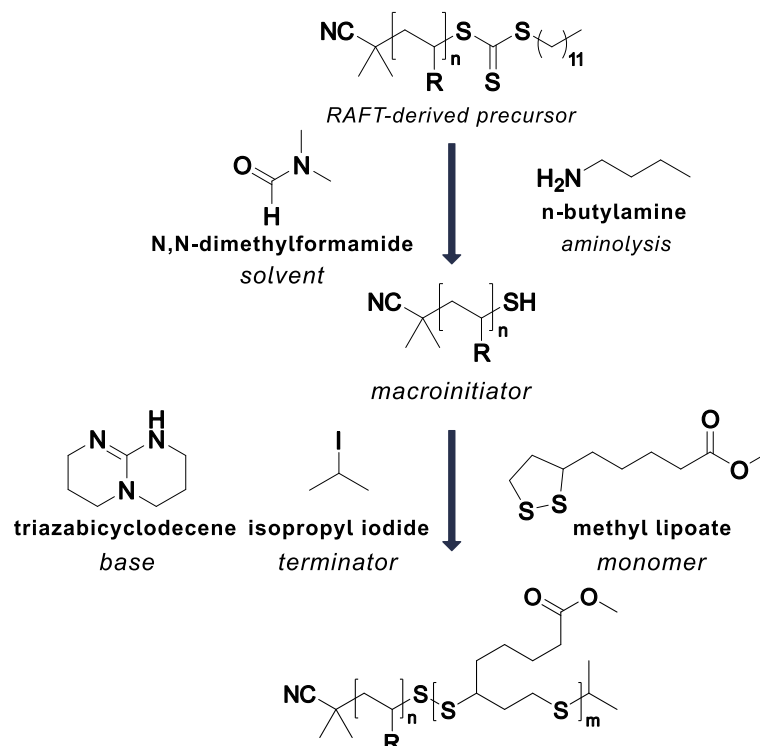


Figure 25 - Reagents for aminolysis of a **RAFT**-derived precursor and consecutive ring-opening polymerization of **LpMe**.

The RAFT-derived precursor is first aminolyzed using *n*-butylamine in **DMF**. After complete aminolysis, **LpMe** is added to the reaction mixture. The now free thiol macroinitiator is then deprotonated by **TBD**, starting the **aROP** of **LpMe**. After 90 minutes of reaction time, the polymerization is quenched by addition of an excess of isopropyl iodide.

First, the progress of the aminolysis could be monitored using UV-Vis spectroscopy, due to the loss of the π - π^* absorption band of the thiocarbonyl group at approximately 310 nm. The UV-Vis spectra of the polymers before and after aminolysis are shown in Figure 26.

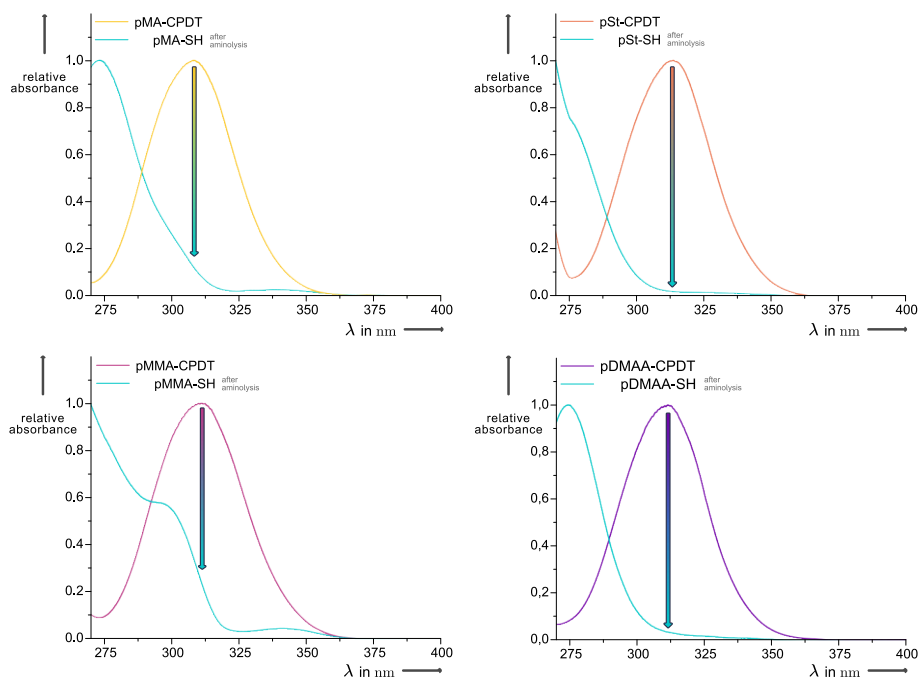


Figure 26 - UV-Vis spectra showing the relative absorbance of the used macroinitiators before and after aminolysis.

As seen in Figure 26, complete loss of the absorption band at 310 nm is observed for three of the four RAFT polymers. Only for pMMA significant absorption was observed at 310 nm due to a nearby absorption band, which could be the π - π^* absorption band of the **CTA** that was blue-shifted because of the change in pH by introduction of the alkaline *n*-butylamine. In that case the aminolysis would not have been carried out to completion. Also, because of the high nucleophilicity of the thiolate chain end, backbiting towards the penultimate ester functionality is feasible due to the high stability of five-membered rings, possibly leading to the formation of thiolactone termini.^[104] However, based on the observation of successful polymerization of the second block, which is discussed later, the presumably incomplete aminolysis of the macroinitiator still was sufficient for the initiation of **aROP**.

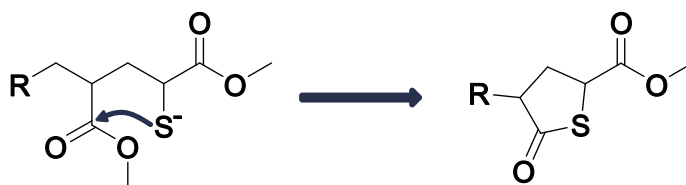


Figure 27 - Mechanism of thiolactone-termini formation of polymers with ester functionality.

For the polymerization, the monomer-to-initiator ratio was set to 50:1 with a monomer concentration $[M]_0$ of 2.5 M. As the first experiment the precursor of the macroinitiator based on styrene was used to investigate the feasibility of the experiment design. For this, after

aminolysis the now thiol-terminated macroinitiator was deprotonated by addition of the **TBD** solution and then **LpMe** was added. After 90 minutes of reaction time followed by termination and purification, the block copolymer was analyzed using **GPC**. The molecular-weight distribution is shown in Figure 28.

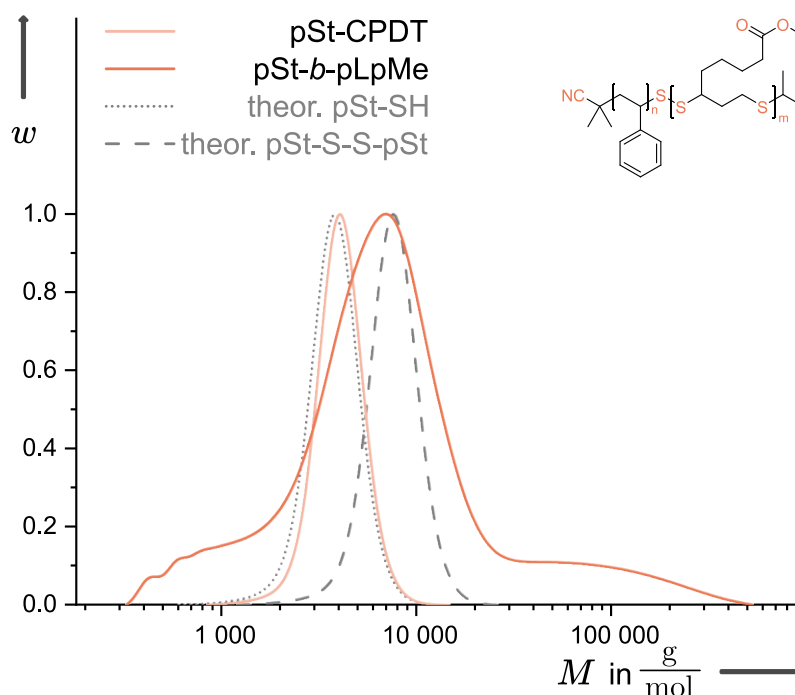


Figure 28 – Molecular-weight distributions (**THF-GPC, UV**) of **pSt-CPDT** (macroinitiator precursor, light red) and **pSt-b-pLpMe** (red) alongside the theoretical molecular-weight distributions of **pSt-SH** (aminolyzed macroinitiator, dotted grey) and their disulfides (dashed grey).

As shown, the molecular weight of the macroinitiator precursor increased from an M_n of $3.9 \text{ kg}\cdot\text{mol}^{-1}$ to M_p of $7.1 \text{ kg}\cdot\text{mol}^{-1}$, with the M_n unchanged at $3.9 \text{ kg}\cdot\text{mol}^{-1}$. Apart from the main population, which is presumed to be the desired block copolymer additional low- and high-molecular-weight fractions can be noticed.

It can be reasoned that because of the high viscosity of the monomer and because the thiol macroinitiators were already deprotonated by **TBD**, due to the high reaction rate of the polymerization, areas of high concentration were formed upon addition of the monomer, in which the polymer chains had the chance to grow much larger than was expected. Therefore, a small fraction of high-molecular-weight polymer chains were formed before the monomer was evenly distributed in the reaction mixture. To remedy this behavior in further experiments the monomer is added and mixed into the reaction mixture, thus ensuring even concentration, before deprotonation by **TBD**.

Also, the formation of the low-molecular-weight fractions may be explained by the formation of cyclic **LpMe** oligomers (due to backbiting of the active thiolate chain end), which could not effectively be removed by precipitation in methanol.

To confirm that the obtained polymer was indeed the desired **LpMe**-based polymer **NMR** spectroscopy was measured, of which the spectrum is shown in Figure 29.

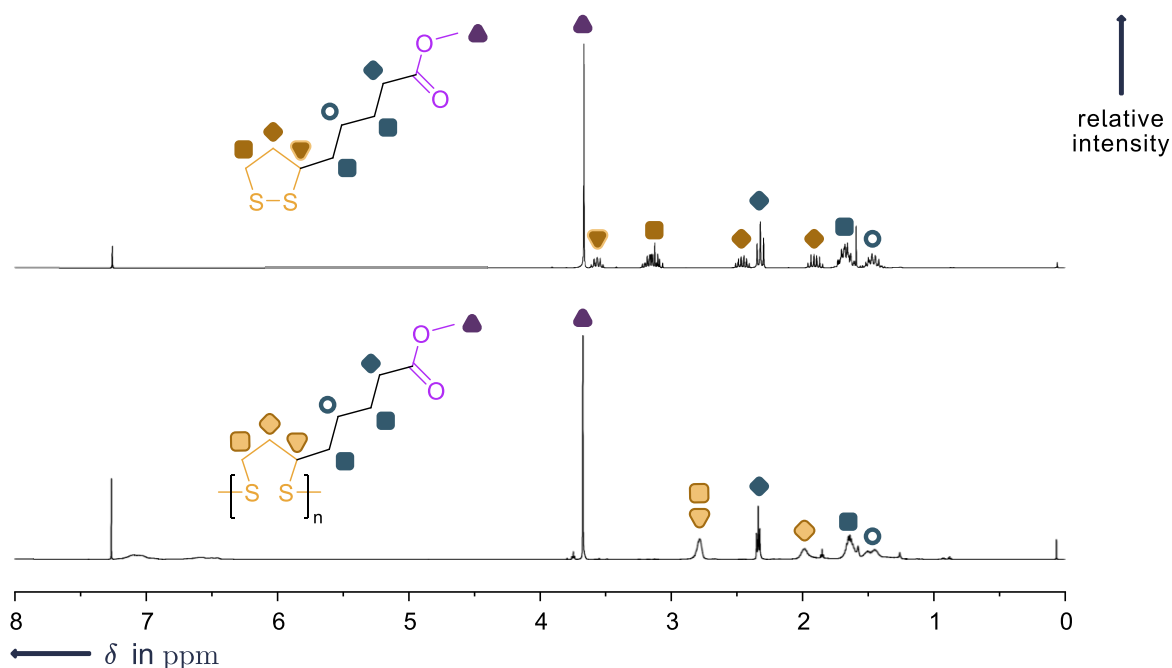


Figure 29 - ^1H NMR spectra of **LpMe** (upper spectrum, 300 MHz, CDCl_3) and **pSt-*b*-pLpMe** (lower spectrum, 600 MHz, CDCl_3).

The loss of the signals at 3.56 ppm, 3.14 ppm, 2.45 ppm, and 1.90 ppm and the significant broadening of the newly appearing signals at 2.78 ppm and 1.98 ppm indicate conversion of the monomer to the polymer. However, based on the appearance of signals from the dodecyl end-group of the **RAFT**-derived macroinitiator (see Figure 30), it can be reasoned that either the aminolysis of the **CTA** was not complete, and, therefore, a small amount of the macroinitiator precursor present in the sample, or the thiocarbamate side product of the aminolysis was precipitated alongside the desired polymer.

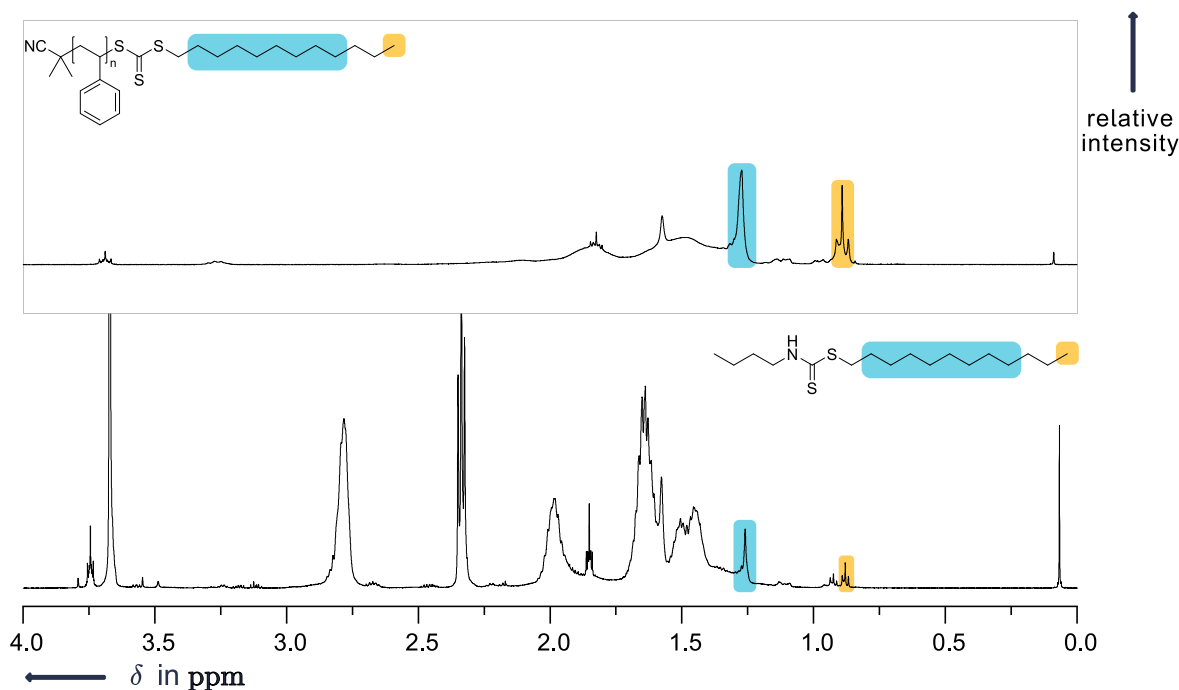


Figure 30 - ^1H NMR spectra of **pSt-CPDT** (upper spectrum, 300 MHz, CDCl_3) and **pSt-*b*-pLpMe** (lower spectrum, 600 MHz, CDCl_3) with signals from the aminolysis side-product highlighted.

The experiment was repeated with the addition of the monomer before **TBD**. Additionally, the other macroinitiators were used to initiate the **aROP** of **LpMe**. The block copolymers were obtained after 90 minutes of reaction time and termination with 2-iodopropane, with the **LpMe** conversions and chain lengths after three precipitations from methanol and one precipitation from diethyl ether shown in Table 3.

Table 3 - Conversions of obtained block copolymers measured using **NMR** spectroscopy.

Block copolymer	Conversion
pSt- <i>b</i> -pLpMe	52 %
pMA- <i>b</i> -pLpMe	—
pMMA- <i>b</i> -pLpMe	52 %
pDMAA- <i>b</i> -pLpMe	53 %

As evident from Table 3 the conversion for all polymerization experiments at a monomer concentration $[\text{M}]_0$ of 2.5 M was limited to around 50 %. **ROP** is known to be strongly influenced by temperature and monomer concentration. The lower the temperature or the higher the concentration the higher conversions can be reached. For the **aROP** of methyl lipoate the ceiling temperature was determined to be around 300 K. However, conversion

was lower than the reported 65 % for polymerization of LpMe initiated by thiophenol, yielding a polymer of cyclic topology.^[73]

To investigate if higher concentration in monomer can increase conversion, an additional experiment was carried out at $[M]_0$ 4 M with a monomer to initiator ratio of 200:1. However, surprisingly the conversion was lower at 32 %. The lower conversion could partly be explained by a large increase in viscosity of the reaction mixture due to higher monomer concentration, which prevented agitation, therefore hindering homogenous growth.

To further investigate, the block copolymers were analyzed using **GPC**. Figure 31 shows the molecular-weight distributions of the RAFT-derived polystyrene macroinitiator and of the block copolymer after 3-fold precipitation in methanol and after final precipitation from diethyl ether.

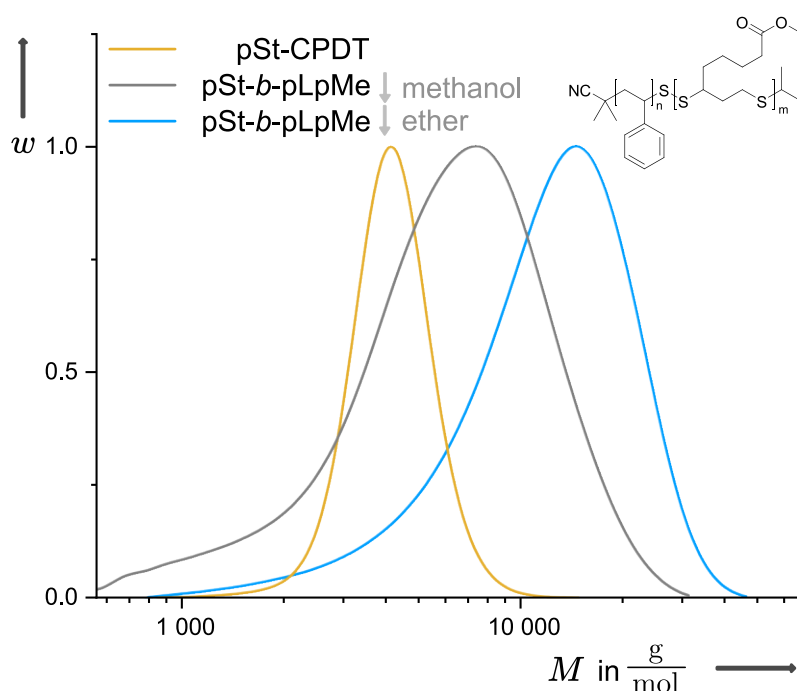


Figure 31 – Molecular-weight distributions (UV) of the styrene-based macroinitiator precursor and of the lipoate block copolymer after precipitation in methanol and diethyl ether, respectively.

As seen in Figure 31, the average molecular weight increases after polymerization and precipitation in methanol from $3.9 \text{ kg} \cdot \text{mol}^{-1}$ to $7.7 \text{ kg} \cdot \text{mol}^{-1}$. Evidently, the molecular-weight distribution of the polymer precipitated from methanol exhibits a tailing containing low-molecular-weight species, which were successfully removed by precipitation in diethyl ether. However, precipitation in ether also led to an overall increase in average molecular weight, by washing out lower-molecular-weight fractions, thus strongly decreasing overall yields from 41 % to only 7 %. The preparation of the other block copolymers showed similar behavior (see Appendix Section 6.3). These low yields can partly be explained by the low

conversion achievable with **LpMe**, which was only around 50 %, or an inefficient purification. In respect to only the polymerized **LpMe** (conversion around 50 %), the yields were 65 % and 12 %, respectively. The achieved yield of around 40 % after precipitation in methanol is consistent with yields reported in literature of 40 %-60 % for the polymerization of *R*-**LpMe** initiated by low-molecular-weight thiols like thiophenol.^[73]

In Figure 32 the molecular-weight distributions of all purified polymers obtained by **aROP**, except that of the block copolymer based on poly(*N,N*-dimethylacrylamide), which could not be analyzed using **THF-GPC**. Indeed, polyacrylamides tend to interact strongly with the column material when **THF** is used as eluent. *N,N*-Dimethylacetamide- (**DMAc**) or **DMF-GPC** should be performed.

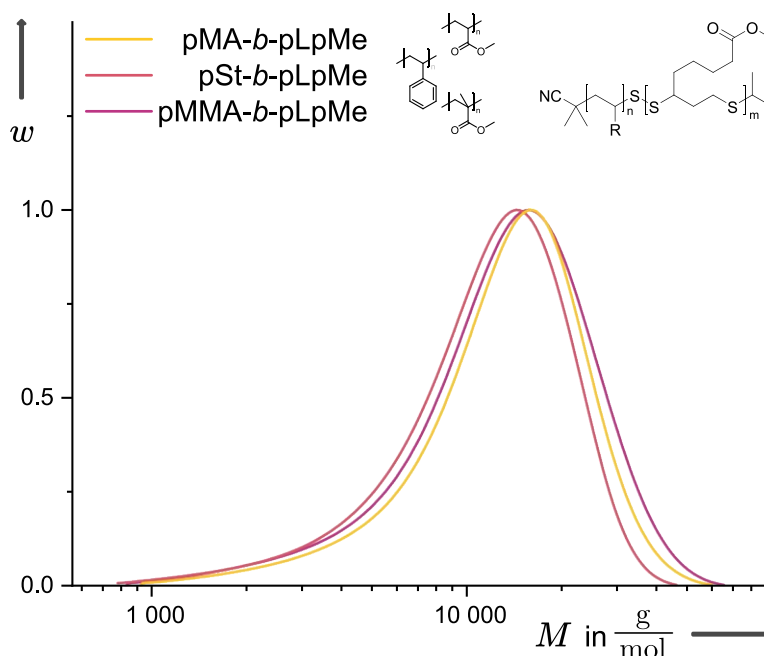


Figure 32 – Molecular-weight distributions (UV) of lipoate block copolymers after precipitation in diethyl ether.

The corresponding average molecular weights and dispersity values are shown in Table 4.

Table 4 – Number- and weight-average molecular weights, and dispersity values of the block copolymers measured using **THF-GPC**.

Block copolymer	M_n in $\text{kg}\cdot\text{mol}^{-1}$ (UV)	DP_n LpMe (GPC)	$\bar{D}$ (UV)
pSt- <i>b</i> -pLpMe	8.8	23	1.5
pMA- <i>b</i> -pLpMe	10.2	32	1.5
pMMA- <i>b</i> -pLpMe	10.0	28	1.6
pDMAA- <i>b</i> -pLpMe	—	—	—

Because of the approximately three-fold increase in molecular weight, it can be assumed that the methyl lipoate was successfully polymerized. Furthermore, the dispersities increased with the introduction of the second block. This observation is in accordance with previous literature stating the dispersity of thiolate-initiated methyl lipoate homopolymers to be about 1.4.^[73] The increase of the dispersity after polymerization of the second block could be explained by disulfide exchange-type chain transfer reactions leading to backbiting and, thus, the formation of low molecular-weight-oligomers.

The block copolymer synthesized with higher initial monomer concentration was also analyzed using **GPC**.

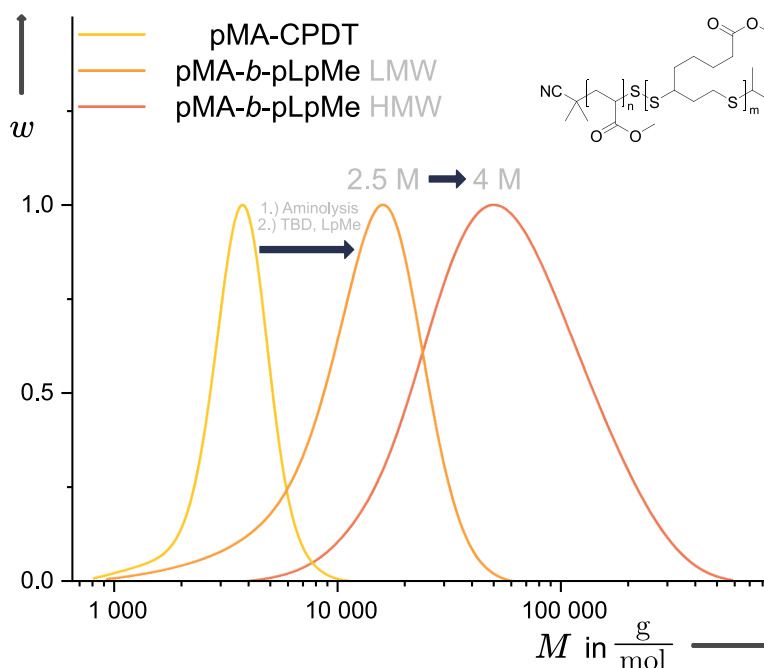


Figure 33 – Molecular-weight distributions (UV) of **pMA** macroinitiator precursor and low- and high-molecular-weight **pMA-*b*-pLpMe** block copolymers.

Table 5 - Comparison of the **GPC** results of **pMA** macroinitiator with low- and high-molecular-weight **pMA-*b*-pLpMe** block copolymers.

Polymer	M_n in kg·mol⁻¹ (UV)	DP_n LpMe (GPC)	$\bar{D}$ (UV)
pMA-CPDT	3.3	—	1.1
pMA- <i>b</i> -pLpMe - LMW	10.2	32	1.5
pMA- <i>b</i> -pLpMe - HMW	41.0	172	1.9

Usually, integration in **NMR** spectroscopy is a reliable way for the determination of chain lengths. However, when the analyzed polymer gets larger the signals for end-groups are getting smaller, leading to higher signal-to-noise ratio, thus increasing inaccuracy in integration. Also, the **NMR** signals overlapped with the broad signals of the polymer, hindering successful integration and, thus, estimation of the chain length. To avoid that, a higher number of scans, higher waiting times, or a different solvent may be necessary to lower signal-to-noise ratio.

The **NMR** spectrum of the purified high-molecular-weight polymer (see Appendix 6.3 Figure 82) still showed significant amounts of **LpMe**. It was also observed that the low-molecular-weight samples regained a faint yellow color after leaving them in solution under room temperature and light exposure, suggesting that these kinds of polymers may be light sensitive and, thus, depolymerize. Also, due to the lack of agitation during polymerization and termination of the high-molecular-weight sample the termination could have been incomplete, leaving chains with thiol end-groups, which could act as monothiol reducing agents for the disulfide bonds in the polymer backbone. The yield of the higher-molecular-weight block copolymer, which is 31 % overall is expected in comparison to the yields of the other block copolymers, because of the low conversion of **LpMe**. However, when considering this low conversion of only 32 %, the yield in respect to only the polymerized LpMe is 85 %, showing that the purification of the block copolymers by precipitation is more effective at higher molecular weights.

To evaluate if block copolymers were formed and if termination occurred as intended the block copolymers were analyzed using **MALDI-TOF-MS**. Figure 34 shows the analysis of the end-groups by linear regression of peaks in the corresponding **MALDI-TOF-MS** spectra.

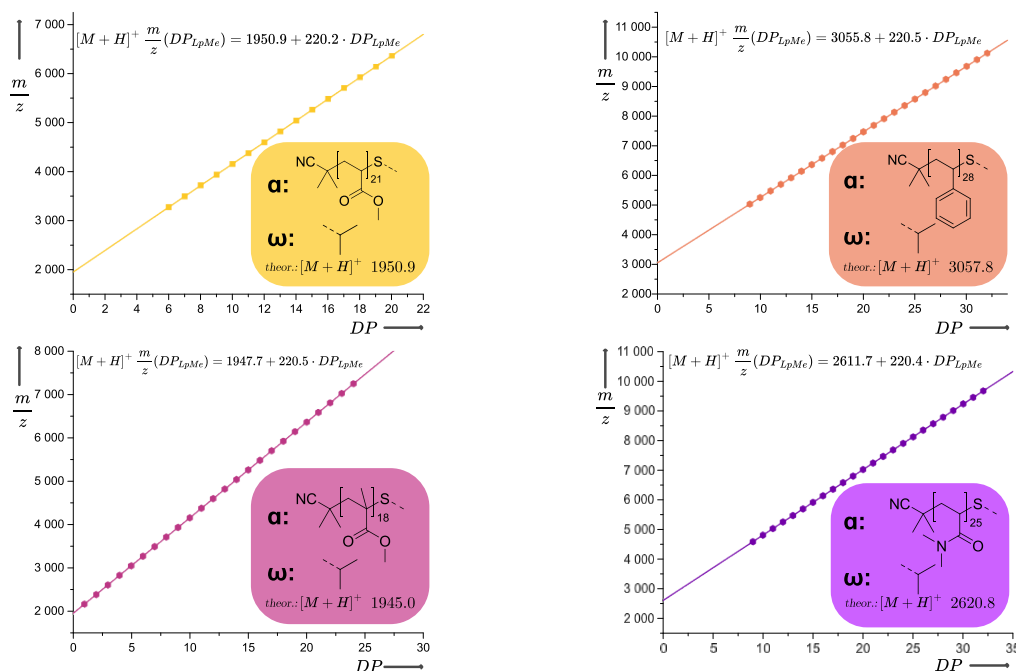


Figure 34 - End-group analysis of the obtained block copolymers by linear regression of peaks in **MALDI-TOF-MS** spectra. The α -end-group corresponds to the specific macroinitiator with a particular degree of polymerization, the ω -end-group is isopropyl.

As shown in Figure 34 the mass-to-charge ratio, which was identified by linear regression of peaks in the **MALDI** spectra, approximately resembles the theoretical mass-to-charge ratios of the end-groups with given chain length of the first block. The mass-to-charge ratio of the block copolymer based on **MA** most closely matches with the theoretical mass of the protonated species, whereas the deviation is largest for the block copolymer based on **DMAA**. Because of the nature of block copolymers, it would be expected that the **MALDI-TOF-MS** spectra would not only show the signals for different chain lengths of the second block, but would show a distribution of signals for every chain length of the first block. Additionally, the mass spectra obtained were of low resolution, because of the measurement done with linear time-of-flight detection rather than reflectron time-of-flight detection, allowing for spread of the flight times of ions with the same mass. The observed deviations of the mass-to-charge ratios to the theoretical values is most likely attributed to this. However, the results obtained hint at the formation of block copolymers and successful termination.

To ensure formation of block copolymers 2D diffusion-ordered **NMR** spectroscopy (**DOSY-NMR**) was measured. Opposite to two different homopolymers, the signals of different blocks in block copolymers should appear to have the same diffusion coefficient, because they belong to the same diffusing object.

Figure 35 shows the 2D-**DOSY-NMR** spectrum of the block copolymer based on polystyrene as an example of the 2D-**DOSY** spectra obtained.

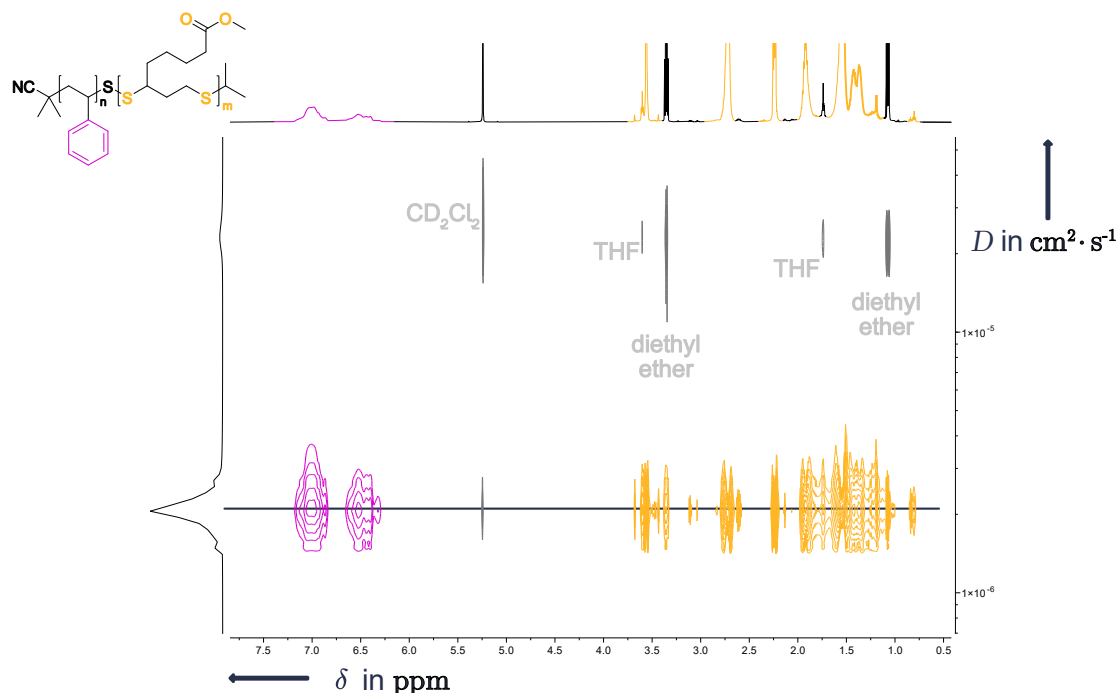


Figure 35 - 2D-**DOSY-NMR** (600 MHz, CD_2Cl_2) spectrum of **pSt-b-pLpMe** with the signals of the polystyrene block colored pink and the poly(methyl lipoate) block colored orange.

Apart from a few contaminants like THF and diethyl ether the **DOSY** spectrum in Figure 35 shows the signals of the first block, which is polystyrene, and the second poly(methyl lipoate) block at the same diffusion coefficient. Since the **DOSY** measurements were carried out with a high sample concentration of $20 \text{ mg} \cdot \text{mL}^{-1}$ the apparent diffusion coefficient significantly differs from the real diffusion coefficients.^[105] Therefore the **DOSY-NMR** measurements are only to be understood as qualitative evidence for the presence of block copolymers. The **DOSY** spectra of the other block copolymers show similar results (Figure 36).

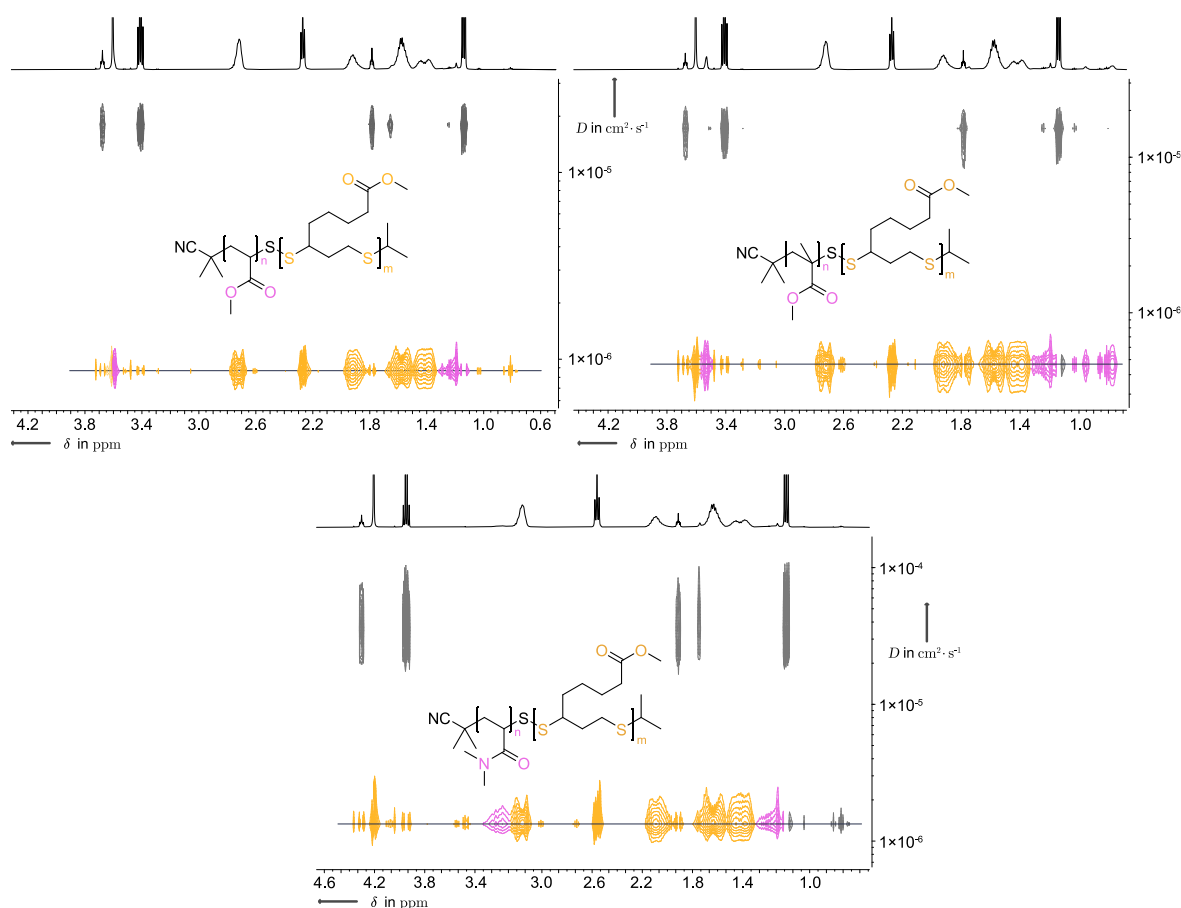


Figure 36 - ^1H -DOSY-NMR (600 MHz, CDCl_3) of **pMA-*b*-pLpMe** (upper left), **pMMA-*b*-pLpMe** (upper right) and **pDMAA-*b*-pLpMe** (below) with the signals corresponding to the first block marked in pink and the signals of the second block colored orange.

As with the **DOSY-NMR** results from the block copolymer based on styrene, the other block copolymer samples also show the signals of both of the blocks within the same diffusion coefficient range. It is noteworthy that all samples are contaminated with the solvents used in the precipitation. Although they were dried in vacuo, because of the highly viscous properties of the final products, the solvent residues are difficult to remove. However, the results from **DOSY-NMR** in combination with results from **NMR** spectroscopy, **GPC**, and **MALDI-TOF-MS** confirm the successful obtention of block copolymers.

3.4 Additional experiments

3.4.1 Synthesis of a High-Molecular Weight pSt-*b*-pLpMe Initiated by a Higher-Molecular Weight Macroinitiator

After the successful establishment of RAFT-derived vinyl polymer initiated **aROP** of **LpMe** additional experiments were performed to investigate the scope of these methods and possibilities to improve on them.

First, a block copolymer based on a higher-molecular-weight polystyrene initiator was targeted to investigate if higher-molecular-weight block copolymers are accessible. The RAFT polymerization was designed to yield a polymer with a chain length of 100 styrene units with a monomer-to-**CTA** ratio of 143:1 and a target conversion of 67 %. The obtained polymer displayed a number-average molecular weight average of $10.6 \text{ kg}\cdot\text{mol}^{-1}$, which is equivalent to a chain length of 98, and a dispersity of 1.1.

Aminolysis and ring-opening polymerization with a monomer concentration of 3 M and a monomer-to-initiator ratio of 80:1 then yielded a block copolymer with a number average molecular weight of $30.1 \text{ kg}\cdot\text{mol}^{-1}$ and a dispersity of 1.5. Based on this result the possibility of obtention of higher-molecular-weight block copolymers seems feasible. However, due to the high viscosity of the polymer solution in the low amounts of solvent needed to ensure high monomer concentrations and thus high monomer conversion, agitation of the reaction mixture was not possible, evidencing a challenge with this kind of polymerization, which could in the future be lessened by the use of a different solvent.

3.4.2 Degradation of High-Molecular Weight pMA-*b*-pLpMe

Secondly, because the synthesized polymers are supposed to be degradable under reducing conditions, the high-molecular-weight block copolymer based on **MA** was chosen to be exposed to reductive conditions using **DTT** as a dithiol reducing agent to show if this polymer is degradable. The corresponding molecular-weight distributions are shown in Figure 37.

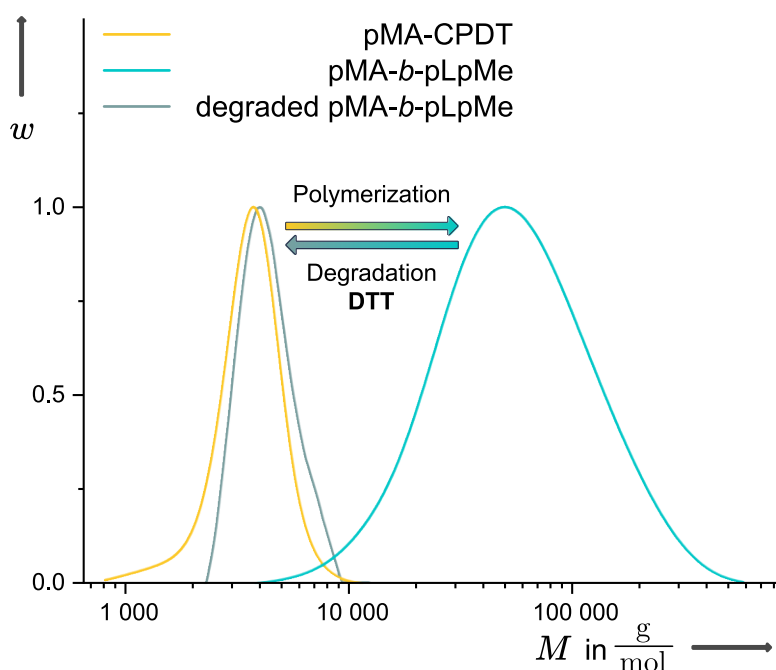


Figure 37 – Molecular-weight distributions of the **RAFT**-derived **MA**-based macroinitiator, the high-molecular-weight lipoate block copolymer based on that, and the corresponding degraded copolymer.

As shown in Figure 37 the molecular weight of the block copolymer decreased upon exposure to **DTT**, almost returning to the molecular weight of the original macroinitiator. However, based on the slightly higher molecular weight it can be assumed that a few methyl lipoate units are still attached to the initiator.

To verify, if indeed the initiator was regained, **NMR** spectroscopy was measured, which is shown in Figure 38.

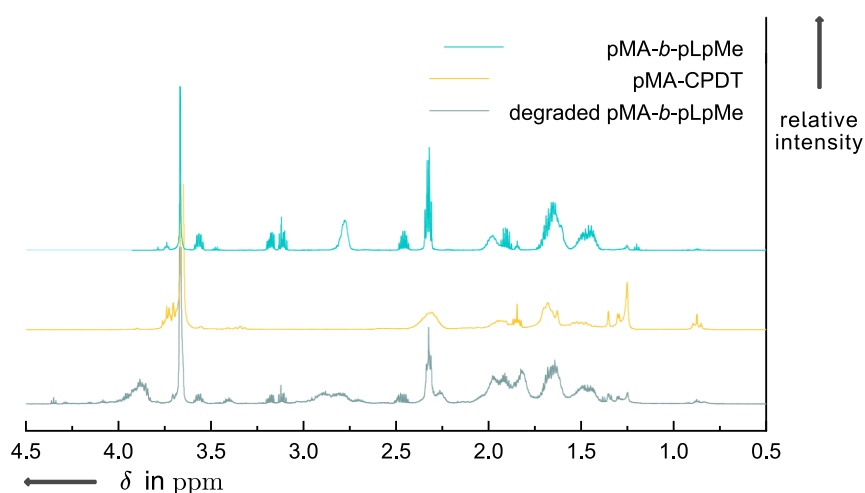


Figure 38 – ^1H NMR (600 MHz, CDCl_3) spectrum of degraded **pMA-b-pLpMe** along with spectra of non-degraded block copolymer and RAFT-derived macroinitiator.

From the **NMR** spectrum in Figure 38 the degradation of the block copolymer is not readily apparent. Although the signals found in the **NMR** spectrum of the block copolymer undergo peak broadening and position shifts, this spectrum does only slightly resemble the **NMR** spectrum of the macroinitiator. However, because of the overlap of most of the signals of the initiator and the polyipoate block or rather the signals of lipoate oligomers, it is difficult to definitively rule out successful degradation. This observation alongside the apparent degradation from the **GPC** results indicate that degradation does occur, which is, however, not complete, with possibly still some methyl lipoate units attached to the initiator and some methyl lipoate oligomers left in the sample, which could not be separated from the initiator by precipitation in hexane and water.

3.4.3 Aminolysis without Oxidation to Disulfides.

Lastly, in an effort to improve the aminolysis step of the polymerization and to purify aminolyzed macroinitiator with a thiol end-group without them oxidizing to disulfides, **RAFT**-derived polystyrene was aminolyzed and later acidified using trifluoroacetic acid. Since the oxidation of thiols to disulfides with oxygen is catalyzed in basic conditions^[106] such as during aminolysis, it was presumed that acidification of the solution after complete aminolysis under inert conditions would inhibit disulfide formation.

After complete aminolysis, which was determined by UV-Vis spectroscopy, acidification, and purification, the obtained polymer was analyzed using **GPC**. The UV-Vis spectra are shown in Figure 39 and the molecular-weight distribution is shown in Figure 40.

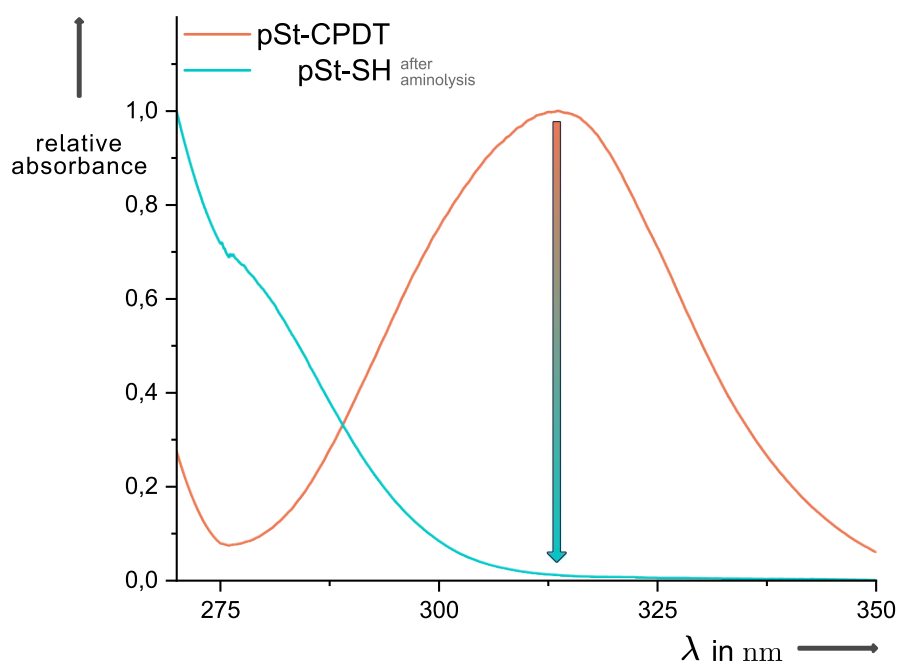


Figure 39 - UV-Vis spectra of **RAFT**-derived styrene-based macroinitiator (red) and aminolyzed macroinitiator (blue).

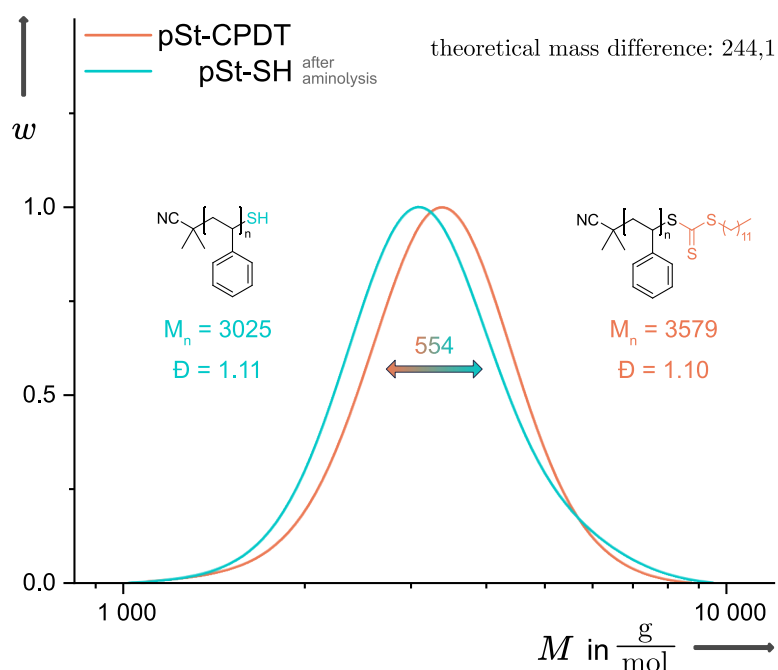


Figure 40 – Molecular-weight distributions of (non-)aminolyzed styrene-based macroinitiator with corresponding number-average molecular weights and dispersity values.

As seen in Figure 39 and Figure 40 the aminolysis was complete, which is indicated by the complete loss of absorption at 311 nm, which is the π - π^* absorption band of the **CTAs** thiocarbonyl group.^[107] Additionally, the apparent number-average molecular weight decreased after aminolysis by 554 $\text{g}\cdot\text{mol}^{-1}$. Most importantly, the molecular-weight distribution remains fully monomodal, hinting at the quantitative suppression of disulfide formation

Figure 41 shows the ^1H NMR spectra of the **RAFT**-derived polystyrene-based precursor and the aminolyzed macroinitiator.

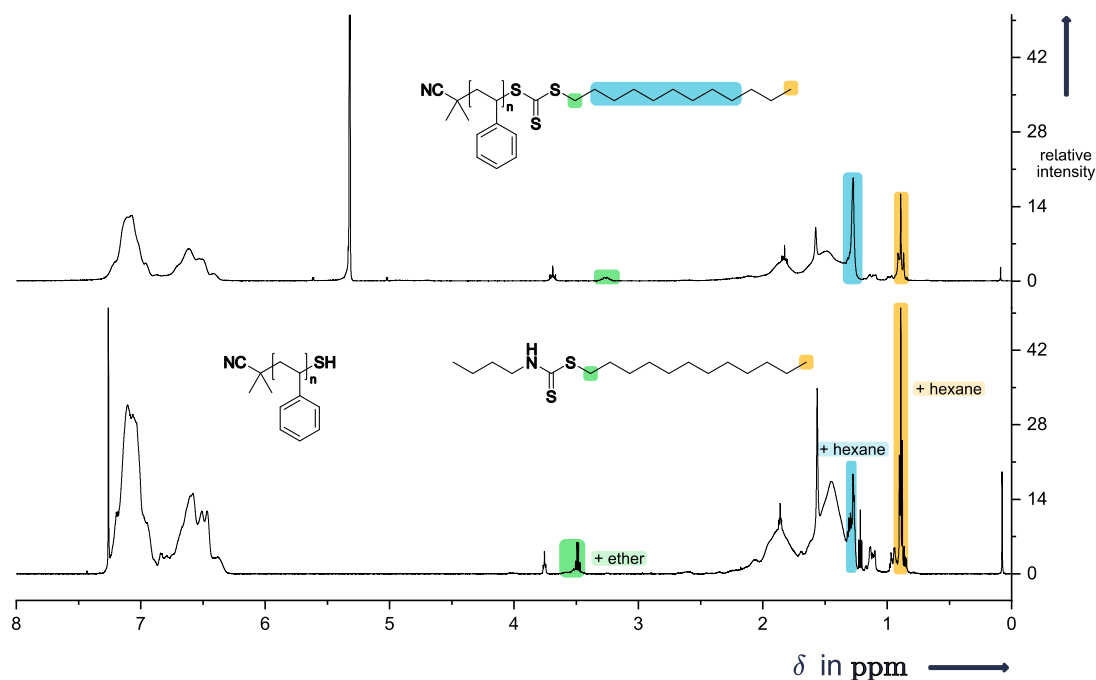


Figure 41 - ^1H NMR spectra of **pSt-CPDT** (upper spectrum, 300 MHz, CD_2Cl_2) and **pSt-SH** (lower spectrum, 600 MHz, CDCl_3) after aminolysis.

As seen in Figure 41 the NMR spectra appear to be very similar. In both spectra the signals for the dodecyl end-group of the **RAFT**-derived precursor are visible, albeit with a decrease in relative intensity. Judging from the results of UV-Vis spectroscopy and **GPC**, this may indicate that the thiocarbamate side product of the aminolysis is still present in the sample after the purification by precipitation in hexane. This could be remedied by increasing the number of precipitations or finding a different solvent, in which the side-product is more soluble in.

To confirm that indeed the thiol-terminated polymer was obtained, the polymer was analyzed using **MALDI-TOF-MS**. The signals in the mass spectra appear with a distance of 104 m/z corresponding to the expected mass of the repeating unit of styrene, which is 104,06 m/z. To analyze the exact structure these signals can be assigned to, end-group analysis was done using linear regression of the signals, which were presumed to be the silver adduct of the polymer species $[\text{M}+\text{Ag}]^+$, because the sample was prepared in dithranol with silver trifluoroacetate as ionization agent. The end-group analysis is shown in Figure 42.

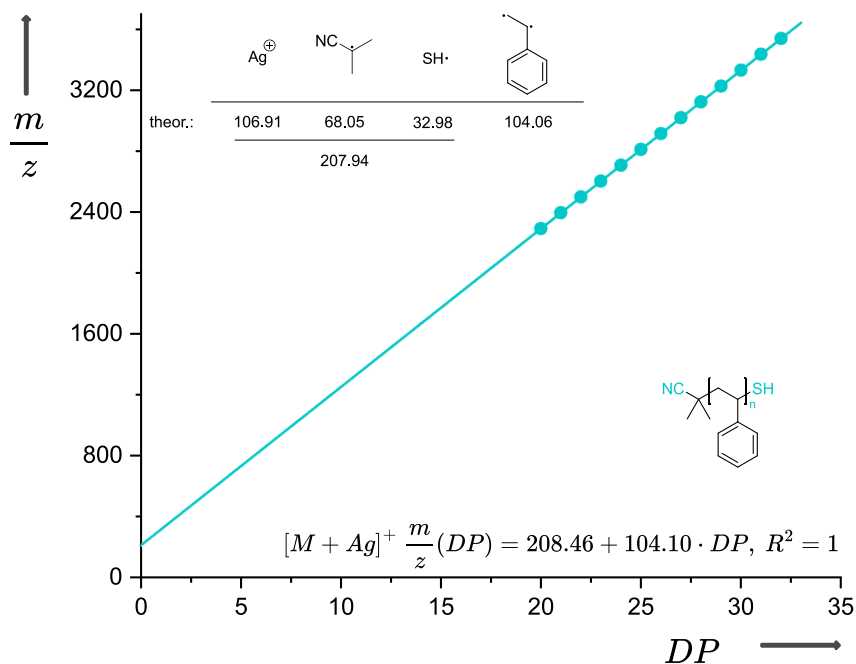


Figure 42 - End-group analysis using **MALDI-TOF-MS** of aminolyzed polystyrene using linear regression.

The intercept with the y-axis then aligns with the mass of the end-group derived from the **CTA** R-group of 68.05 m/z , the thiol end-group of 32.98 m/z , and a silver cation. To more securely confirm the obtention of thiol-terminated polystyrene, **MALDI-TOF-MS** in negative-ion-mode may show further evidence in the form of signals corresponding to the deprotonated, thiolate end-group polymer species.

Based on the observations from UV-Vis spectroscopy, **GPC**, and **MALDI-TOF-MS** it can be assumed that the free thiol polymers could successfully obtained by acidification of the reaction mixture after complete aminolysis.

4 Conclusion and Outlook

The goal of this thesis was to establish the initiation of the polymerization of lipoates by **RAFT**-derived macroinitiators. For that several strategies were investigated. Owing to the nucleophilicity of thiolate anions, it was reasoned that aminolyzed **RAFT** polymers could be deprotonated and then initiate **aROP** of lipoate derivatives by attacking the disulfide bond of the 1,2-dithiolane moiety. First the obtained **RAFT**-derived precursors to the macroinitiators were synthesized and aminolyzed. It was found that the aminolyzed macroinitiators readily oxidized during purification in the basic reaction conditions that are needed for the aminolysis, forming disulfide containing dimers and, thus, deactivating the macroinitiators. To remedy this a “one-pot” solution was devised, in which there would be no need for the purification of the aminolyzed macroinitiators and where the polymerization could be initiated directly, thus preventing the deactivation of the initiators. This resulted in the obtention of block copolymers, which were investigated using **GPC**, **NMR**, **MALDI-TOF-MS** and **DOSY-NMR**. It was found that block copolymers ranging from approximately 10 kg·mol⁻¹ to 40 kg·mol⁻¹ with dispersities of 1.5 to 1.8 and the desired end-groups could be synthesized. **DOSY-NMR** revealed that indeed block copolymers instead of homopolymers were formed. However, yields of the block copolymers were found to be quite limited, with the maximum yield of 41 % after three precipitations in methanol. Additionally, it could be shown by **GPC** measurements that a high-molecular-weight sample of a block copolymer based on poly(methyl acrylate) could be degraded by reduction with dithiothreitol to almost the same mass as the original macroinitiator. Lastly, it could be shown that thiol-terminated macroinitiators are accessible via aminolysis and consecutive acidification without the occurrence of oxidation to disulfide dimers.

Since this thesis only showed that the concept of polymerization of lipoates by initiation with macroinitiators is viable and that the synthesis could successfully be established, more work has to be done in characterization of the obtained polymers. Differential scanning calorimetry could be measured to obtain glass transition and other phase change temperatures, as well as thermogravimetric analysis to study thermal decomposition. In addition to that, quantitative **DOSY-NMR** could provide additional data, especially average molecular weights, complementing the **GPC** results.

In the future, several routes to further improve and build on the foundation of this thesis can be pursued. Because of the versatility of **RAFT** polymerization other monomers can easily be used to obtain different macroinitiators with other functional groups and therefore other properties, e.g., water solubility or stimuli-responsive behavior like thermoresponsivity. Also, for the polymerization of the second block the synthesis and properties of other monomers

that are either based on bio-based 1,2-dithiolanes like α -lipoic acid or asparagusic acid or synthetic monomers that contain a 1,2-dithiolane moiety can be explored. Because of the rather low ring strain of **ALA** derivatives investigation into molecules with higher ring strain through sterically demanding substituents, bridging, or different substitution patterns may enable higher conversions, yields, and higher molecular weights. This could also enable higher regioselectivity of the polymerization reaction. Another area of improvement of the polymerization is the high viscosity of the monomer and the reaction solution. For that, monomers that are either solids that are easily dissolved in the given solvent or lower viscosity liquids can be explored. Additionally, the change in substitution pattern of the monomer could alter the depolymerization properties, thus, activating or deactivating decomposition.

Lastly, possible applications of these kinds of partly decomposable block copolymers can be explored. Owing to the diblock copolymer structure and their potential for self-assembly, if the monomers of the different blocks are chosen correctly, the synthesis of nanoparticles by, for example, polymerization induced self-assembly (**PISA**), would enable the obtention of partly degradable nanoparticles that could be used for the encapsulation and release of drugs, DNA or RNA. Furthermore, these block copolymers could be used for triggerable surface coatings and adhesives, which could degrade under reductive conditions.

5 Experimental

5.1 Materials

The materials used, which, unless otherwise stated, were used without further processing, were purchased from the following manufacturers in the stated purities:

1-(methoxycarbonyl)ethyl benzodithioate (**MCEBD**, CAS 474746-06-0, Sigma-Aldrich, $\geq$ 97%), 1,5,7-triazabicyclo[4.4.0]dec-5-ene (**TBD**, CAS 5807-14-7, Sigma-Aldrich, 98%), 2-(dodecylthiocarbonothioylthio)-2-methylpropionic acid (**DDMAT**, CAS 461642-78-4, Sigma-Aldrich, 98% (HPLC)), 2,2'-azobis(2-methylpropionitrile) (**AIBN**, CAS 78-67-1), 2-cyano-2-propyl dodecyl trithiocarbonate (**CPDT**, CAS 870196-83-1, BLDpharm, 97 %), 2-iodopropane (CAS 75-30-9, Acros Organics, > 98%, stabilized), 4-cyano-4-[(dodecylthio)carbonothioyl]thio]pentanoic acid (**DCPT**, CAS 870196-80-8, BLDpharm, 0,95), 4-dimethylaminopyridine (**DMAPI**, CAS 1122-58-3, Sigma-Aldrich, $\geq$ 99%), acetone-d₆ (CAS 666-52-4, EurIsotop, 99.8% D), benzylmercaptane (**BM**, CAS 100-53-8, Alfa Aesar, 99%), chloroform-d (**CDCl₃**, CAS 865-49-6, Sigma-Aldrich, 99.8 atom % D), chloroform-d (**CDCl₃**, CAS 865-49-6, Deutero, 99.8%), dichloromethane (**DCM**, CAS 75-09-2, Sigma-Aldrich, puriss., p.a., > 99.9%), dichloromethane-d₂ (CAS 1665-00-5, Deutero, 99.6%), diethylether (CAS 60-29-7, Sigma Aldrich, Reag. Ph. Eur., contains BHT as inhibitor), dithiothreitol (**DTT**, CAS 3483-12-3, Carl Roth, p.a., $\geq$ 99%), DL- α -lipoic acid (**ALA**, CAS 1077-28-7, ABCR, 98%), DL- α -lipoic acid (**ALA**, CAS 1077-28-7, Fluorochem, 95%), ethanolamine (CAS 141-43-5, Sigma-Aldrich, > 99%), ethyl acetate (CAS 141-78-6, Merck, Reag. Ph. Eur.), iodomethane (CAS 74-88-4, Acros Organics, 99%, stabilized with copper), methanol (CAS 67-56-1, VWR, Reag. Ph. Eur.), Methyl acrylate (**MA**, CAS 96-33-3, Acros Organics, 99%, stabilized), methyl methacrylate (**MMA**, CAS 80-62-6, Merck, $\geq$ 99%, stabilized, distilled and stored in fridge), *N,N'*-dicyclohexylcarbodiimide (**DCC**, CAS 538-75-0, Sigma-Aldrich, $\geq$ 99%), *N,N*-diisopropylethylamine (**DIPEA**, CAS 7087-68-5, Carl Roth, $\geq$ 99%), *N,N*-dimethylacrylamide (**DMAA**, CAS 2680-03-7, Fluorochem, 98.0%, stabilized with MEHQ, distilled and stored in fridge), *N,N*-dimethylformamide (**DMF**, CAS 68-12-2, Carl Roth, $\geq$ 99.8%), *n*-butylamine (**BA**, CAS 109-73-9, Alfa Aesar, 99%), *n*-hexane (CAS 110-54-3, VWR, 97 %, for HPLC), styrene (St, CAS 100-42-5, VWR, $\geq$ 99.5%, stabilized, distilled and stored in fridge), tetrahydrofurane (**THF**, CAS 109-99-9, Biosolve, HPLC, unstabilized), tetrahydrofurane (**THF**, CAS 109-99-9, Sigma-Aldrich, Reag. Ph. Eur., contains 250 ppm BHT as inhibitor), toluene (CAS 108-88-3, VWR, Reag. Ph. Eur.), trifluoroacetic Acid (**TFA**, CAS 76-05-1, Acros Organics, 99.5%).

5.2 Methods

Gel Permeation Chromatography (GPC)

THF-GPC measurements were performed with a modular system by hs-GmbH. An S 3245 UV/Vis detector (Sykam) was equipped with one pre-column (AppliChrom StyDiViBe, 50 mm x 8 mm) and one column (polystyrene/divinylbenzene, 300 mm x 8 mm, 5 µm). A S5250 sample injector (Sykam) as auto sampler was utilized. The pump used was the Intelligent Pump AI-12 by Flow. Toluene was used for calibration and **THF** as solvent. The measurements were carried out with an injection volume of 100 µL with a flowrate of 1 mL/min. Samples were prepared with a concentration of 3.75 mg/mL and left for at least 12 h to dissolve and swell. The molecular weights were determined with the software Chromatographica (hs GmbH).

Nuclear Magnetic Resonance (NMR) Spectroscopy

¹H **NMR** spectroscopy measurements were performed on a Bruker Advance III – 300 for 300 MHz measurements, on a Bruker Advance III – 600 for 600 MHz and for Diffusion Ordered Spectroscopy (**DOSY**) measurements at room temperature. All **NMR** measurements were performed by the Center for Molecular and Structural Analytics of the Heinrich-Heine-University Düsseldorf (**CeMSA@HHU**). The spectra were evaluated using MestReNova software, version 14.2.0.

Matrix-assisted Laser Desorption/Ionization – Time of Flight – Mass Spectrometry (MALDI-TOF-MS)

MALDI-TOF-MS measurements were performed on a Bruker Daltonics MALDI-TOF/TOF UltrafleXtreme. Various matrices were used, which are specified for each spectrum. All mass spectrometry measurements were performed by the Center for Molecular and Structural Analytics of the Heinrich-Heine-University Düsseldorf (**CeMSA@HHU**). Origin 2024b was used for the evaluation.

Thin-Layer Chromatography (TLC)

TLC was performed on ready-made **TLC** plates ALUGRAM® Xtra SIL G/UV₂₅₄ from Macherey-Nagel.

UV-VIS-Spectroscopy

UV-VIS measurements were performed on an analytikjena Specord 210 plus. Origin 2024b was used for the evaluation. The spectra were smoothed using Savitzky-Golay smoothing with 50 points.

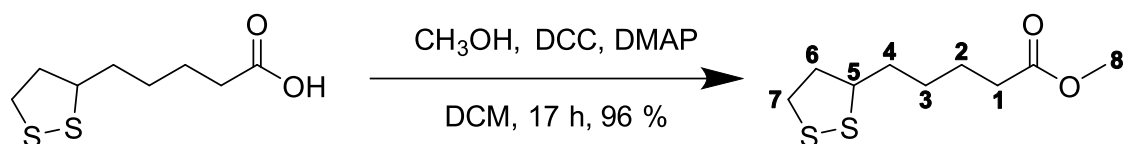
Preparation of stock solutions

Stock solutions are prepared to ensure dry and inert conditions for the polymerization of **LpMe**. For that, the desired reagent is added under inert gas flux to a Schlenk flask with activated molecular sieves (3 Å) that was evacuated, heated, and backfilled with an inert gas, i.e., nitrogen or argon, three times. The flask is then evacuated and backfilled again three times. The reagent is then dissolved in dry and degassed solvent. The stock solutions are stored in the freezer and warmed up to room temperature before use.

5.3 Syntheses

Methyl lipoate (LpMe)

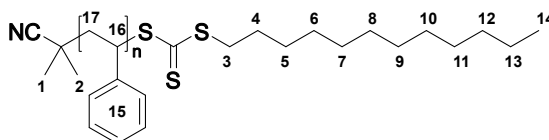
The Steglich esterification of α -lipoic acid was carried out according to Liu et al.^[73]



α -Lipoic acid (**ALA**, 10.0 g, 48.5 mmol, 1.0 eq.) is dissolved in dry dichloromethane (150 mL) and chilled using an ice bath. Then *N,N'*-dicyclohexylcarbodiimide (**DCC**, 12.0 g, 58.2 mmol, 1.2 eq.) and 4-dimethylaminopyridine (**DMAP**, 592.6 mg, 4.85 mmol, 0.1 eq.) are added. A white solid precipitates from solution. Methanol (13.0 mL, 321 mmol, 6.6 eq.) is then slowly added dropwise and the solution is left to warm to room temperature and stirred for 17 h. To ensure completion of the reaction the conversion is monitored using thin-layer chromatography (TLC) with a mixture of hexane and ethyl acetate in a volume ratio of 20:1 as the eluent ($R_F = 0.3$). The reaction suspension is then filtered and the filtrate is concentrated. The raw product is then purified using column chromatography. For that, the raw product is adsorbed onto 40 g of celite and then further dried. The column chromatography is carried out with a mixture of hexane and ethyl acetate with a volume ratio of 20:1 as the eluent. The product fractions are then combined, the solvent is evaporated, and the product is dried in vacuo. The product is obtained as a yellow oil.

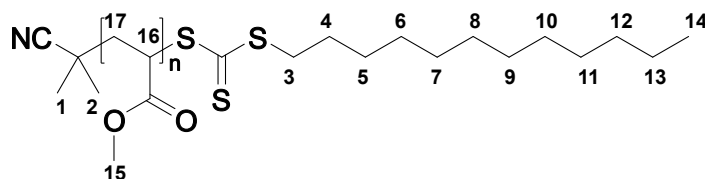
Yield	10.3 g (46.6 mmol), 96 %
¹H NMR	3.60 (s, 3H, 8), 3.58–3.43 (m, 1H, 5), 3.19–2.97 (m, 2H, 7), 2.48–2.33 (dtd, J = 13.0, 6.6, 5.4, 1H, 6), 2.26 (t, J = 7.4, 2H, 1), 1.84 (ddt, J = 12.8, 6.9, 6.9, 1H, 6'), 1.75–1.55 (m, 4H, 2,4), 1.51–1.26 (m, 2H, 3)
R_F	0.3 (hexane : ethyl acetate volume ratio 20:1)

Poly(styrene) macroinitiator – pSt-CPDT



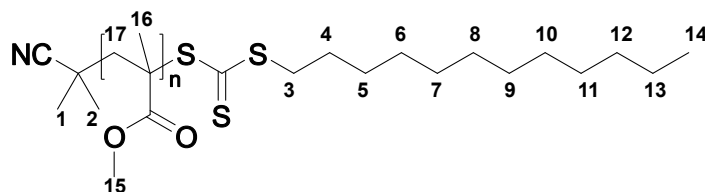
2-Cyanopropan-2-yl dodecyl trithiocarbonate (**CPDT**, 277 μ L, 0.801 mmol, 1.00 eq.) and azoisobutyronitril (**AIBN**, 13.1 mg, 0.080 mmol, 0.100 eq) are dissolved in styrene (2.8 mL, 24.4 mmol, 30.5 eq.) and degassed for 15 minutes by bubbling under argon flux. The reaction mixture is then heated to 70 °C for 15 hours until the solution gels and agitation is no longer possible. The obtained polymer is dissolved in **THF**, precipitated in 10-fold hexane, and centrifuged at 10,000 rcf and –10 °C three times. Lastly, the purified polymer is dried in vacuo to obtain a yellow powder.

Yield	761,0 mg, 27 %
Conversion	80%
¹H NMR	7.45–6.09 (m, 131H, 15), 3.74–3.64 (m, 2H, 3), 2.38–1.22 (m, 127H, 1,2,4-13,16,17 , THF), 0.89 (t, 11H, 14 , hexane)
M_n (GPC)	3885 (UV)
$\bar{D}$	1.09
MALDI-TOF-MS	theoretical m/z for n = 21: [M+H] ⁺ 2531.484 found: [M+Na] ⁺ 2554.541

Poly(methyl acrylate) macroinitiator – pMA-CPDT

2-Cyanopropan-2-yl dodecyl trithiocarbonate (**CPDT**, 277 μL , 0.801 mmol, 1.00 eq.) and azoisobutyronitril (**AIBN**, 13.1 mg, 0.080 mmol, 0.100 eq) are dissolved in methyl acrylate (2.2 mL, 24.4 mmol, 30.5 eq.) and degassed for 15 minutes by bubbling under argon flux. The reaction mixture is then heated to 70 °C for 60 minutes until the solution gels and agitation is no longer possible. The obtained polymer is dissolved in **THF**, precipitated in 10-fold hexane, and centrifuged at 10,000 rcf and –10 °C three times. Lastly, the purified polymer is dried in vacuo to obtain a yellow highly viscous liquid.

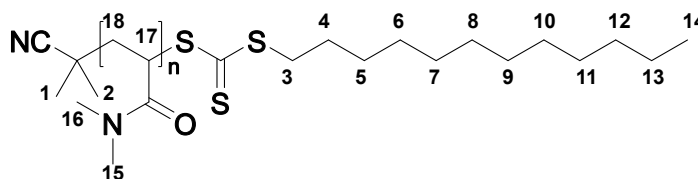
Yield	2291.3 mg, 96 %
Conversion	87%
^1H NMR	3,66 (s, 71H, $-\text{OCH}_3$ 15), 3,34 (t, 2H, 3), 2.47–2.22 (m, 25H, aliphatic pMA backbone, $-\text{CH}-$), 2.21–1.18 (m, 96H, aliphatic pMA backbone $-\text{CH}_2-$, 1,2,4-12 , THF) 0.88 (t, 5H, 14)
M_n (GPC)	3331 (UV)
$\bar{D}$	1.09
MALDI-TOF-MS	Theoretical m/z for $n = 34$: $[\text{M}+\text{H}]^+$ 3271.420 Found: $[\text{M}+\text{H}]^+$ 3272.848, $[\text{M}+\text{Na}]^+$ 3291.622, $[\text{M}+\text{K}]^+$ 3308.106

Poly(methyl methacrylate) macroinitiator – pMMA-CDPT

2-Cyanopropan-2-yl dodecyl trithiocarbonate (**CPDT**, 277 μL , 0.801 mmol, 1.00 eq.) and azoisobutyronitril (**AIBN**, 0.080 mmol, 0.100 eq) are dissolved in methyl methacrylate

(2.6 mL, 24.4 mmol, 30.5 eq.) and degassed for 15 minutes by bubbling under argon flux. The reaction mixture is then heated to 70 °C for 3.5 hours until the solution gels and agitation is no longer possible. The obtained polymer is dissolved in **THF**, precipitated in 10-fold hexane, and centrifuged at 10,000 rcf and –10 °C three times. Lastly, the purified polymer is dried in vacuo to obtain a yellow powder.

Yield	2418.8 mg, 90%
Conversion	69%
¹H NMR	3.59 (s, 73H, -OCH ₃ 15), 3.23 (m, 2H, 3), 2.13–0.72 (m, 298H, aliphatic pMMA backbone, 1,2,4-14,16,17 , THF)
M_n (GPC)	3914 (UV)
Đ	1.20
MALDI-TOF-MS	Theoretical <i>m/z</i> for n = 30: [M+H] ⁺ 3347.420, [M+Na] ⁺ 3369.724, [M+K] ⁺ 3385.698 Found: [M+H] ⁺ 3348.551, [M+Na] ⁺ 3365.555, [M+K] ⁺ 3391.142

Poly(*N,N*-dimethyl acrylamide) macroinitiator – pDMAA-CPDT

2-Cyanopropan-2-yl dodecyl trithiocarbonate (**CPDT**, 277 μL , 0.801 mmol, 1.00 eq.) and azoisobutyronitril (**AIBN**, 13.1 mg, 0.080 mmol, 0.100 eq) are dissolved in *N,N*-dimethylacrylamide (2.5 mL, 24.4 mmol, 30.5 eq.) and degassed for 15 minutes by bubbling under argon flux. The reaction mixture is then heated to 70 $^{\circ}\text{C}$ for 40 minutes until the solution gels and agitation is no longer possible. The obtained polymer is dissolved in **THF**, precipitated in 10-fold diethyl ether, and centrifuged at 10,000 rcf and -10°C three times. Lastly, the purified polymer is dried in vacuo to obtain a yellow powder.

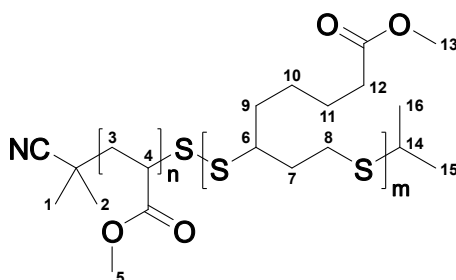
Yield 1547.1 mg, 57%

Conversion 84%

^1H NMR 3.33 (t, $J = 7.6$, 2H, $-\text{OCH}_3$ **3**),
 3.25–2.74 (m, 162H, **15**, **16**),
 2.75–2.17 (m, 31H, **17**),
 2.07 (s, 12H, *not assigned*),
 1.98–1.06 (m, 92H, **1**, **2**, **4-13**, **18**, THF, EtOEt),
 0.87 (t, $J = 6.5$, 3H, **14**)

MALDI-TOF-MS Theoretical m/z for $n = 30$: $[\text{M}+\text{H}]^+$ 2327.537, $[\text{M}+\text{Na}]^+$ 2349.519,
 $[\text{M}+\text{K}]^+$ 2365.493

Found: $[\text{M}+\text{H}]^+$ 2329.726, $[\text{M}+\text{Na}]^+$ 2349.734, $[\text{M}+\text{K}]^+$ 2365.688

Methyl acrylate-based Lipoate Block Copolymer – pMA-*b*-pLpMe

A 0.14 M solution of **pMA-CPDT** in dry **DMF** (0.45 mL, 0.062 mmol, 1.0 eq.) is added to an evacuated and dried Schlenk flask under nitrogen atmosphere. To that a 1.1 M solution of *n*-butylamine in dry **DMF** (0.07 mL, 0.075 mmol, 1.2 eq.) is added. The solution is left stirring for 5 h. The loss of the yellow color is observed. Then, methyl lipoate (**LpMe**, 0.61 mL, 3.1 mmol, 50 eq.) is added and stirred until a homogenous reaction mixture is obtained. The polymerization is then initiated by adding a 0.6 M solution of triazabicyclodecene (**TBD**) in dry **DMF** (0.13 mL, 0.078 mmol, 1.3 eq.) to the reaction mixture. The reaction is stirred for 90 minutes and then quenched by the addition of isopropyl iodide (0.07 mL, 0.70 mmol, 11 eq.). The polymer is then precipitated in methanol, centrifuged at 10.000 rcf at -10°C and redissolved in THF. This procedure is repeated three times. After that the polymer is finally precipitated from diethyl ether. The supernatant is decanted, and the polymer is dried in vacuo. The polymer is obtained as a clear, colorless viscous gel.

Yield 282.8 mg (32 %, after three precipitations from methanol)

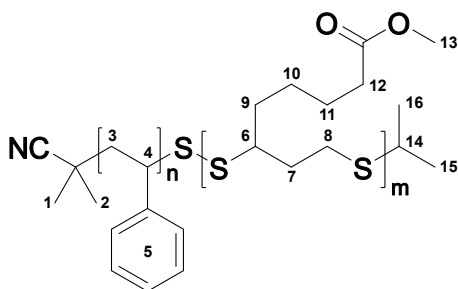
127.1 mg (14 %, after precipitation from diethyl ether)

^1H NMR 3.65 (s, 3H, **5**, **13**),
 2.87–2.70 (br s, 3H, **6**, **8**),
 2.32 (t, 2H, **12**),
 2.06–1.87 (br, 2H, **7**),
 1.74–1.55 (br, 5H, **9**, **11**),
 1.55–1.38 (br, 2H, **10**)

M_n (GPC) 10.2 kg·mol $^{-1}$

$\bar{D}$ 1.5

MALDI-TOF-MS repeating unit: 220.2 , theor.: 220.3 m/z
 theoretical m/z for $n = 21$, $m = 12$: $[\text{M}+\text{H}]^+$ 4592.5 m/z
 found: 4591.6 (peak)

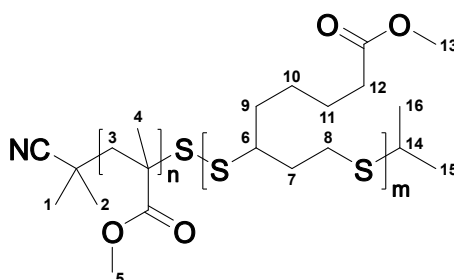
Styrene-based Lipoate Block Copolymer – pSt-*b*-pLpMe

A 0.14 M solution of **pSt-CPDT** (0.45 mL, 0.062 mmol, 1.0 eq.) in dry **DMF** is added to an evacuated and dried Schlenk flask under nitrogen atmosphere. To that a 1.1 M solution of *n*-butylamine (0.07 mL, 0.075 mmol, 1.2 eq.) in dry **DMF** is added. The solution is left stirring for 2 h. The loss of the yellow color is observed. Then, methyl lipoate (**LpMe**, 0.61 mL, 3.1 mmol, 50 eq.) is added and stirred until a homogenous reaction mixture is obtained. The polymerization is then initiated by adding a 0.6 M solution of triazabicyclodecene (**TBD**, 0.13 mL, 0.078 mmol, 1.3 eq.) in dry **DMF** to the reaction mixture. The reaction is stirred for 90 minutes and then quenched by the addition of isopropyl iodide (0.07 mL, 0.70 mmol, 11 eq.). The polymer is then precipitated in methanol, centrifuged at 10.000 rcf at $-10\text{ }^{\circ}\text{C}$ and redissolved in THF. This procedure is repeated three times. After that the polymer is finally precipitated from diethyl ether. The supernatant is decanted, and the polymer is dried in vacuo. The polymer is obtained as a clear, colorless viscous gel.

Yield	382.2 mg (42% overall, 65% with 52% LpMe conversion, after three precipitations from methanol)
	69.4 mg (7.6% overall, 12% with 52% LpMe conversion after precipitation from diethyl ether)
LpMe conversion (NMR)	52%

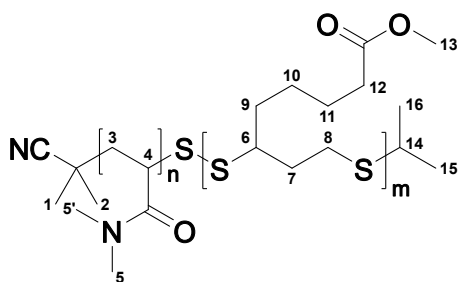
^1H NMR	7.26–6.20 (br, 2H, 5), 3.56 (s, 3H, 13), 2.80–2.66 (br s, 3H, 6 , 8), 2.24 (t, 2H, 12), 1.99–1.21 (br, 11H, 7 , 9 , 10 , 11)
M_n (GPC)	8.8 kg·mol ⁻¹
D	1.53
MALDI-TOF-MS	m/z repeating unit: 220.5, theor.: 220.3 theoretical m/z for $n = 28$, $m = 17$: $[\text{M}+\text{H}]^+$ 6798.8 found: 6803.2 (peak)

Methyl methacrylate-based Lipoate Block Copolymer – pMMA-*b*-pLpMe



A 0.14 M solution of **pMMA-CPDT** (0.45 mL, 0.062 mmol, 1.0 eq.) in dry **DMF** is added to an evacuated and dried Schlenk flask under nitrogen atmosphere. To that a 1.1 M solution of *n*-butylamine (0.07 mL, 0.075 mmol, 1.2 eq.) in dry **DMF** is added. The solution is left stirring for 1 h. The loss of the yellow color is observed. Then, methyl lipoate (**LpMe**, 0.61 mL, 3.1 mmol, 50 eq.) is added and stirred until a homogenous reaction mixture is obtained. The polymerization is then initiated by adding a 0.6 M solution of triazabicyclodecene (**TBD**, 0.13 mL, 0.078 mmol, 1.3 eq.) in dry **DMF** to the reaction mixture. The reaction is stirred for 90 minutes and then quenched by the addition of isopropyl iodide (0.07 mL, 0.70 mmol, 11 eq.). The polymer is then precipitated in methanol, centrifuged at 10.000 rcf at $-10\text{ }^{\circ}\text{C}$ and redissolved in THF. This procedure is repeated three times. After that the polymer is finally precipitated from diethyl ether. The supernatant is decanted, and the polymer is dried in vacuo. The polymer is obtained as a clear, colorless viscous gel.

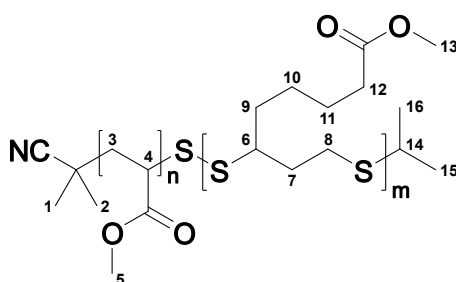
Yield	231.2 mg (25% overall, 39% with 52% LpMe conversion, after three precipitations from methanol) 169.9 mg (18%, 28% with 52% LpMe conversion, after precipitation from diethyl ether)
LpMe conversion	52%
¹H NMR	3.62 (s, 3H, 13), 3.55 (s, 1H, 5), 2.82–2.67 (br, 3H, 6, 8), 2.29 (t, 2H, 12), 2.01–1.84 (br, 2H, 7), 1.68–1.51 (br, 4H, 9, 11), 1.51–1.35 (br, 2H, 10)
M_n (GPC)	9976 kg·mol ⁻¹
Đ	1.6
MALDI-TOF-MS	<i>m/z</i> repeating unit: 220.5, theor.: 220.3 theoretical <i>m/z</i> for n = 16, m = 11: [M+H] ⁺ 4385.6 found: 4374.6 (peak)

Dimethyl acrylamide-based Lipoate Block Copolymer – pDMAA-*b*-pLpMe

A 0.14 M solution of **pMA-CPDT** (0.45 mL, 0.062 mmol, 1.0 eq.) in dry **DMF** is added to an evacuated and dried Schlenk flask under nitrogen atmosphere. To that of a 1.1 M solution *n*-butylamine (0.07 mL, 0.075 mmol, 1.2 eq.) in dry **DMF** is added. The solution is left stirring for 1 h. The loss of the yellow color is observed. Then, methyl lipoate (**LpMe**, 0.61 mL, 3.1 mmol, 50 eq.) is added and stirred until a homogenous reaction mixture is obtained. The polymerization is then initiated by adding a 0.6 M solution of triazabicyclodecene (**TBD**, 0.13 mL, 0.078 mmol, 1.3 eq.) in dry **DMF** to the reaction mixture. The reaction is stirred for 90 minutes and then quenched by the addition of isopropyl iodide (0.07 mL, 0.70 mmol, 11 eq.). The polymer is then precipitated in methanol, centrifuged at 10.000 rcf at $-10\text{ }^{\circ}\text{C}$ and redissolved in THF. This procedure is repeated three times. After that the polymer is finally precipitated from diethyl ether. The supernatant is decanted, and the polymer is dried in vacuo. The polymer is obtained as a clear, colorless viscous gel.

Yield	241.1 mg (26% overall, 46% with 53% LpMe conversion, after three precipitations from methanol) 118.2 mg (13%, 22% with 53% LpMe conversion, after precipitation from diethyl ether)
LpMe conversion	53%
¹H NMR	3.60 (s, 2H, 13), 2.99–2.79 (br, 1H, 5 , 5'), 2.79–2.64 (br, 3H, 6 , 8), 2.27 (t, 2H, 12), 2.03–1.82 (br, 2H, 7), 1.71–1.50 (br, 5H, 9 , 11), 1.50–1.32 (br, 2H, 10)
MALDI-TOF-MS	<i>m/z</i> repeating unit: 220.4, theor.: 220.3 theoretical <i>m/z</i> for <i>n</i> = 16, <i>m</i> = 13: [M+H] ⁺ 5481.6 found: 5474.9 (peak)

High-molecular-weight Methyl acrylate-based Lipoate Block Copolymer – pMA-*b*-pLpMe – HMW

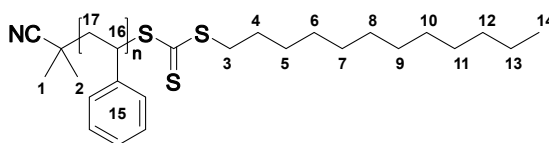


A 0.14 M solution of **pMA-CPDT** (0.45 mL, 0.062 mmol, 1.0 eq.) in dry **DMF** is added to an evacuated and dried Schlenk flask under nitrogen atmosphere. To that a 1.1 M solution of *n*-butylamine (0.08 mL, 0.085 mmol, 1.4 eq.) in dry **DMF** is added. The solution is left stirring for 3 h. The loss of the yellow color is observed. Then, methyl lipoate (**LpMe**, 2.4 mL, 12.3 mmol, 198.6 eq.) is added and stirred until a homogenous reaction mixture is obtained. The polymerization is then initiated by adding of a 0.6 M solution of triazabicyclodecene (**TBD**, 0.16 mL, 0.096 mmol, 1.5 eq.) in dry **DMF** to the reaction mixture. The reaction is stirred for 90 minutes and then quenched by the addition of isopropyl iodide (0.07 mL, 0.70 mmol, 11 eq.). Due to the high viscosity of the reaction

mixture the solution is agitated manually for 15 min. The polymer is then precipitated from methanol, centrifuged at 10.000 rcf at $-10\text{ }^{\circ}\text{C}$, and decanted three times. The polymer is dried in vacuo. The polymer is obtained as a clear, yellow gum.

Yield	913.8 mg (31% overall, 86% with 32% LpMe conversion, after three precipitations from methanol)
LpMe conversion	32%
^1H NMR	3.66 (s, 5H, 5 , 13 , 24), 3.56 (dq, $J = 8.8, 6.5$, 1H, 19), 3.18 (ddd, $J = 11.0, 7.1, 5.3$, 1H, 17), 3.11 (dt, $J = 11.0, 6.9$, 1H, 17), 2.87–2.72 (br, 3H, 6 , 8), 2.46 (dtd, $J = 13.0, 6.6, 5.3$, 1H, 18), 2.33 (td, $J = 7.4, 5.8$, 4H, 12 , 23), 2.05–1.94 (br, 2H, 7), 1.94–1.86 (m, 1H, 18'), 1.76–1.35 (br, 12H, 9-11 , 20-22)
M_n (GPC)	41.0 $\text{kg}\cdot\text{mol}^{-1}$
$\bar{D}$	1.85

High-molecular-weight polystyrene – pSt-CPDT-HMW

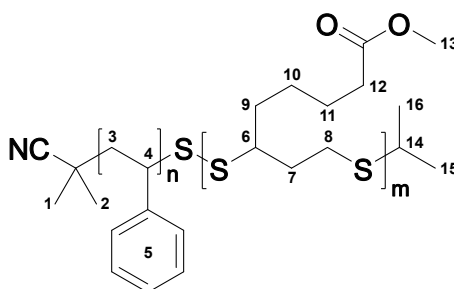


2-Cyanopropan-2-yl dodecyl trithiocarbonate (**CPDT**, 34.6 μL , 0.100 mmol, 1.00 eq.)) and azoisobutyronitril (**AIBN**, 1.6 mg, 0.010 mmol, 0.100 eq.) are dissolved in styrene (1.6 mL, 1.5 g, 14 mmol, 140 eq.)) and degassed for 15 minutes by bubbling under argon flux. The reaction mixture is then heated to $70\text{ }^{\circ}\text{C}$ for 24 hours until the solution gels and agitation is no longer possible. The obtained polymer is dissolved in **THF**, precipitated in 10-fold hexane and centrifuged at 10,000 rcf and $-10\text{ }^{\circ}\text{C}$ three times. Lastly, the purified polymer is dried in vacuo to obtain a yellow solid.

Yield	694,3 mg (45%)
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Conversion	52%
^1H NMR	7.16–6.19 (br, 242H, 5), 3.27–3.14 (br, 2H, 3), 2.26–1.11 (br, 176H, 4-13), 0.4–0.79 (br, 3H, 14)
M_n (GPC)	10588 (UV)
$\bar{D}$	1.09
MALDI-TOF-MS	theoretical m/z for $n = 73$: $[\text{M}+\text{H}]^+$ 7941.732 found: $[\text{M}+\text{H}]^+$ 7939.299

High-molecular-weight styrene-based block copolymer – pSt-*b*-pLpMe–DHMW



A 66 mM solution of **pSt-CPDT** (0.45 mL, 0.029 mmol, 1.0 eq.) in dry **DMF** is added to an evacuated and dried Schlenk flask under nitrogen atmosphere. To that a 1.1 M solution of *n*-butylamine (0.08 mL, 0.088 mmol, 3.0 eq.) in dry **DMF** is added. The solution is left stirring for 3 h. The loss of the yellow color is observed. Then, methyl lipoate (**LpMe**, 1.0 mL, 5.1 mmol, 177 eq., $[\text{M}]_0 = 3.0 \text{ M}$) is added and stirred until a homogenous reaction mixture is obtained. The polymerization is then initiated by adding of a 0.6 M solution of triazabicyclodecene (**TBD**, 0.16 mL, 0.096 mmol, 3.3 eq.) in dry **DMF** to the reaction mixture. The reaction is stirred for 90 minutes and then quenched by the addition of isopropyl iodide (0.07 mL, 0.70 mmol, 24 eq.). The polymer is then precipitated from methanol centrifuged at 10.000 rcf at -10°C three times. The supernatant is decanted, and the polymer is dried in vacuo. The polymer is obtained as a clear, yellow viscous gum.

Yield	592.5 mg (41%, after three precipitations from methanol)
LpMe conversion (NMR)	32%
¹H NMR	7.22–6.15 (br, 4H, 5), 3.56 (s, 8H, 13 , 24), 3.53–3.45 (m, 2H, 19), 3.14–2.98 (m, 4H, 17), 2.84–2.62 (br, 3H, 6 , 8), 2.43–2.34 (m, 2H, 18), 2.29–2.17 (m, 6H, 12 , 23), 1.99–1.86 (br, 2H, 7), 1.86–1.78 (m, 2H, 18'), 1.67–1.25 (br, 20H, 9-11 , 20-22)
M_n (GPC)	30.0 kg·mol ⁻¹
Đ	1.5

Degradation of pMA-*b*-pLpMe-HMW with DTT

pMA-*b*-pLpMe (615.1 mg, 0.015 mmol, 1.0 eq.) is dissolved in THF (3.085 mL, ([Polymer] = 4.9 μM). To that dithiothreitol (300 mg, 1.94 mol, 130 eq.) and a drop of *N,N'*-diisopropylethylamine are added. The reaction is then stirred for 18 h, after which the now yellow reaction mixture is precipitated from 5-fold –20 °C hexane, centrifuged at 10,000 rcf at –10 °C and the solvent is decanted. The sample is then dissolved in THF and precipitated from 10-fold 4 °C water and centrifuged and decanted again. The obtained precipitate is then dried in vacuo.

Aminolysis of pSt-CPDT without the formation of disulfides

A 0.14 M solution of **pSt-CPDT** (0.45 mL, 0.062 mmol, 1.0 eq.) in dry, degassed **DMF** is added to an evacuated and dried Schlenk flask under nitrogen atmosphere. To that, a 1.1 M solution of *n*-butylamine (0.14 mL, 0.15 mmol, 2.4 eq.) in dry, degassed **DMF** is added. The solution is left stirring for 1 h. The loss of the yellow color is observed. The reaction mixture is then acidified with a few drops of trifluoroacetic acid (**TFA**), precipitated from 10-fold $-20\text{ }^{\circ}\text{C}$ methanol and centrifuged at 10,000 rcf at $-10\text{ }^{\circ}\text{C}$. The supernatant is then decanted and the polymer again dissolved in **THF**. This procedure is repeated three times after which the obtained polymer is dried in vacuo. The polymer is obtained as a colorless solid.

^1H NMR	7.24–6.25 (br, 77.3H, 5), 2.71–1.25 (br, 60H, 1-4)
M_n (GPC)	3.0 kg·mol $^{-1}$
$\bar{D}$	1.1
MALDI-TOF-MS	m/z repeating unit: found: 104.0, theor.: 104.1 theoretical m/z for $n = 24$: $[\text{M}+\text{Ag}]^+$ 2706.7 found: $[\text{M}+\text{Ag}]^+$ 2705.4

6 Appendix

6.1 RAFT polymerization of Methyl Acrylate with DDMAT

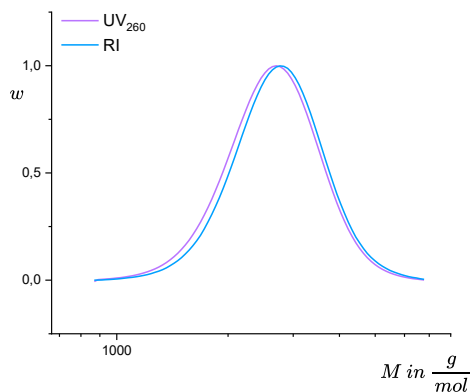


Figure 43 - GPC (THF) trace and molecular-weight distribution of **pMA-DDMAT** (UV purple, RI blue).

M_n	2518 (UV), 2614 (RI)
M_w	2752 (UV), 2844 (RI)
$\bar{D}$	1.09

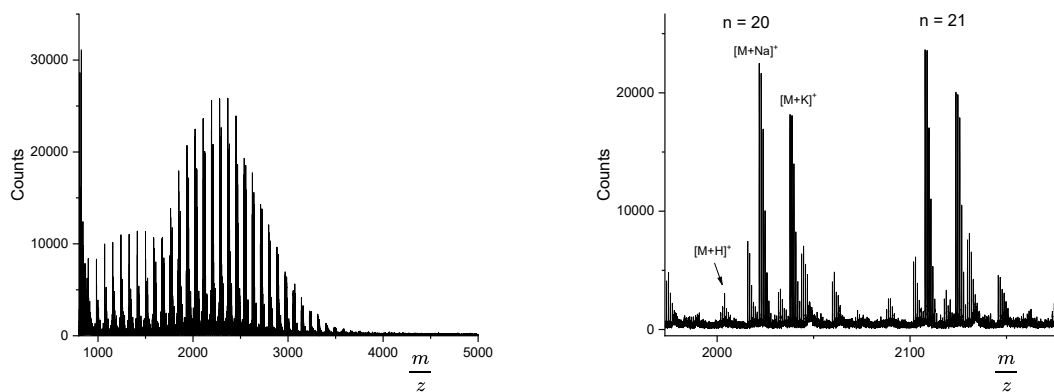


Figure 44 - MALDI-TOF-MS spectrum (HABA 2:1, reflectron) of **pMA-DDMAT** with an enlarged section for assignment of the signals.

MALDI-TOF-MS	m/z repeating unit: 86.0, theor.: 86.0
	theoretical m/z for $n = 19$: $[M+K]^+$ 2037.8
	found: $[M+K]^+$ 2037.8

6.2 RAFT Polymerizations of Styrene, Methyl Acrylate, Methyl Methacrylate, and Dimethylacrylamide with CPDT as the CTA as Macroinitiators for aROP

CPDT

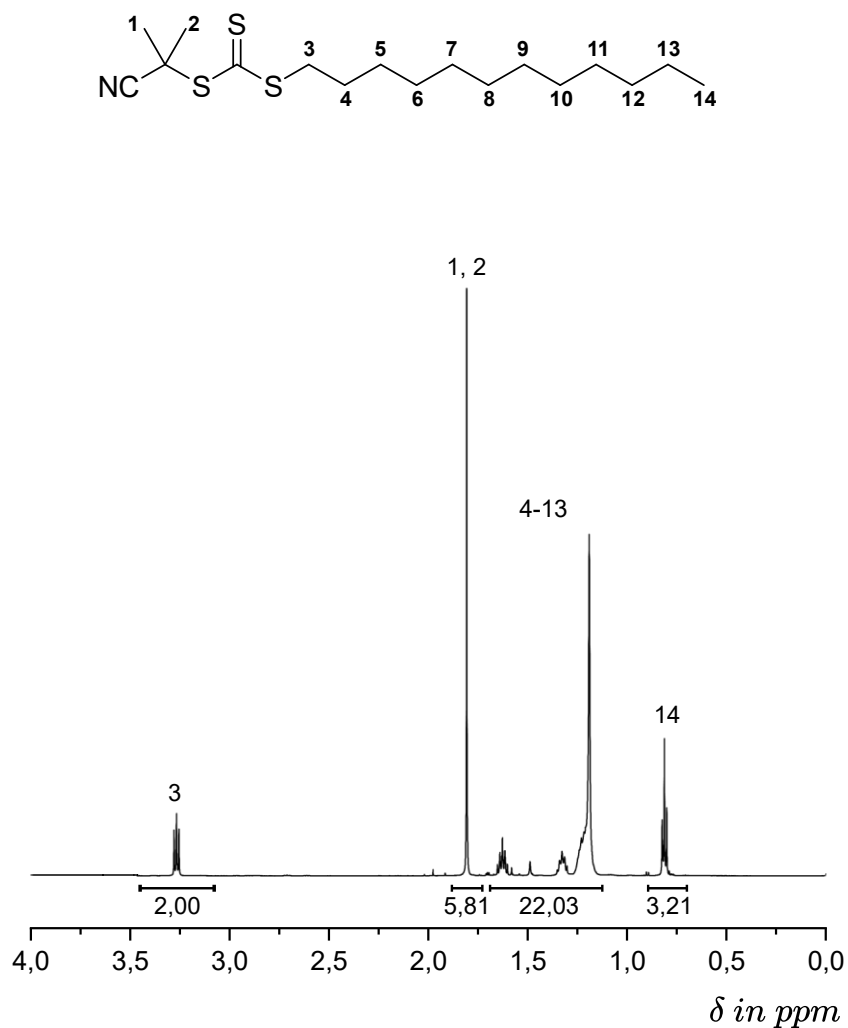
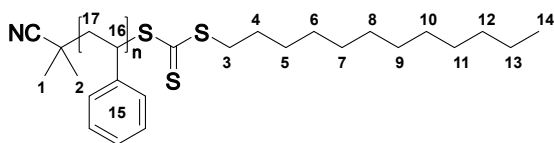


Figure 45 - ^1H NMR (600 MHz, CDCl_3) spectrum of CPDT.

pSt-CPDT



Yield 761.0 mg, 33%

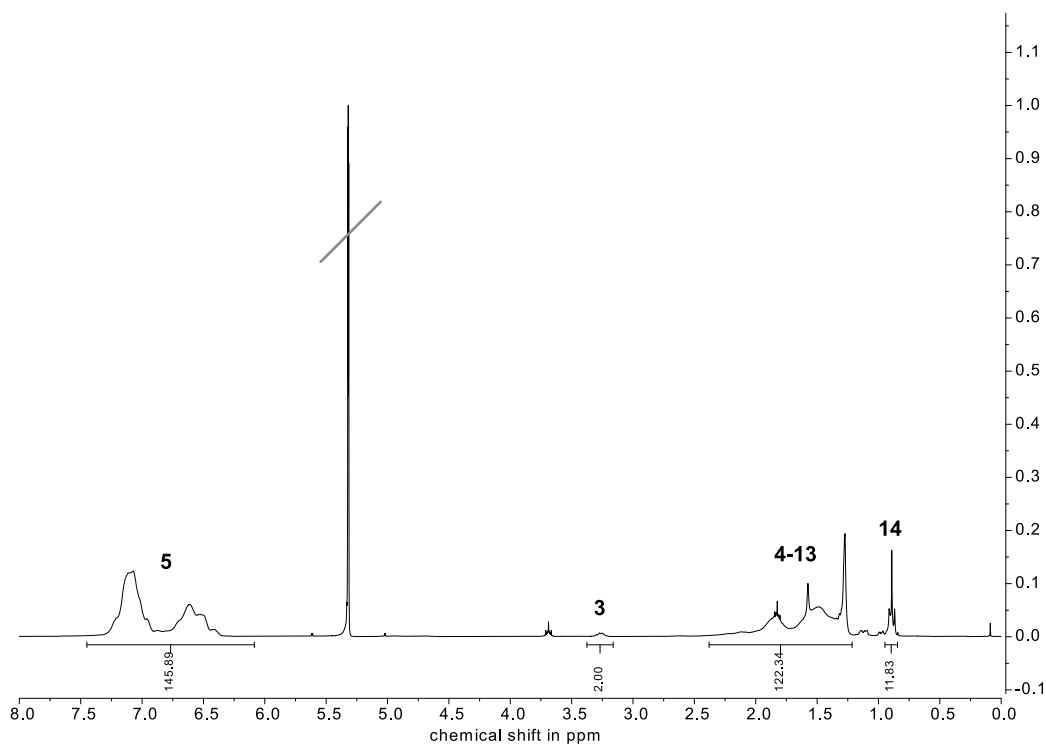


Figure 46 - ^1H NMR (300 MHz; CD_2Cl_2) spectrum of pSt-CPDT.

^1H NMR 7,45–6,09 (m, 131H, **15**),
 3,74–3,64 (m, 2H, **3**),
 2,38–1,22 (m, 127H, **4-13**, THF),
 0,89 (m, 11H, **14**, hexane)

DP_n (NMR) $\frac{131}{5} \approx 26$

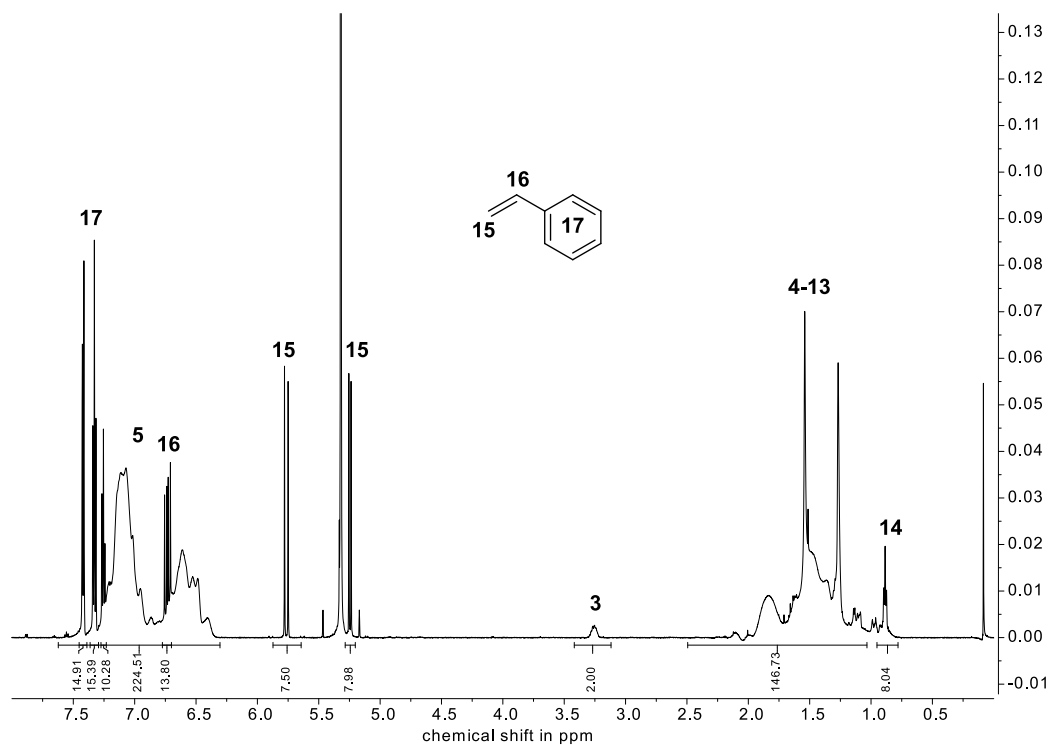


Figure 47 - ^1H NMR (600 MHz; CD_2Cl_2) spectrum of non-purified RAFT polymerization mixture of styrene.

Conversion $\frac{(25,92-6)/5}{(25,92-6)/5+1} = 0.80$ (all signal divided by 7.50)

Time 15 h

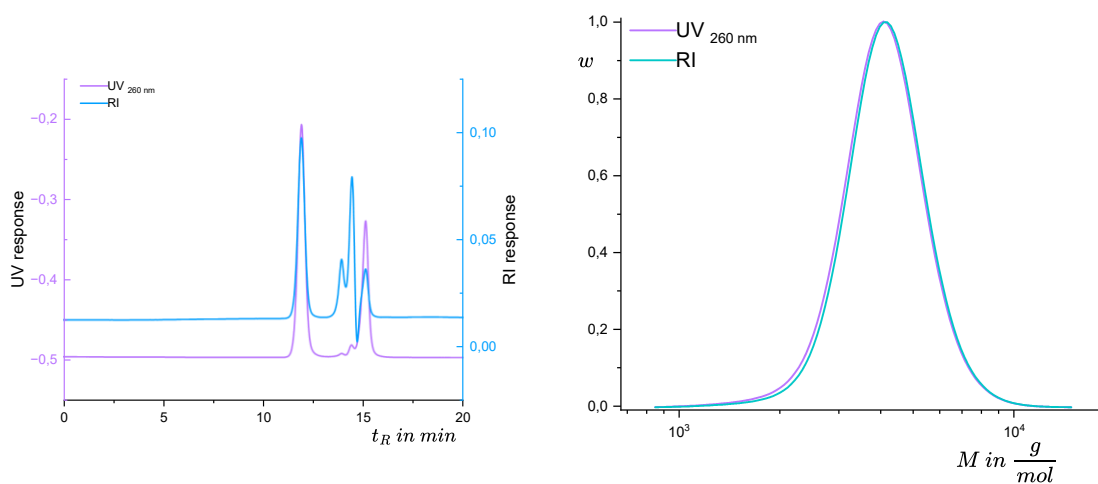


Figure 48 - GPC (THF) trace and molecular-weight distribution of **pSt-CPDT** (UV purple, RI blue).

t_R	11.9 min
M_n	3885 (UV), 3976 (RI)
M_w	4243 (UV), 4319 (RI)
$\bar{D}$	1.09

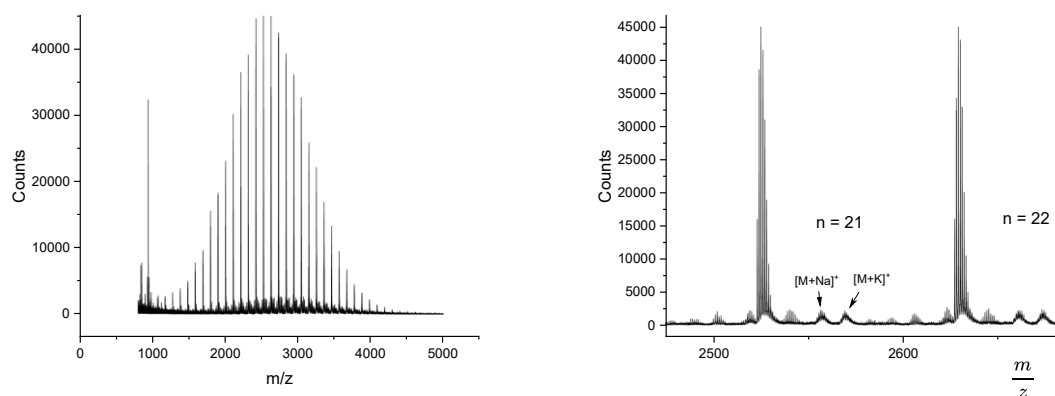
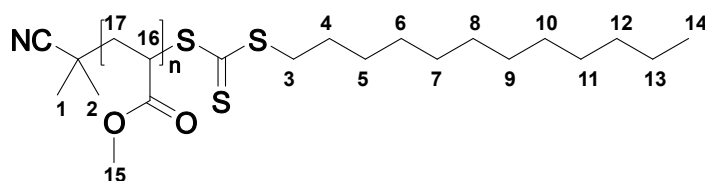


Figure 49 - **MALDI-TOF-MS** spectrum (DCBT 2:1) of **pSt-CPDT** with an enlarged section for assignment of the signals.

MALDI-TOF-MS m/z repeating unit: 104.0 , theor.: 104.1
theoretical m/z for $n = 21$: $[M+K]^+$ 2569.4
found: $[M+K]^+$ 2569.2

pMA-CPDT



Yield 2291.3 mg, 96%

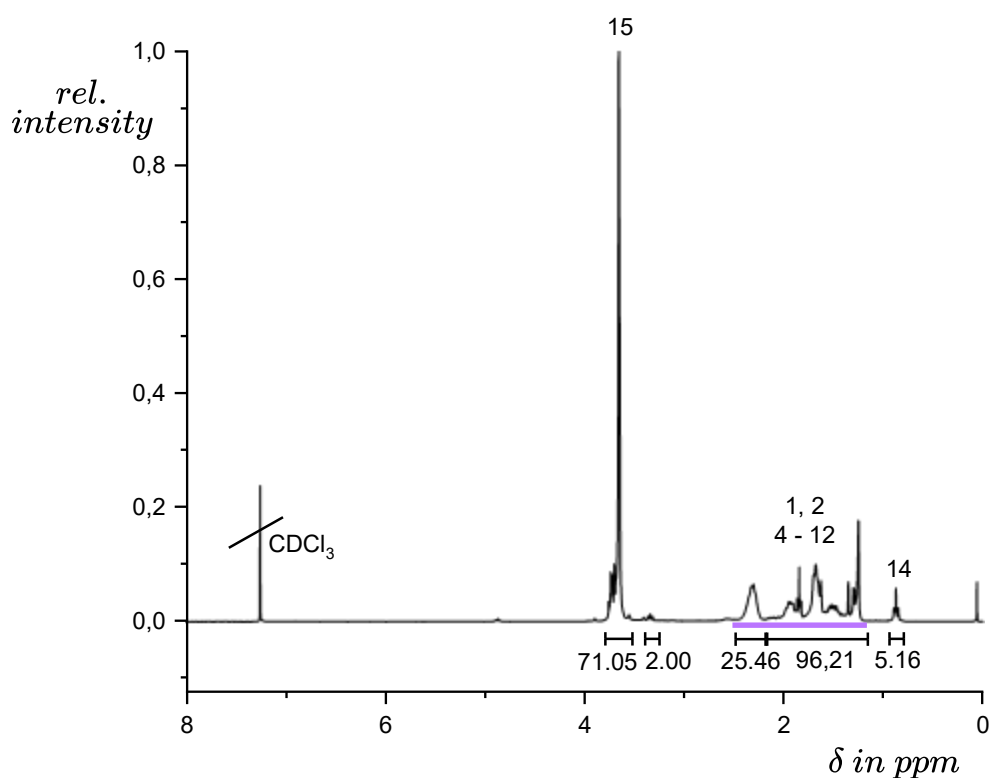


Figure 50 - ^1H NMR (300 MHz, CDCl_3) spectrum of pMA-CPDT.

^1H NMR 3,66 (s, 71H, $-\text{OCH}_3$ **15**),
 3,34 (t, 2H, **3**),
 2.47–2.22 (m, 25H, aliphatic pMA backbone, $-\text{CH}-$),
 2.21–1.18 (m, 96H, aliphatic pMA backbone $-\text{CH}_2-$, **1,2,4-12**, THF)
 0.88 (t, 5H, **14**)

DP_n (NMR) ~ 25 (from aliphatic backbone, $-\text{CH}-$)

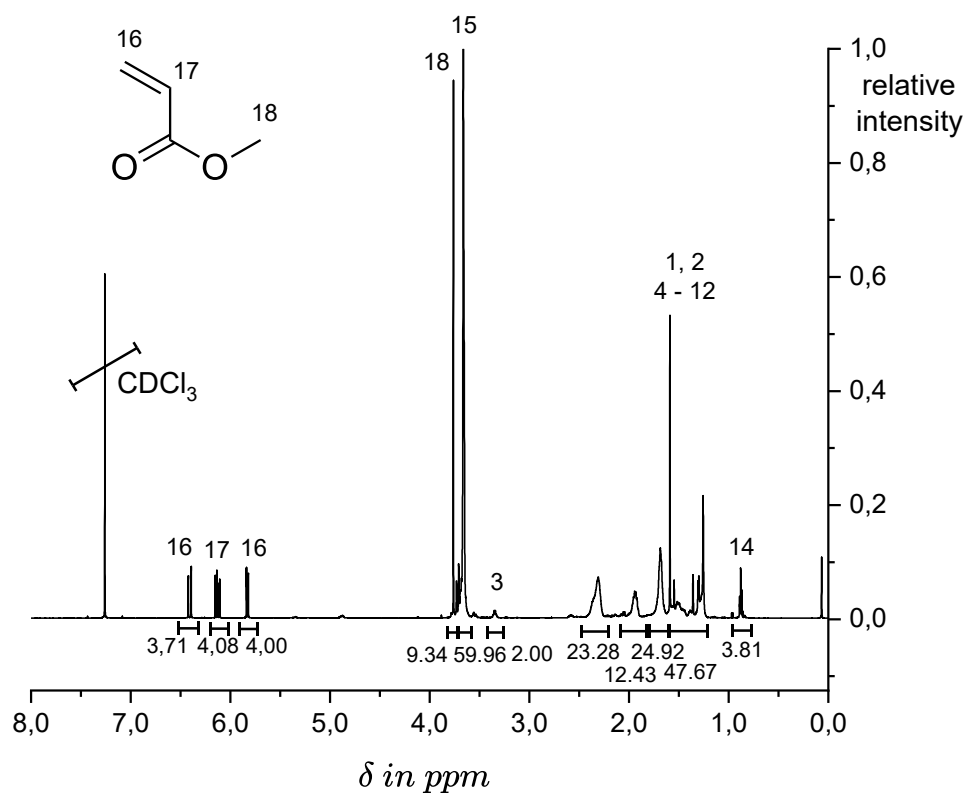


Figure 51 - ^1H NMR (600 MHz, CDCl_3) spectrum of non-purified **RAFT** polymerization mixture of methyl acrylate.

Conversion $\frac{59.96}{59.96 + 9.34} = 0.87$

Time 55 min

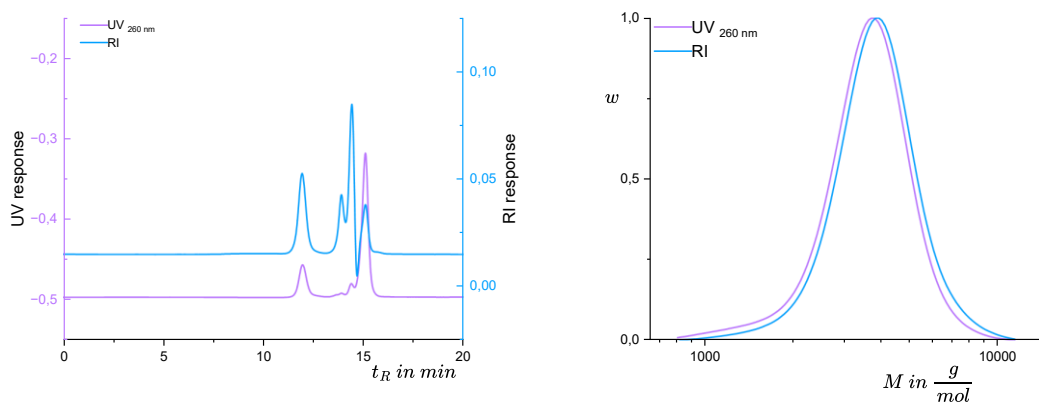


Figure 52 - **GPC** (THF) trace and molecular-weight distribution of **pMA-CPDT** (UV purple, RI blue).

t_R	11.9 min
M_n	3331 (UV), 3588 (RI)
M_w	3794 (UV), 4051 (RI)
$\bar{D}$	1.14

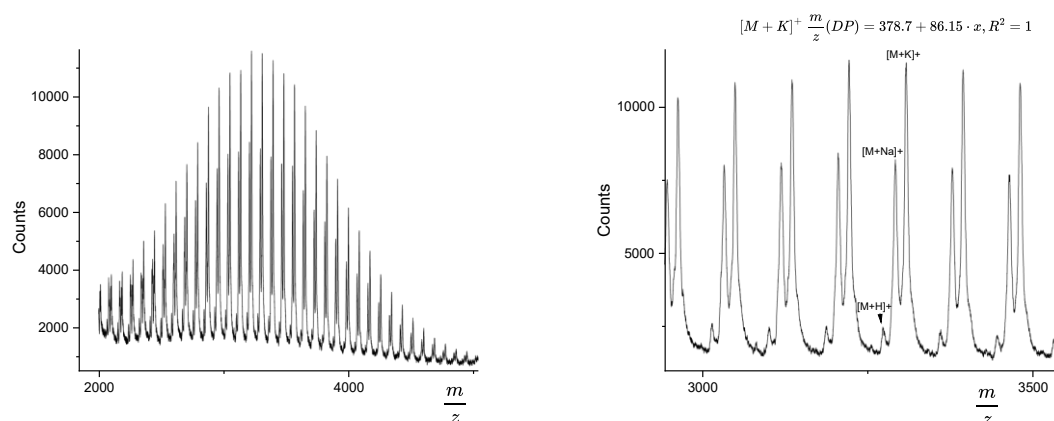


Figure 53 - **MALDI-TOF-MS** spectrum (HABA 1:2) of **pMA-CPDT** with an enlarged section for assignment of the signals.

MALDI-TOF-MS m/z repeating unit: 86.1 , theor.: 86.0
theoretical m/z for $n = 34$: $[M+K]^+$ 3309.4
found: $[M+K]^+$ 3308.1

pMMA-CPDT

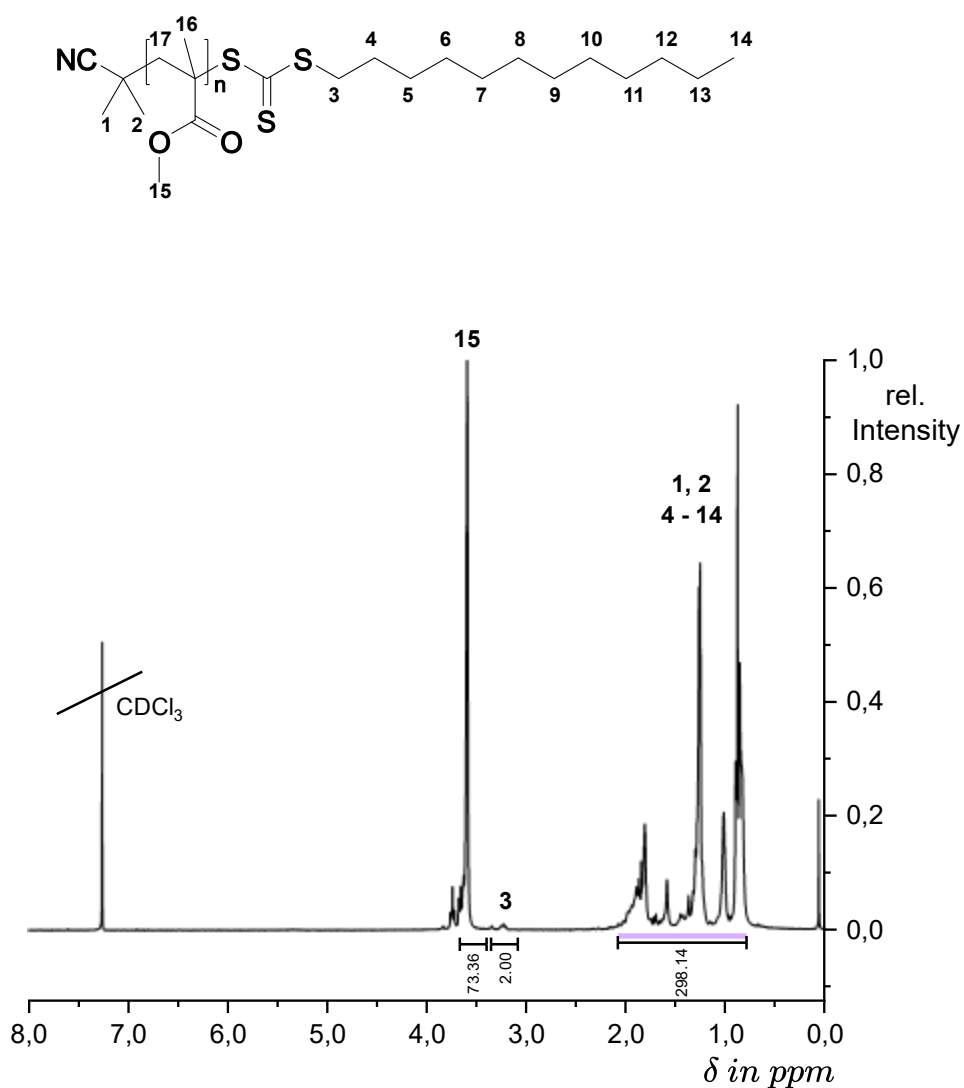


Figure 54 - ^1H NMR (300 MHz, CDCl_3) spectrum of pMMA-CPDT.

^1H NMR	3.59 (s, 73H, $-\text{OCH}_3$ 15), 3.23 (m, 2H, 3), 2.13–0.72 (m, 298H, aliphatic pMMA backbone, 1,2,4-14 , THF)
DP_n (NMR)	$\frac{73,36}{3} = 24$

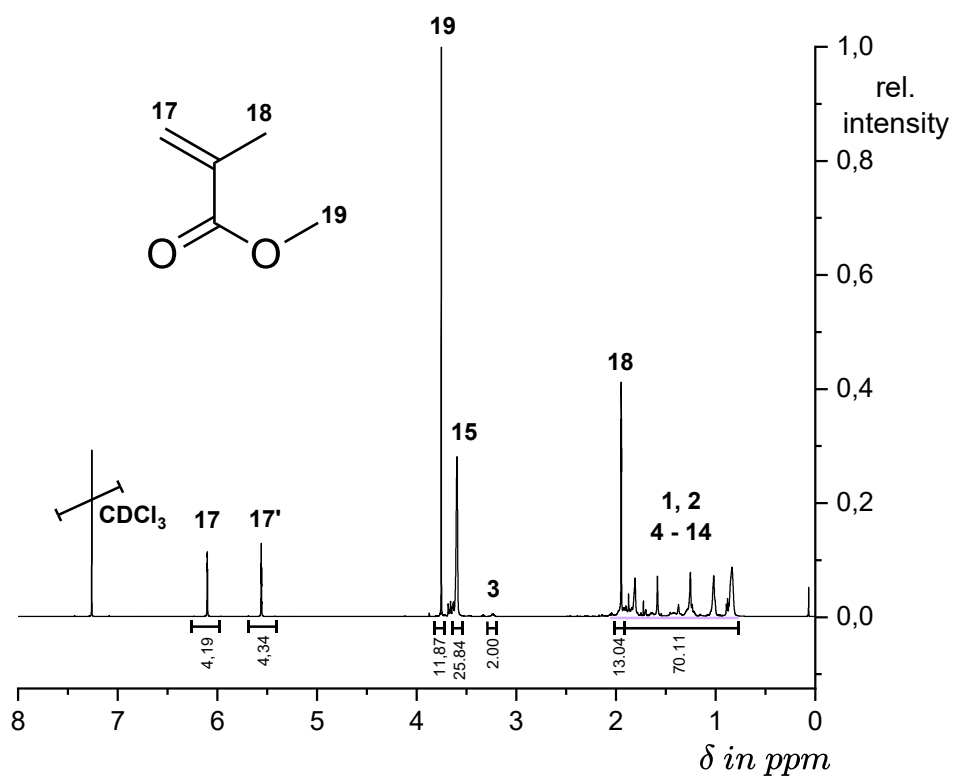


Figure 55 - ^1H NMR (600 MHz, CDCl_3) spectrum of non-purified **RAFT** polymerization mixture of methyl methacrylate.

Conversion $\frac{25.84}{25.84 + 11.87} = 0,69$
Time 3.5 h

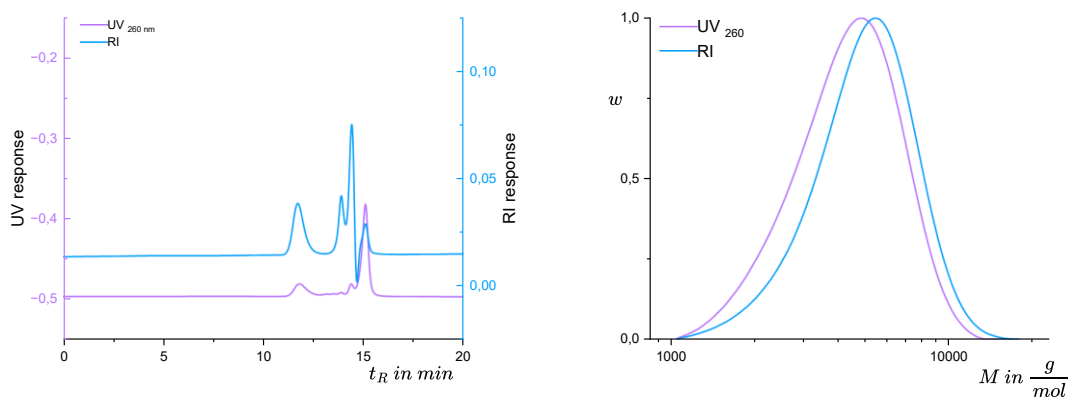


Figure 56 - GPC (THF) trace and molecular-weight distribution of **pMMA-CPDT** (UV purple, RI blue).

t_R	11.7 min
M_n	3914 (UV), 4510 (RI)
M_w	4711 (UV), 5381 (RI)
$\bar{D}$	1.20

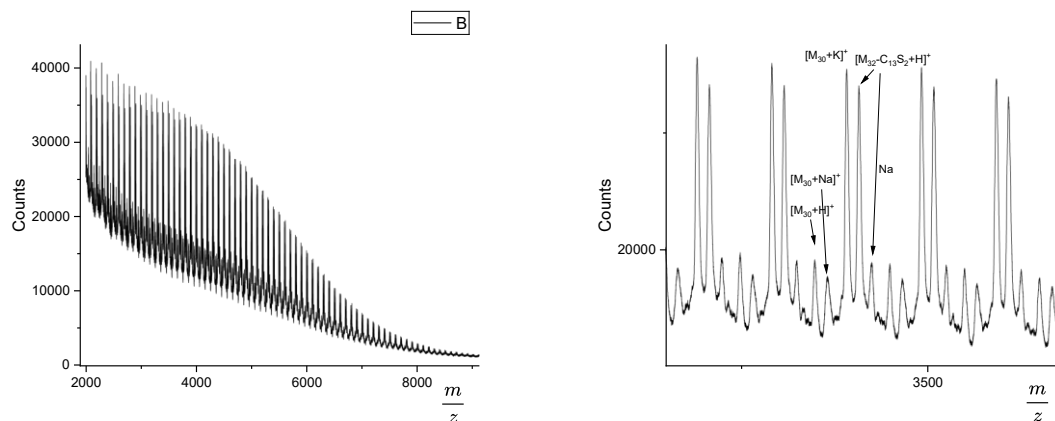


Figure 57 - MALDI-TOF-MS spectrum (HABA 1:2) of **pMMA-CPDT** with an enlarged section for assignment of the signals.

MALDI-TOF-MS m/z repeating unit: 100.2 , theor.: 100.1
theoretical m/z for $n = 30$: $[M+K]^+$ 3391.1
found: $[M+K]^+$ 3385.7

pDMAA-CPDT

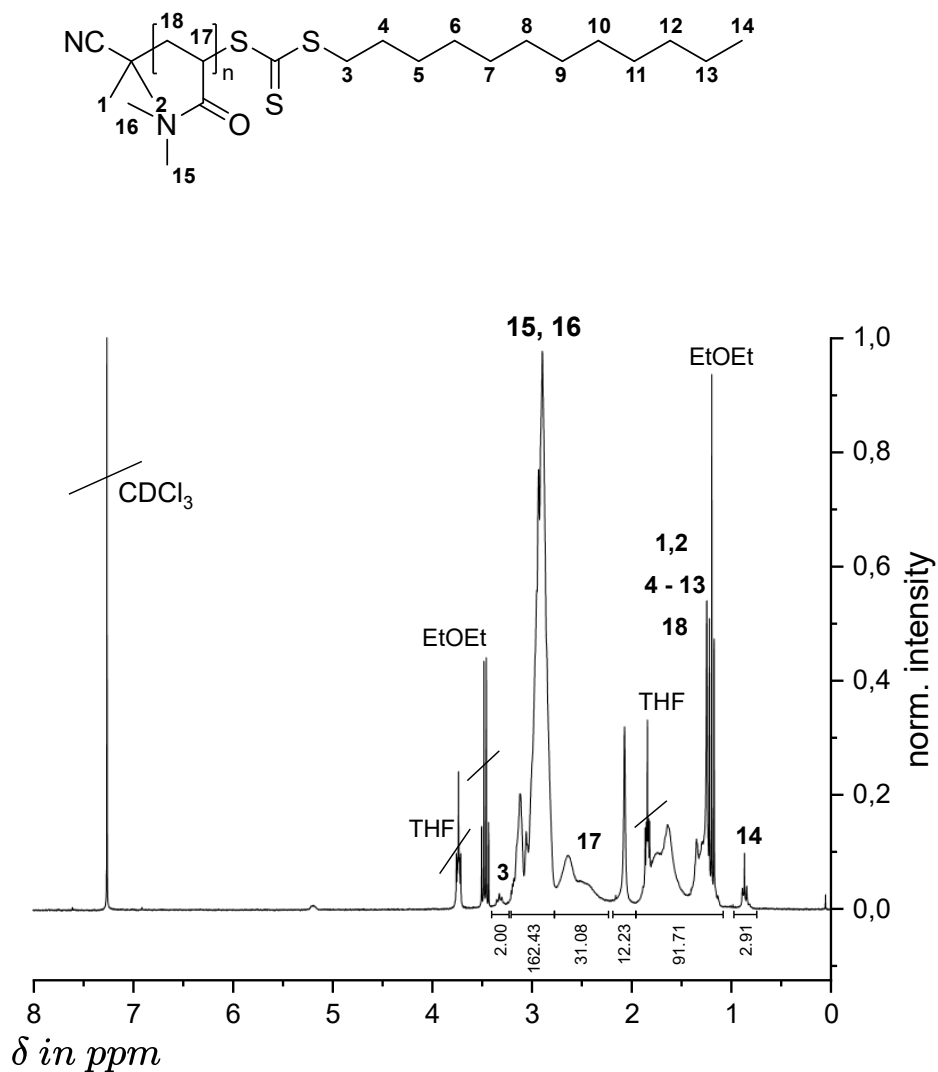


Figure 58 - ^1H NMR (300 MHz; CDCl_3) spectrum of pDMAA-CPDT.

^1H NMR 3.33 (t, $J = 7.6$, 2H, $-\text{OCH}_3$ **3**),
 3.25–2.74 (m, 162H, **15**, **16**),
 2.75–2.17 (m, 31H, **17**),
 2.07 (s, 12H, *not assigned*),
 1.98–1.06 (m, 92H, **1**, **2**, **4–13**, **18**, THF, EtOEt),
 0.87 (t, $J = 6.5$, 3H, **14**)

DP_n (NMR) $\frac{162,43}{6} = 27$

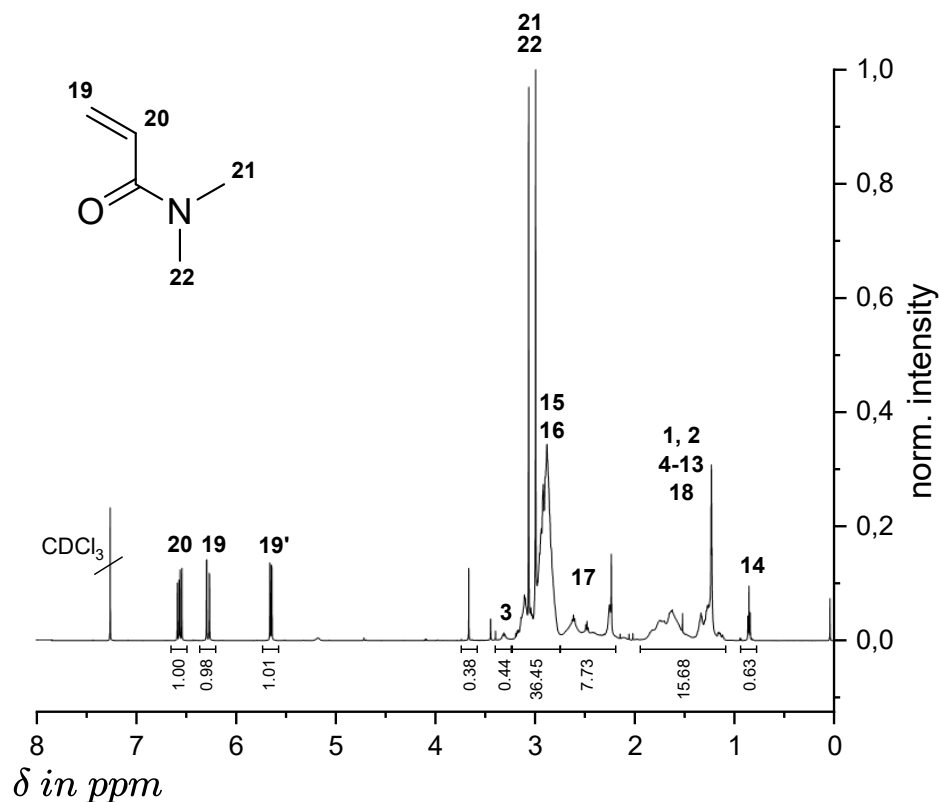


Figure 59 - ^1H NMR (600 MHz, CDCl_3) spectrum of non-purified RAFT polymerization of *N,N*-dimethylacrylamide.

^1H NMR	6.56 (dd, $J = 16.8, 10.5$, 1H, 20),
	6.28 (dd, $J = 16.8, 2.0$, 1H, 19),
	5.65 (dd, $J = 10.5, 2.0$, 1H, 19'),
	3.67 (s, <i>not assigned</i>),
	3.36–3.26 (m, 3),
	3.24–2.75 (m, 36H, 15, 16, 21, 22),
	2.75–2.19 (m, 8H, 17),
	1.97–1.09 (m, 16H, 1, 2, 4-13, 18),
	0.86 (t, $J = 7.0$, 0.6, 14)

Conversion $\frac{36.45 - 6}{36.45} = 0.84$

Time 40 min

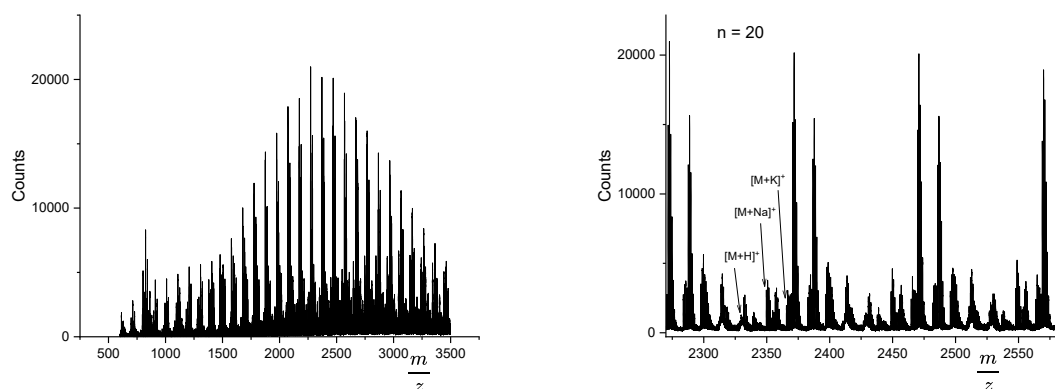


Figure 60 - **MALDI-TOF-MS** spectrum (HABA 1:2) of **pDMAA-CPDT** with an enlarged section for assignment of the signals.

MALDI-TOF-MS m/z repeating unit: 99.1 , theor.: 99.1
theoretical m/z for $n = 20$: $[M+K]^+$ 2365.7 m/z
found: $[M+K]^+$ 2365.5

pSt-CPDT-HMW

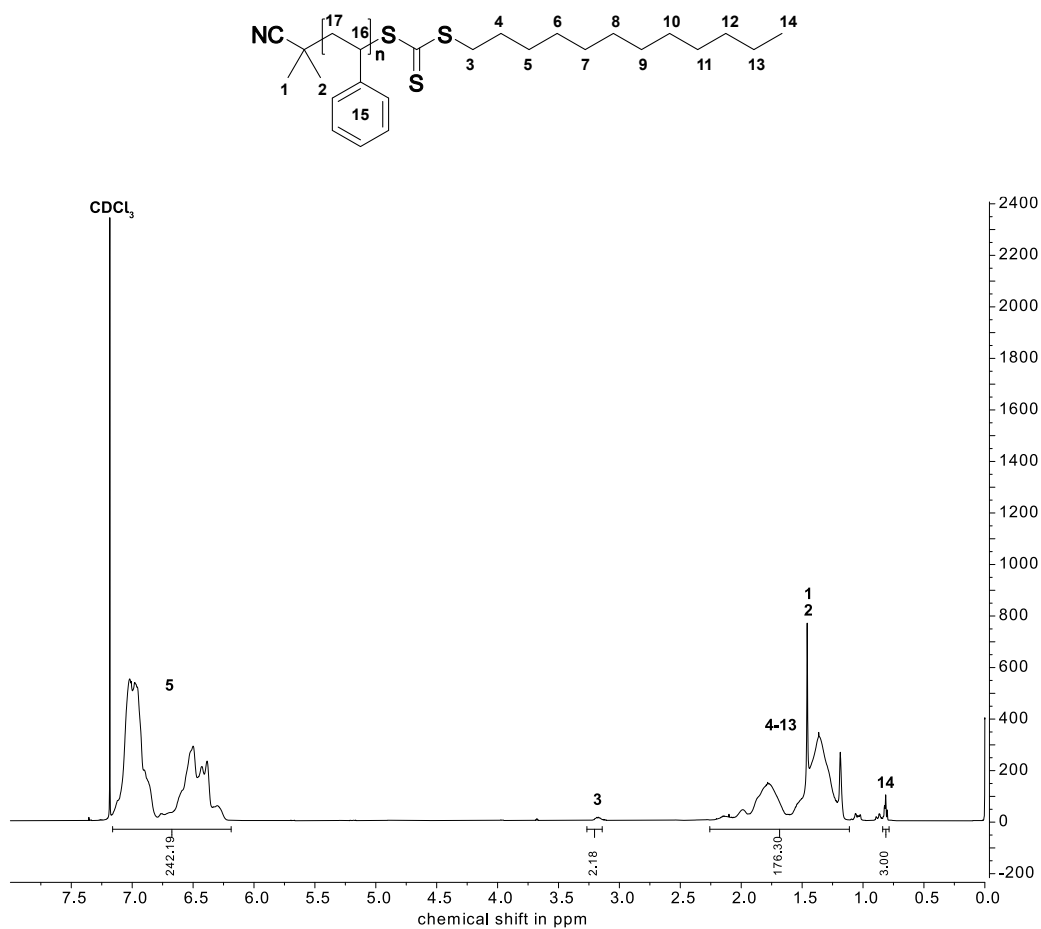


Figure 61 - ^1H NMR (600 MHz; CDCl_3) spectrum of pSt-CPDT-HMW.

^1H NMR 7.16–6.19 (br, 242H, **5**),
 3.27–3.14 (br, 2H, **3**),
 2.26–1.11 (br, 176H, **4-13**),
 0.4–0.79 (br, 3H, **14**)

DP_n (NMR) $\frac{I_5}{5} = \frac{242.19}{5} \approx 48$

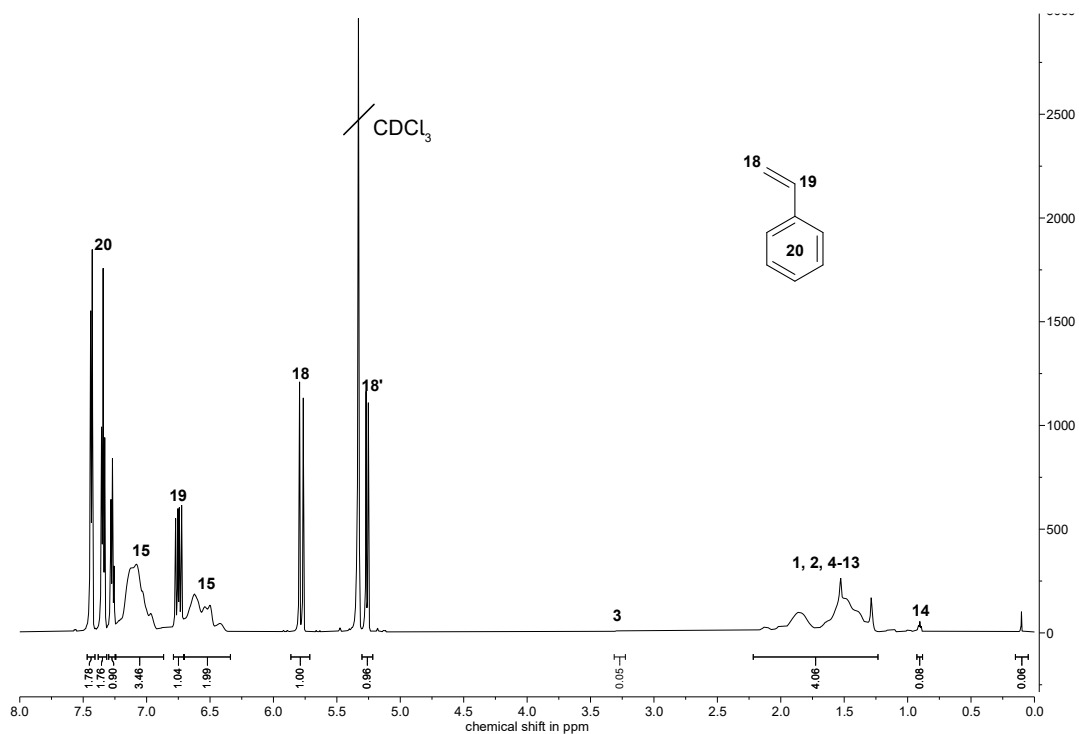


Figure 62 - ^1H NMR (600 MHz, CD_2Cl_2) of non-purified high-molecular-weight **pSt-CPDT**.

Conversion

$$\frac{(3.46+1.99)/5}{(3.46+1.99)/5+1} = 0,52$$

Time

24 h

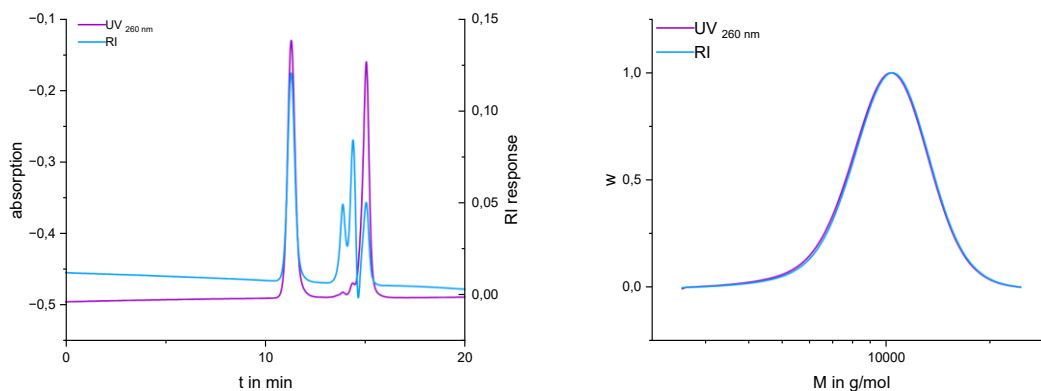


Figure 63 - **GPC** (THF) trace and molecular-weight distribution of high-molecular-weight **pSt-CPDT** (UV purple, RI blue).

t_R	11.3 min
M_n	10588 (UV), 10758 (RI)
M_w	11587 (UV), 11720 (RI)
$\bar{D}$	1.09

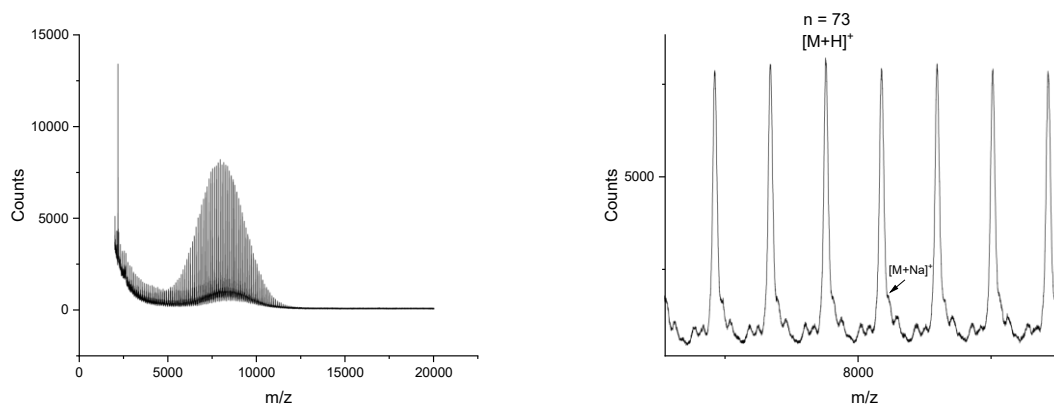


Figure 64 - **MALDI-TOF-MS** spectrum (DCBT 2:1) of high-molecular-weight **pSt-CPDT** with an enlarged section for assignment of the signals.

MALDI-TOF-MS m/z repeating unit: 104.1 , theor.: 104.1
theoretical m/z for $n = 73$: $[M+H]^+$ 7941.732
found: $[M+H]^+$ 7939.299

6.3 Syntheses of LpMe Block Copolymers

pSt-*b*-pLpMe

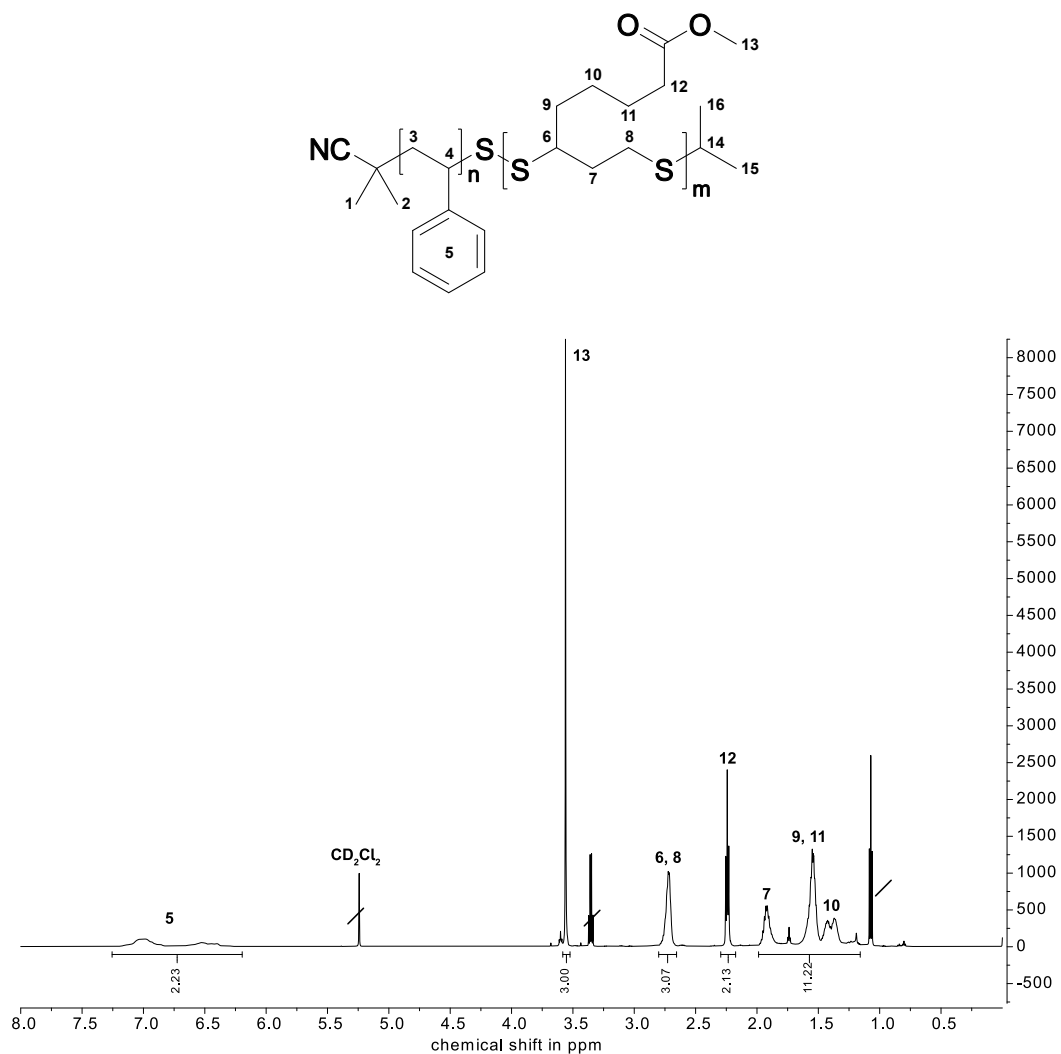


Figure 65 - ^1H NMR (600 MHz, CD_2Cl_2) of pSt-*b*-pLpMe.

^1H NMR	7.26–6.20 (br, 83H, 5),
	3.56 (s, 112H, 13),
	2.80–2.66 (br s, 114H, 6, 8),
	2.24 (t, 79H, 12),
	1.99–1.21 (br, 403H, 7, 9, 10, 11),
	1.19 (s, 6H, 1, 2)

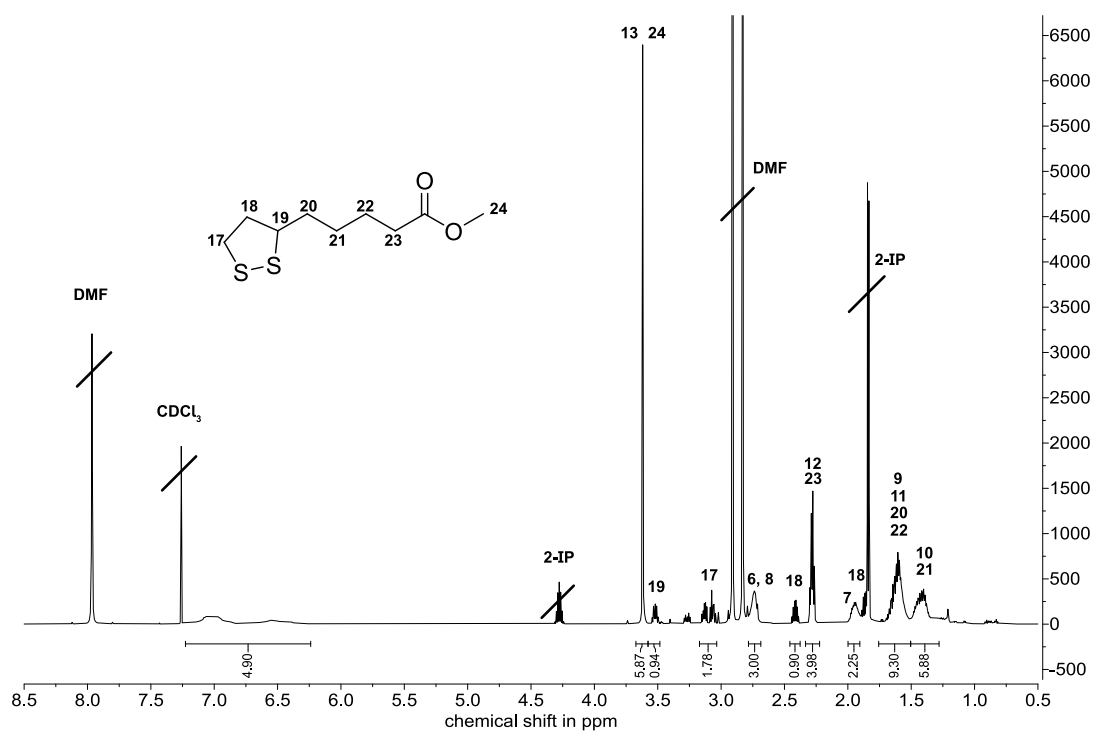


Figure 66 - ^1H NMR (600 MHz, CDCl_3) of the reaction mixture for the polymerization of **pSt-*b*-pLpMe** before purification.

Conversion

$$\frac{\frac{I_{6,8}}{3}}{\frac{I_{6,8}}{3} + I_{18}} = \frac{\frac{3.0}{3}}{\frac{3.0}{3} + 0.9} = 0.52$$

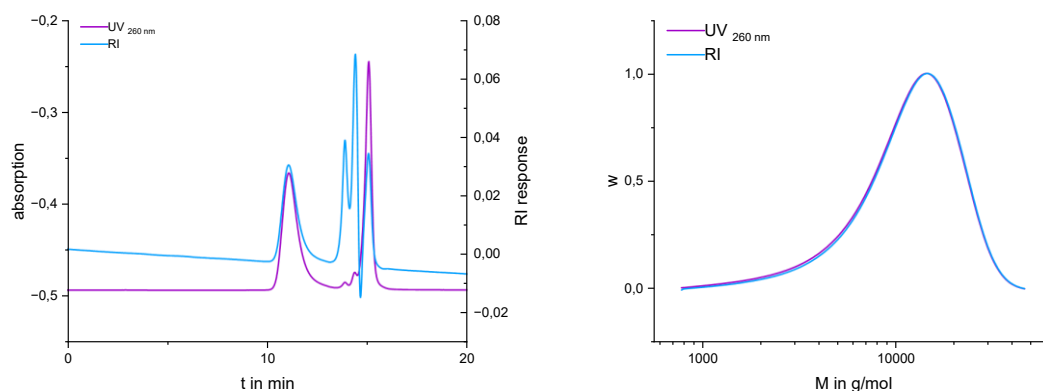


Figure 67 - GPC (THF) trace and molecular-weight distribution of **pSt-*b*-pLpMe** (UV purple, RI blue).

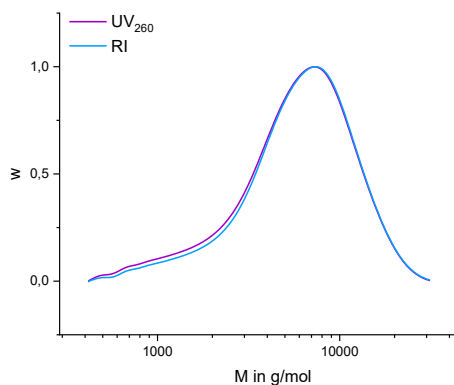


Figure 68 - Molecular-weight distribution from **GPC (THF)** of **pSt-*b*-pLpMe** after precipitation from methanol, before precipitation from diethyl ether (UV purple, RI blue).

t_R	11.1 min
M_n	8844 (UV), 9340 (RI)
M_w	13499 (UV), 13768 (RI)
$\bar{D}$	1.53 (UV), 1.47 (RI)

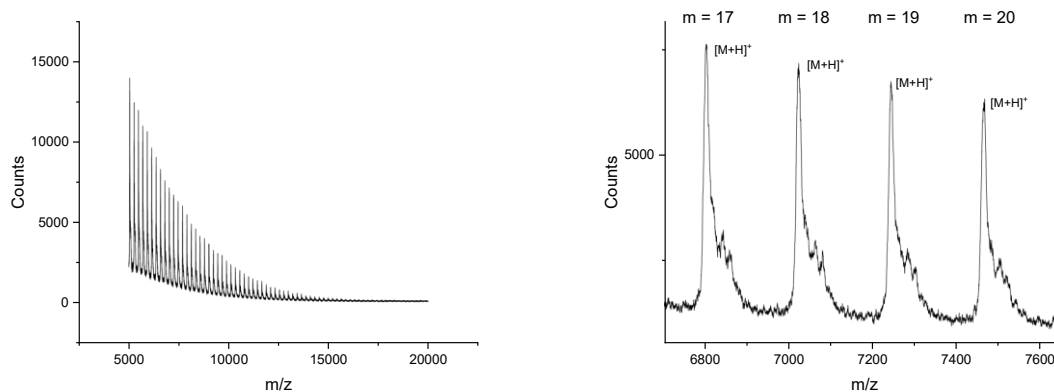
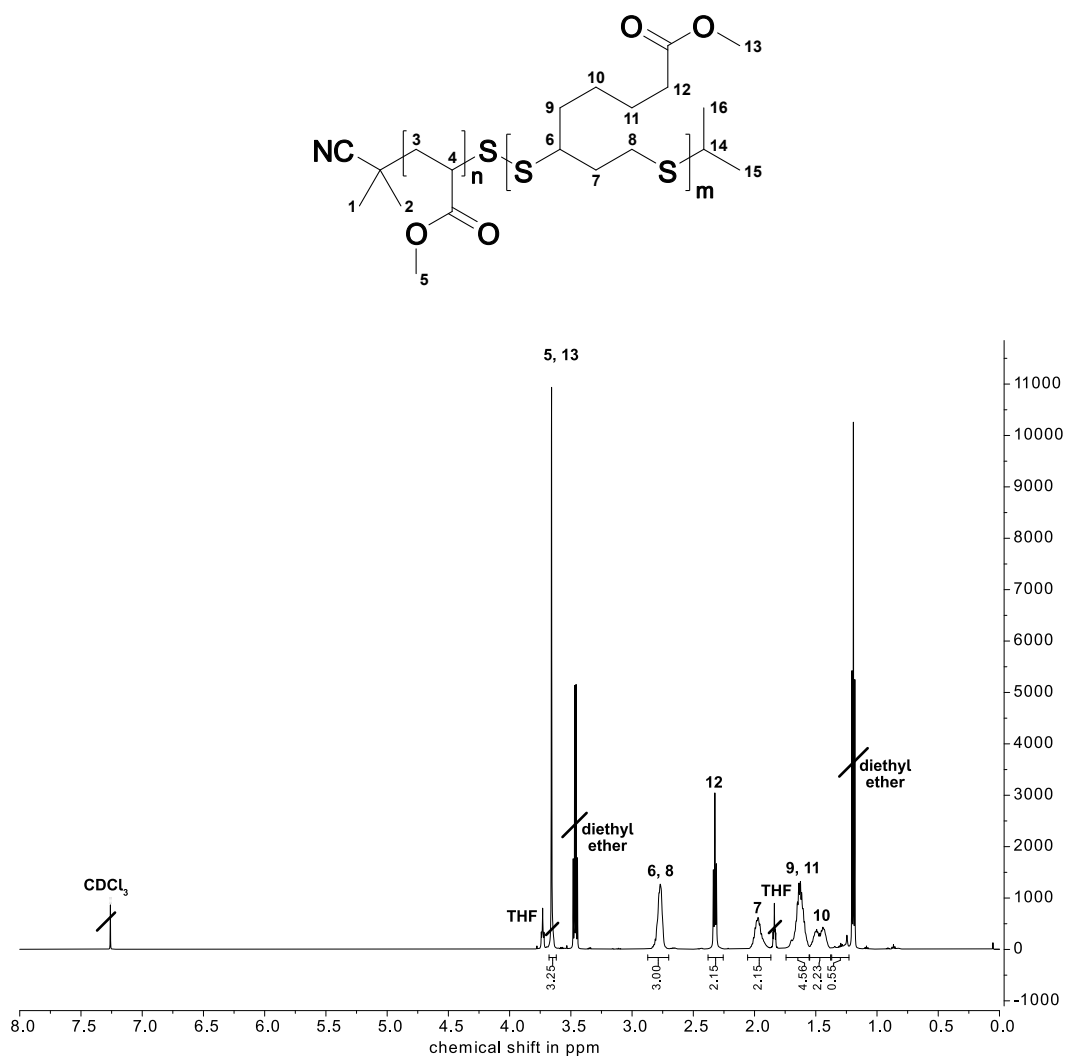
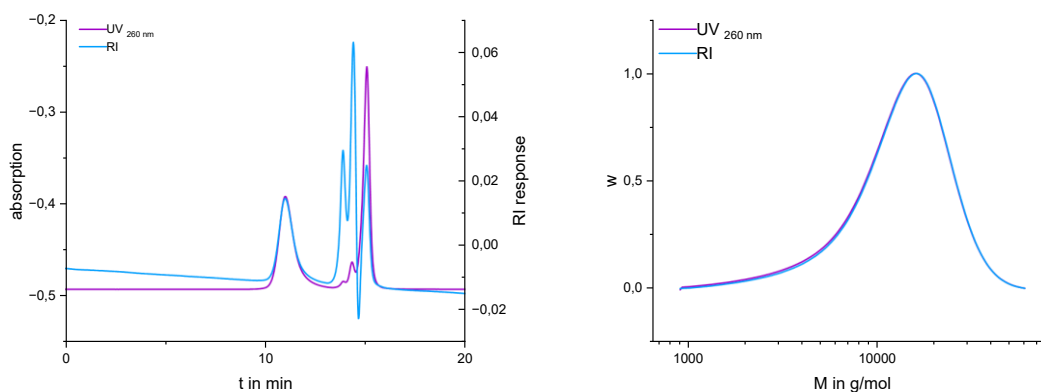
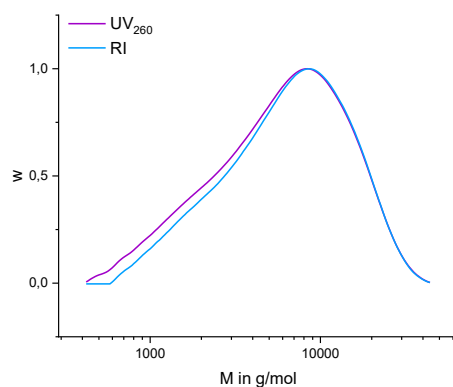


Figure 69 - **MALDI-TOF-MS** spectrum (DCBT 1:2, linear) of **pSt-*b*-pLpMe** with an enlarged section for assignment of the signals.

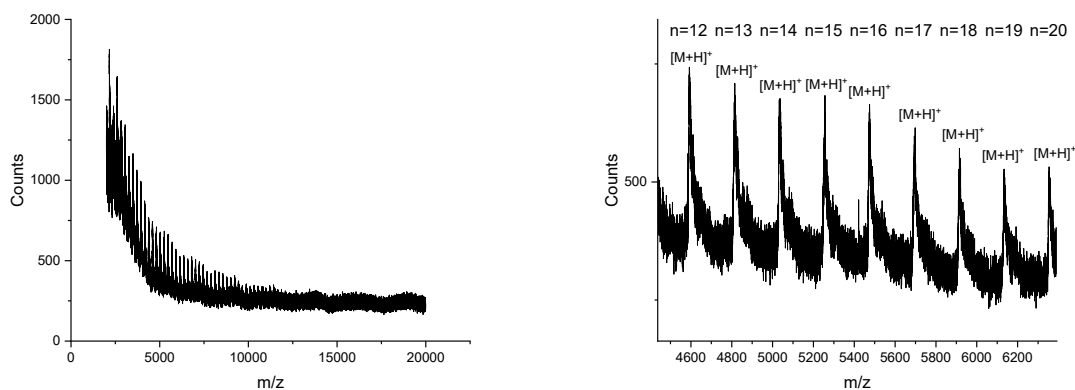
MALDI-TOF-MS	m/z repeating unit: 220.5 , theor.: 220.3
	theoretical m/z for $n = 28$, $m = 17$: $[M+H]^+$ 6798.8
	found: 6803.2 (peak)

pMA-*b*-pLpMeFigure 70 - ^1H NMR (600 MHz, CDCl_3) of pMA-*b*-pLpMe.

^1H NMR	3.65 (s, 88H, 5, 13),
	2.87–2.70 (br s, 81H, 6, 8),
	2.32 (t, 58H, 12),
	2.06–1.87 (br, 58H, 7),
	1.74–1.55 (br, 124H, 9, 11),
	1.55–1.38 (br, 60H, 10),
	1.25 (s, 6H, 1, 2)

Figure 71 - GPC (THF) trace and molecular-weight distribution of **pMA-*b*-pLpMe** (UV purple, RI blue).Figure 72 - Molecular-weight distribution from **GPC (THF)** of **pMA-*b*-pLpMe** after precipitation from methanol, before precipitation from diethyl ether (UV purple, RI blue).

t_R	11.0 min
M_n	10167 (UV), 10668 (RI)
M_w	15470 (UV), 15684 (RI)
D	1.52 (UV), 1.47 (RI)

Figure 73 - **MALDI-TOF-MS** spectrum (DCBT 1:2, linear) of **pMA-*b*-pLpMe** with an enlarged section for assignment of the signals.

MALDI-TOF-MS m/z repeating unit: 220.2 , theor.: 220.3
 theoretical m/z for $n = 21$, $m = 12$: $[M+H]^+$ 4592.5
 found: 4591.6 (peak)

pMMA-*b*-pLpMe

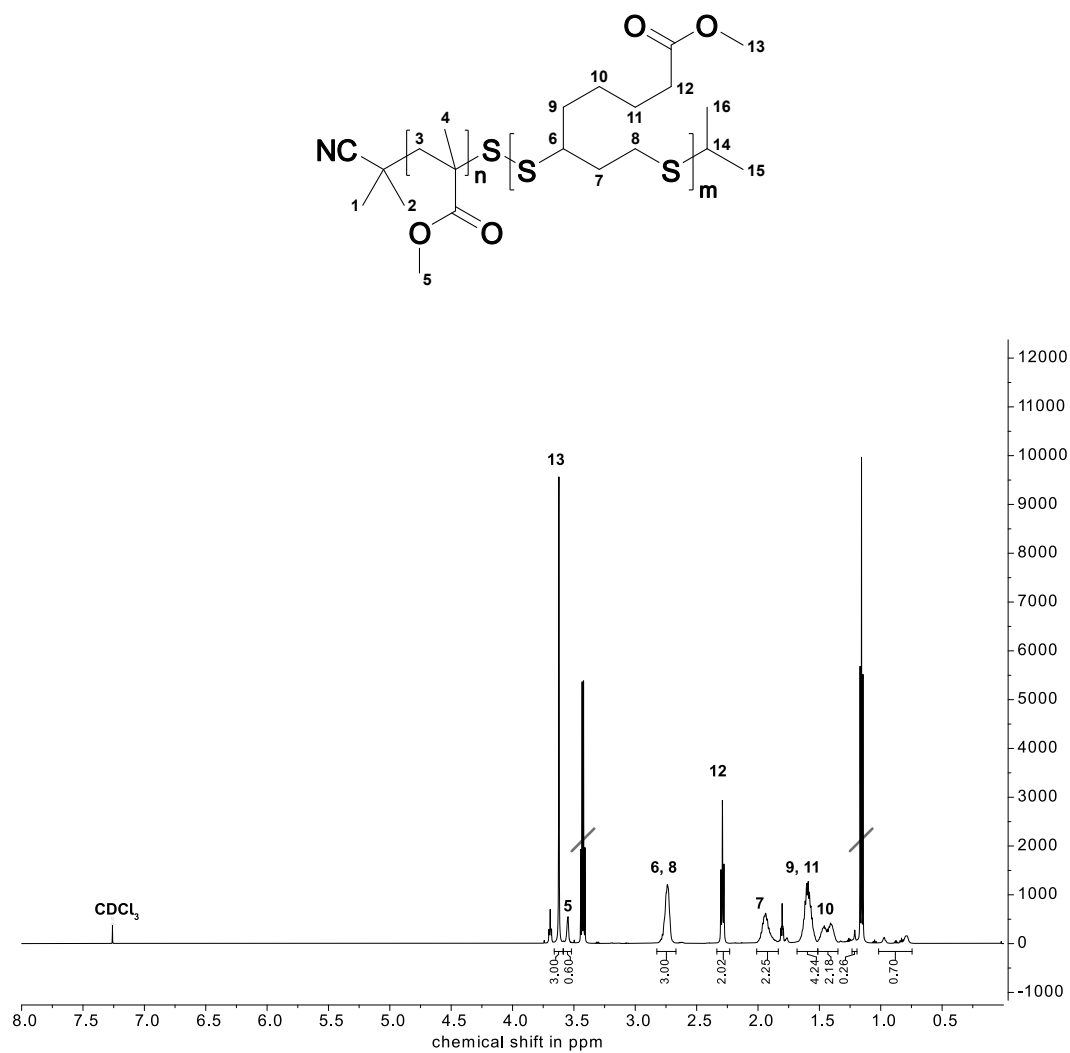


Figure 74 - 1H NMR (600 MHz, $CDCl_3$) of pMMA-*b*-pLpMe.

¹H NMR

3.62 (s, 89H, **13**),
 3.55 (s, 18H, **5**),
 2.82–2.67 (br, 89H, **6, 8**),
 2.29 (t, 60H, **12**),
 2.01–1.84 (br, 67H, **7**),
 1.68–1.51 (br, 126H, **9, 11**),
 1.51–1.35 (br, 65H, **10**),
 1.21 (s, 6H, **1, 2**),
 0.99–0.95 (br, 5H, **4**),
 0.82–0.75 (br, 10H, **4**)

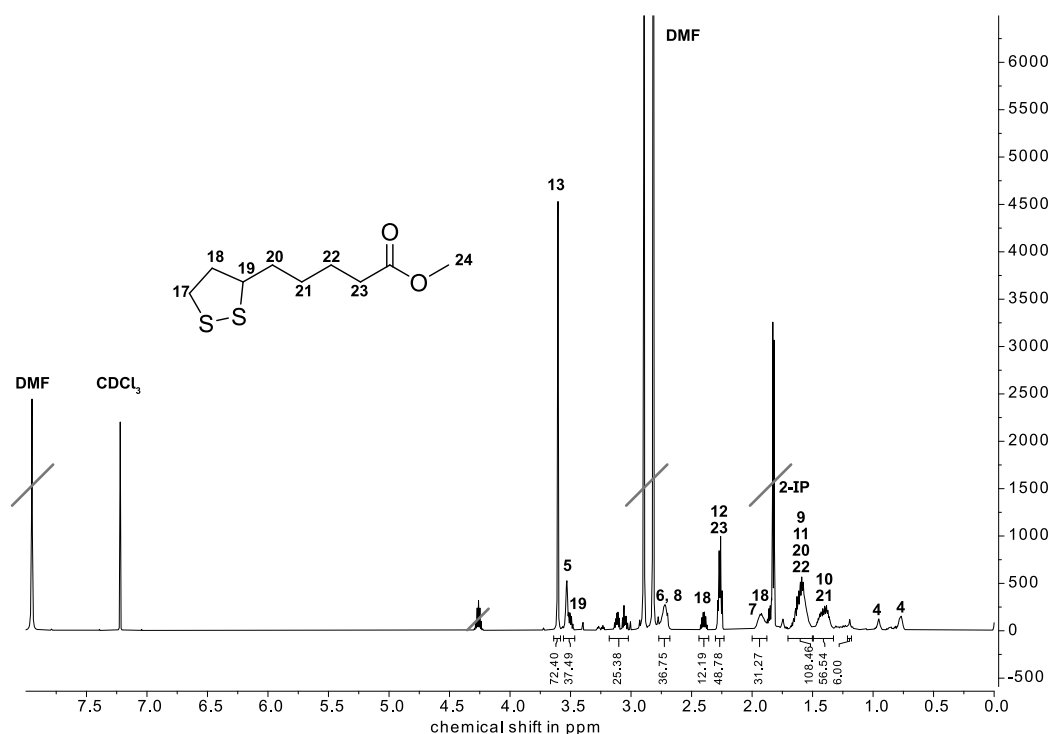


Figure 75 - ¹H NMR (600 MHz, CDCl₃) of the reaction mixture for the polymerization of **pMMA-*b*-pLpMe** before purification.

Conversion

$$\frac{\frac{I_{6,8}}{3}}{\frac{I_{6,8}}{3} + I_{18}} = \frac{\frac{36.75}{3}}{\frac{36.75}{3} + 12.19} = 0.52$$

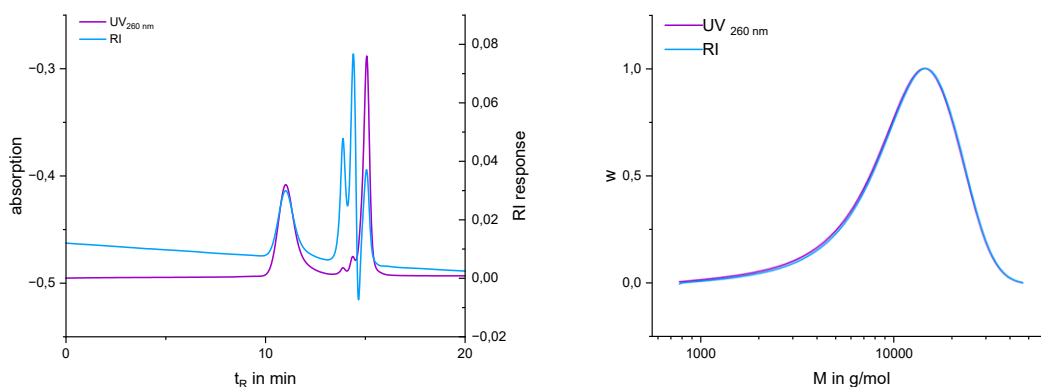


Figure 76 - GPC (THF) trace and molecular-weight distribution of **pMMA-*b*-pLpMe** (UV purple, RI blue).

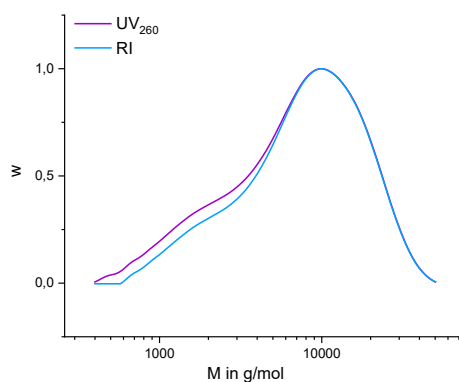


Figure 77 - Molecular-weight distribution from GPC (THF) of **pMMA-*b*-pLpMe** after precipitation from methanol, before precipitation from diethyl ether (UV purple, RI blue).

t_R	11.0 min
M_n	9972 (UV), 10499 (RI)
M_w	15937 (UV), 16166 (RI)
$\bar{D}$	1.60 (UV), 1.54 (RI)

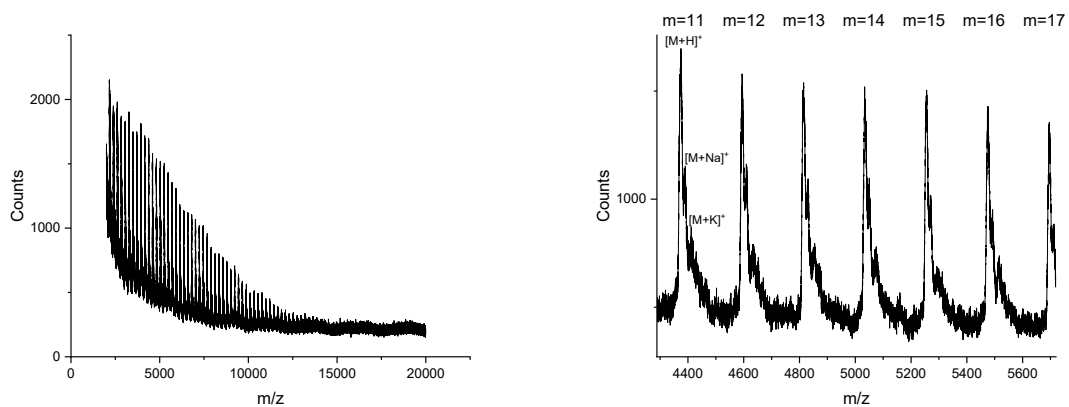
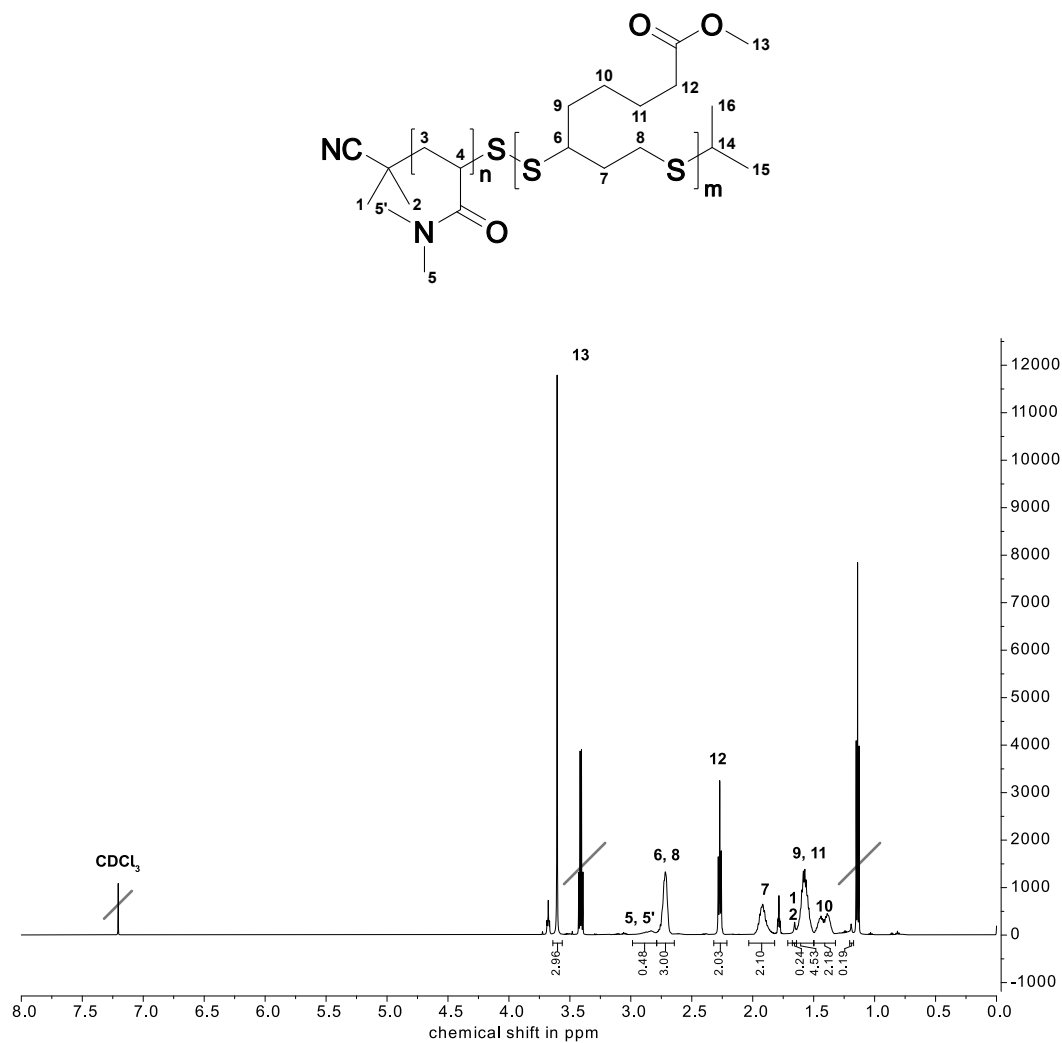


Figure 78 - **MALDI-TOF-MS** spectrum (DCBT 1:2, linear) of **pMMA-*b*-pLpMe** with an enlarged section for assignment of the signals.

MALDI-TOF-MS m/z repeating unit: 220.5 , theor.: 220.3
theoretical m/z for $n = 16$, $m = 11$: $[M+H]^+$ 4385.6
found: 4374.6 (peak)

pDMAA-*b*-pLpMeFigure 79 - ^1H NMR (600 MHz, CDCl_3) of pDMAA-*b*-pLpMe.

^1H NMR	3.60 (s, 81H, 13),
	2.99–2.79 (br, 13H, 5 , 5'),
	2.79–2.64 (br, 82H, 6 , 8),
	2.27 (t, 55H, 12),
	2.03–1.82 (br, 57H, 7),
	1.71–1.50 (br, 123H, 9 , 11),
	1.50–1.32 (br, 59H, 10),
	1.19 (s, 6H, 1 , 2)

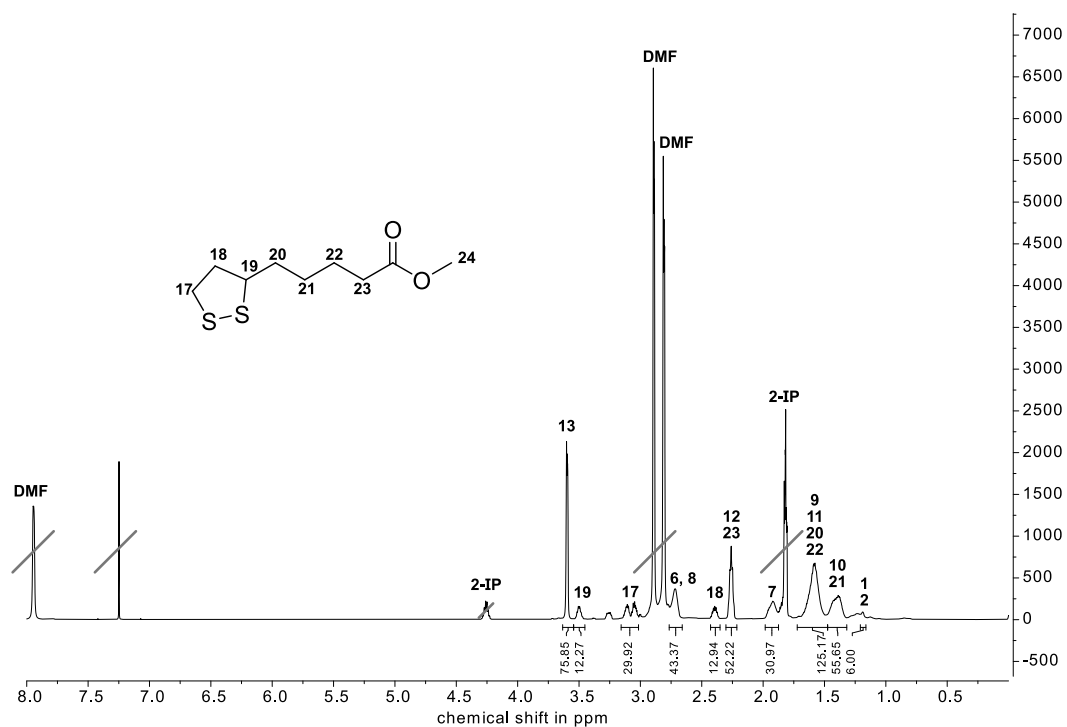


Figure 80 - ^1H NMR (600 MHz, CDCl_3) of the reaction mixture for the polymerization of **pDMAA-*b*-pLpMe** before purification (2-IP = 2-iodopropane).

Conversion $\frac{\frac{I_{6,8}}{3}}{\frac{I_{6,8}}{3} + I_{18}} = \frac{\frac{43.37}{3}}{\frac{43.37}{3} + 12.94} = 0.53$

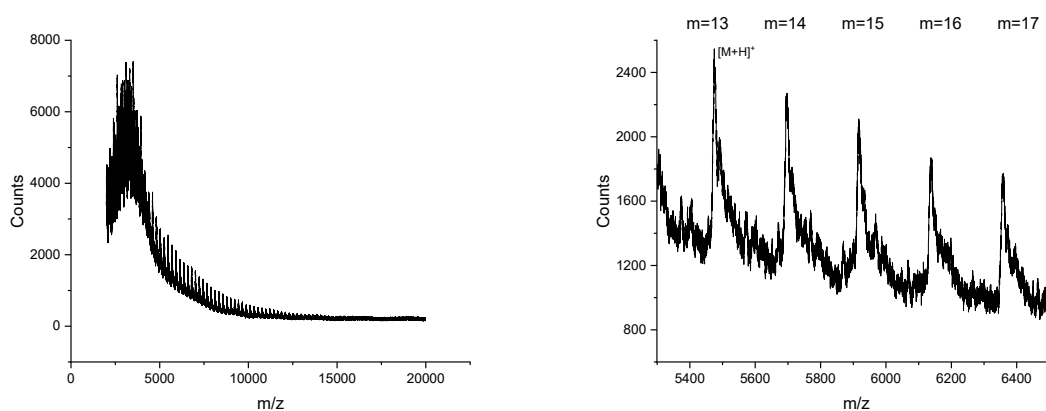
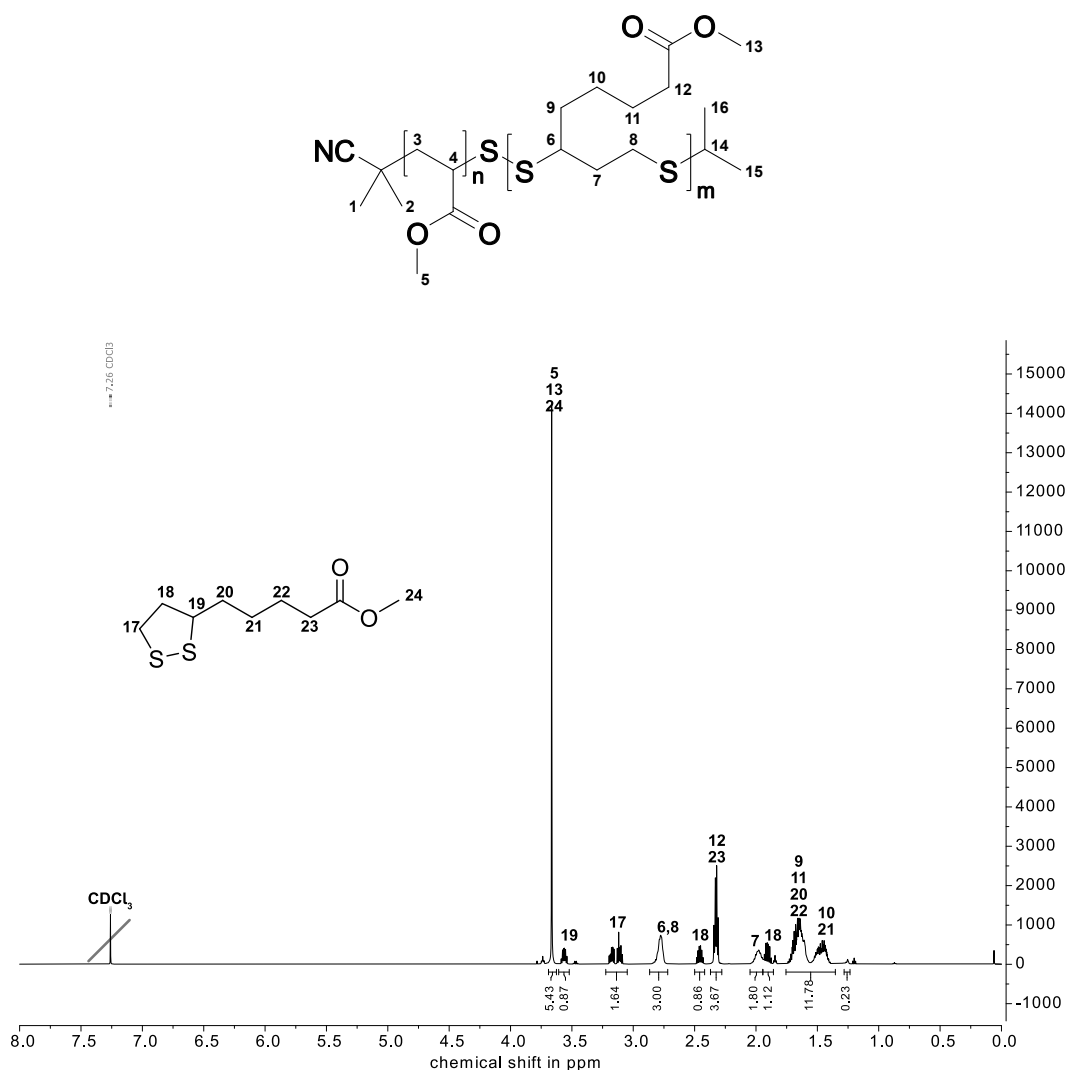


Figure 81 - **MALDI-TOF-MS** spectrum (DCBT 1:2, linear) of **pDMAA-*b*-pLpMe** with an enlarged section for assignment of the signals.

MALDI-TOF-MS m/z repeating unit: 220.4 , theor.: 220.3
theoretical m/z for $n = 16$, $m = 13$: $[M+H]^+$ 5481.6
found: 5474.9 m/z (peak)

pMA-*b*-pLpMe-HMWFigure 82 - ^1H NMR (600 MHz, CDCl_3) of higher-molecular-weight **pMA-*b*-pLpMe**.

^1H NMR	3.66 (s, 164H, 5, 13, 24),
	3.56 (dq, $J = 8.8, 6.5$, 31H, 19),
	3.18 (ddd, $J = 11.0, 7.1, 5.3$, 59H, 17),
	3.11 (dt, $J = 11.0, 6.9$, 1H, 17),
	2.87–2.72 (br, 108H, 6, 8),
	2.46 (dtd, $J = 13.0, 6.6, 5.3$, 31H, 18),
	2.33 (td, $J = 7.4, 5.8$, 132H, 12, 23),
	2.05–1.94 (br, 65H, 7),
	1.94–1.86 (m, 40H, 18'),
	1.76–1.35 (br, 424H, 9-11, 20-22),
	1.26–1.23 (br, 6H, 1, 2)

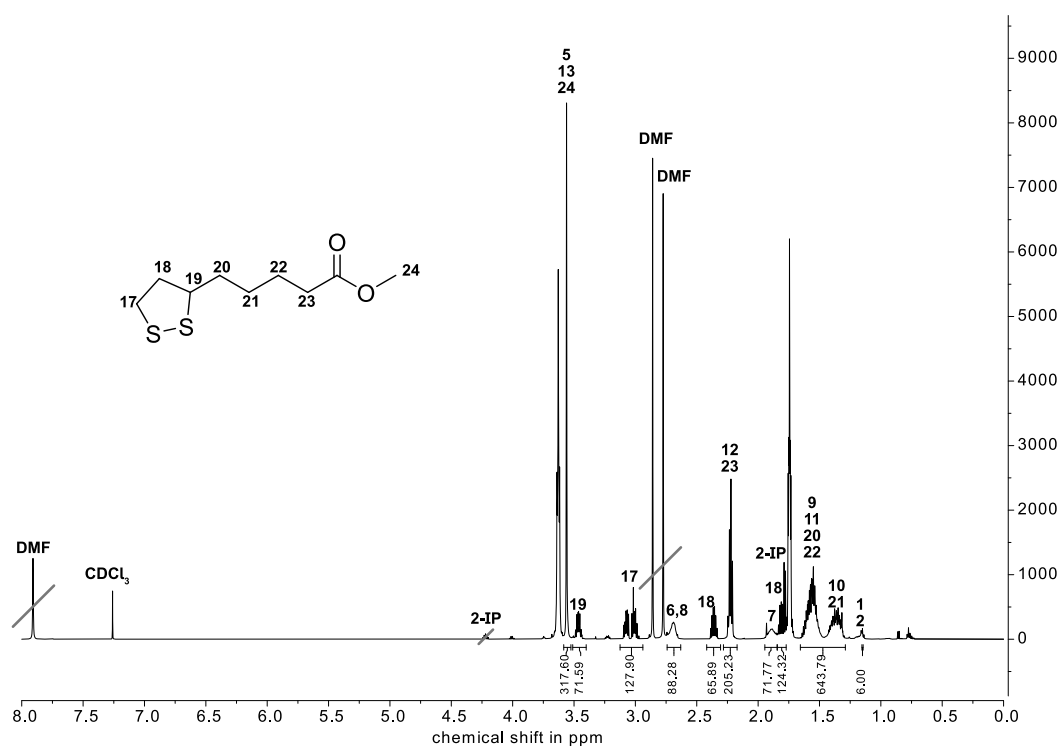


Figure 83 - ^1H NMR (600 MHz, CDCl_3) of the reaction mixture for the polymerization of higher-molecular-weight **pMA-b-pLpMe** before purification (2-IP = 2-iodopropane).

Conversion

$$\frac{\frac{I_{6,8}}{3}}{\frac{I_{6,8}}{3} + I_{18}} = \frac{\frac{88.28}{3}}{\frac{88.28}{3} + 65.89} = 0.31$$

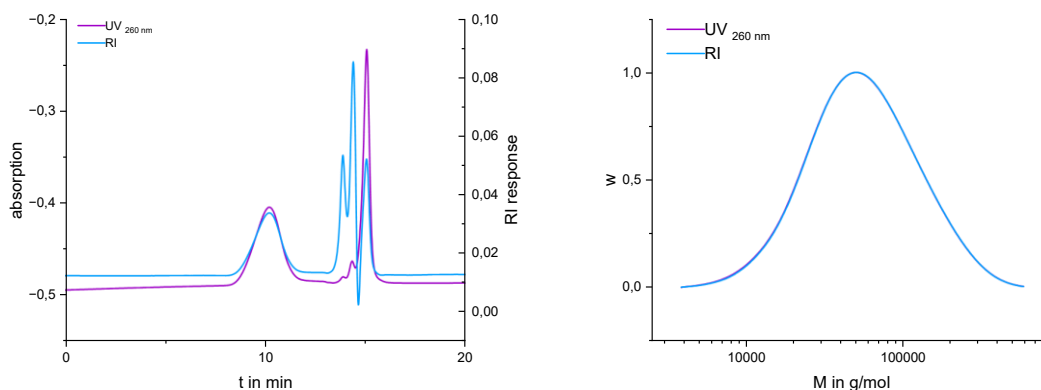
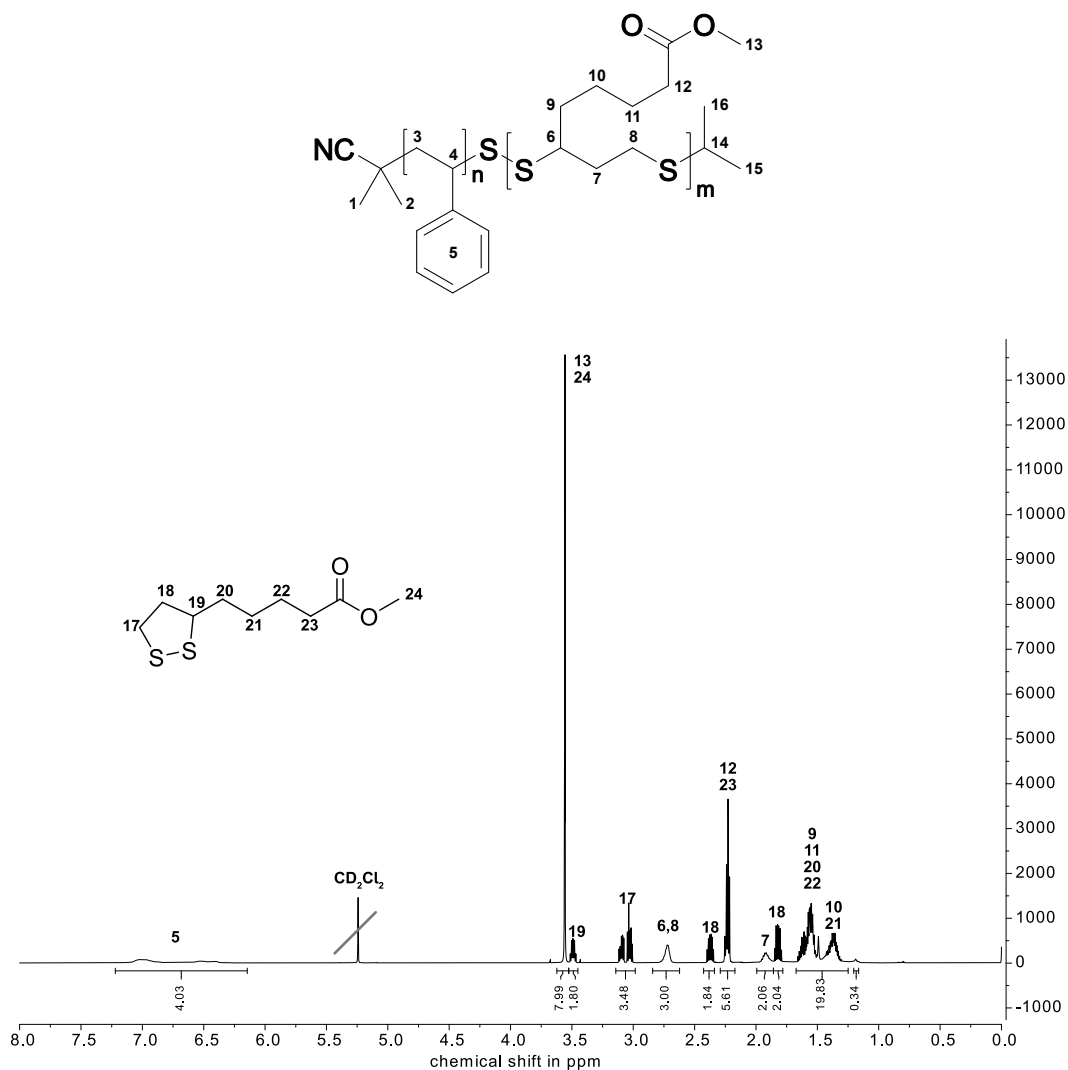


Figure 84 - GPC (THF) trace and molecular-weight distribution of higher-molecular-weight **pMA-b-pLpMe** (UV purple, RI blue).

t_R	10.2 min
M_n	41 004 (UV), 41 438 (RI)
M_w	75 735 (UV), 75 844 (RI)
$\bar{D}$	1.85 (UV), 1.83 (RI)

pSt-*b*-pLpMe-DHMWFigure 85 - 1H NMR (600 MHz, CD_2Cl_2) of pSt-*b*-pLpMe-DHMW.

^1H NMR 7.22–6.15 (br, 83H, **5**),
 3.56 (s, 164H, **13**, **24**),
 3.53–3.45 (m, 37H, **19**),
 3.14–2.98 (m, 72H, **17**),
 2.84–2.62 (br, 62H, **6**, **8**),
 2.43–2.34 (m, 38H, **18**),
 2.29–2.17 (m, 116H, **12**, **23**),
 1.99–1.86 (br, 43H, **7**),
 1.86–1.78 (m, 42H, **18'**),
 1.67–1.25 (br, 409H, **9-11**, **20-22**),
 1.20–1.17 (br, 6H, **1**, **2**)

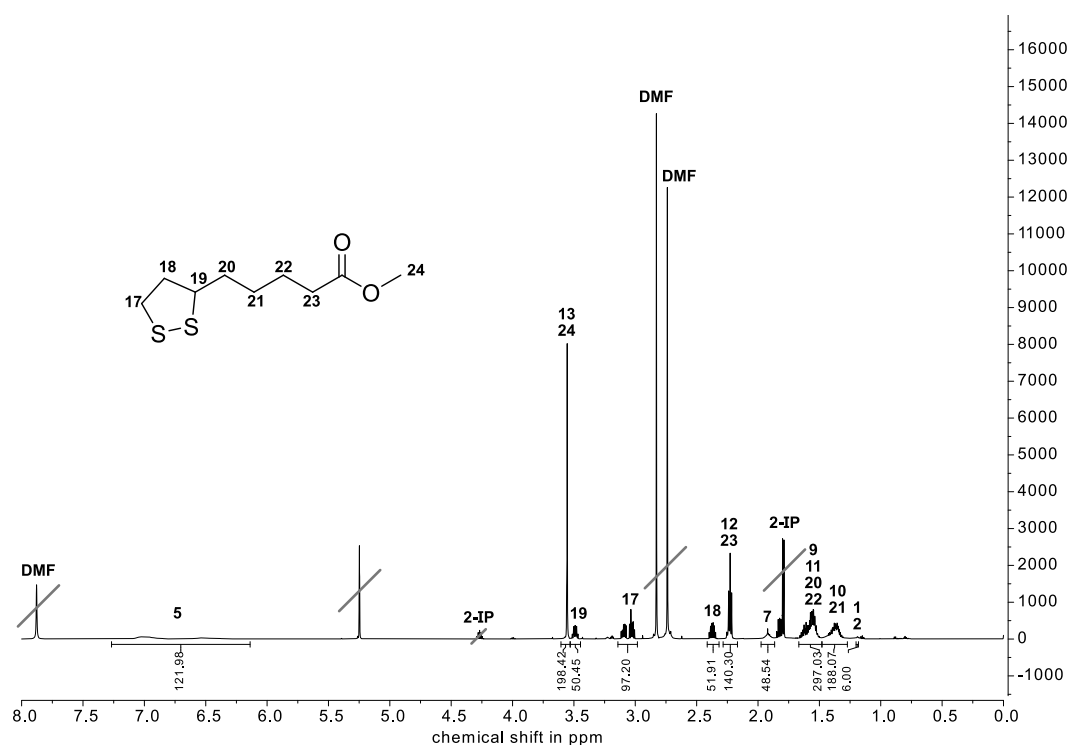


Figure 86 - ^1H NMR (600 MHz, CD_2Cl_2) of the reaction mixture for the polymerization of **pSt-*b*-pLpMe-DHMW** before purification (2-IP = 2-iodopropane).

Conversion

$$\frac{\frac{I_7}{2}}{\frac{I_7}{2} + I_{18}} = \frac{\frac{48,54}{2}}{\frac{48,54}{2} + 51.91} = 0.32$$

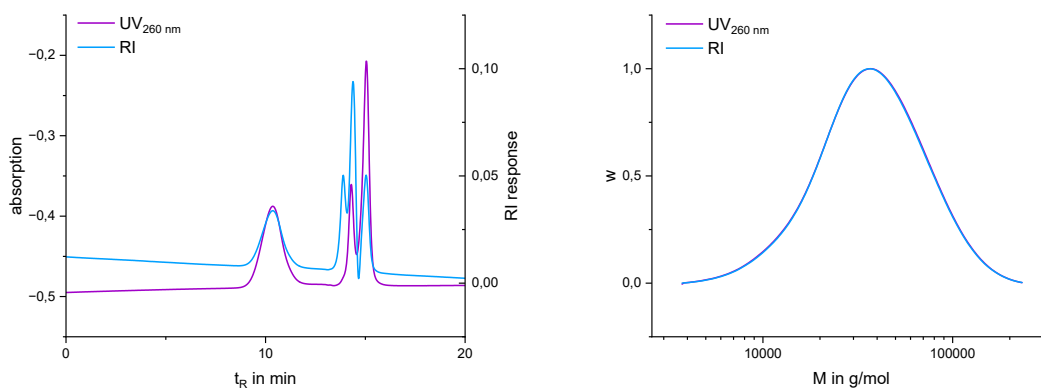
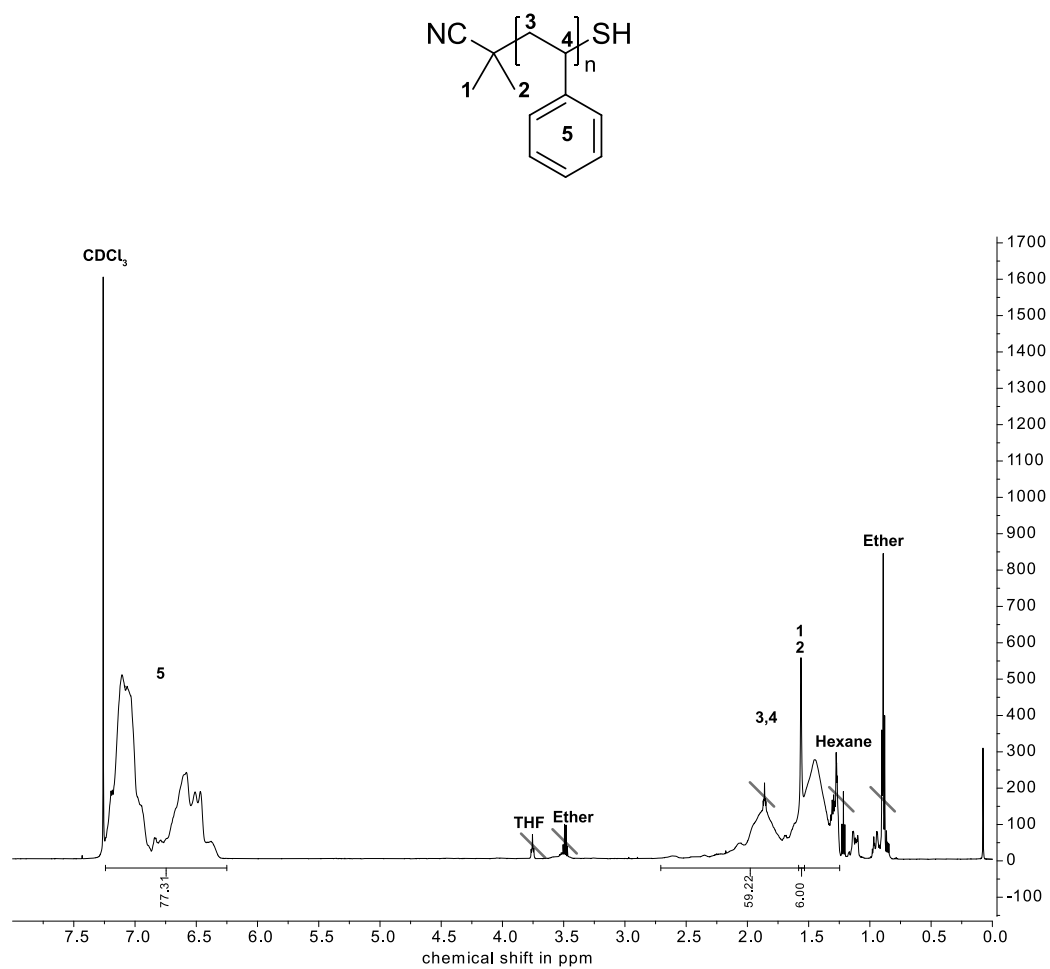


Figure 87 - **GPC** (THF) trace and molecular-weight distribution of higher-molecular-weight **pSt-*b*-pLpMe** (UV purple, RI blue).

t_R	10.4 min
M_n	30 060 (UV), 30 182 (RI)
M_w	45 793 (UV), 45 655 (RI)
$\bar{D}$	1.52 (UV), 1.51 (RI)

6.4 Aminolysis of pSt-CPDT

Figure 88 - ^1H NMR (600 MHz, CDCl_3) of pSt-SH.

^1H NMR 7.24–6.25 (br, 77.3H, **5**),
 2.71–1.25 (br, 60H, **1-4**)

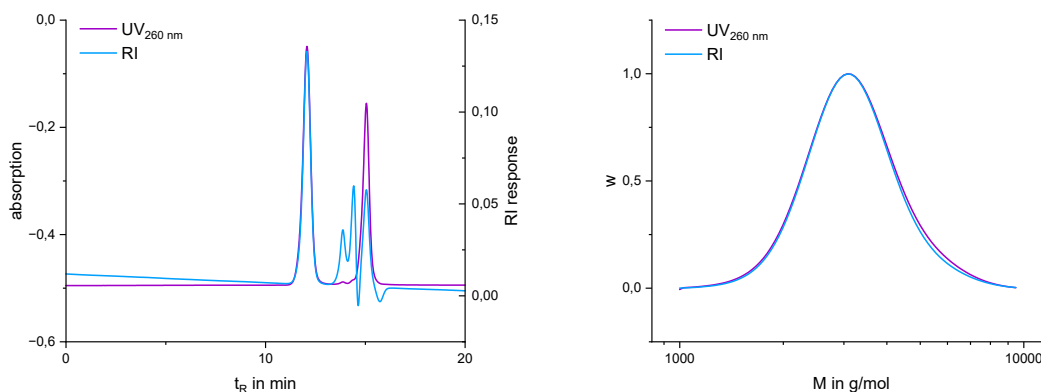
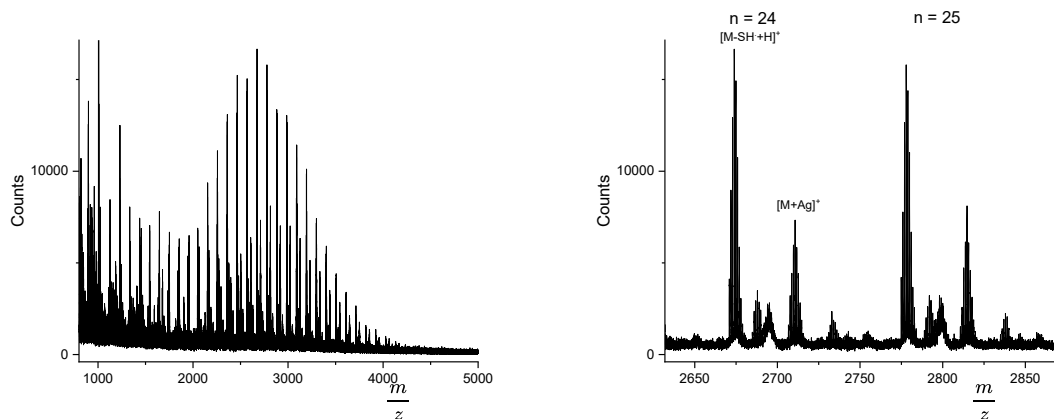


Figure 89 - GPC (THF) trace and molecular-weight distribution of **pSt-SH** obtained after aminolysis of **pSt-CPDT** (UV purple, RI blue).

t_R	12.1 min
M_n	3025 (UV), 3019 (RI)
M_w	3369 (UV), 3338 (RI)
$\bar{D}$	1.11 (UV), 1.11 (RI)



MALDI-TOF-MS m/z repeating unit: found: 104.0, theor.: 104.1
 theoretical m/z for $n = 24$: $[M+Ag]^+$ 2706.7
 found: $[M+Ag]^+$ 2705.4

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